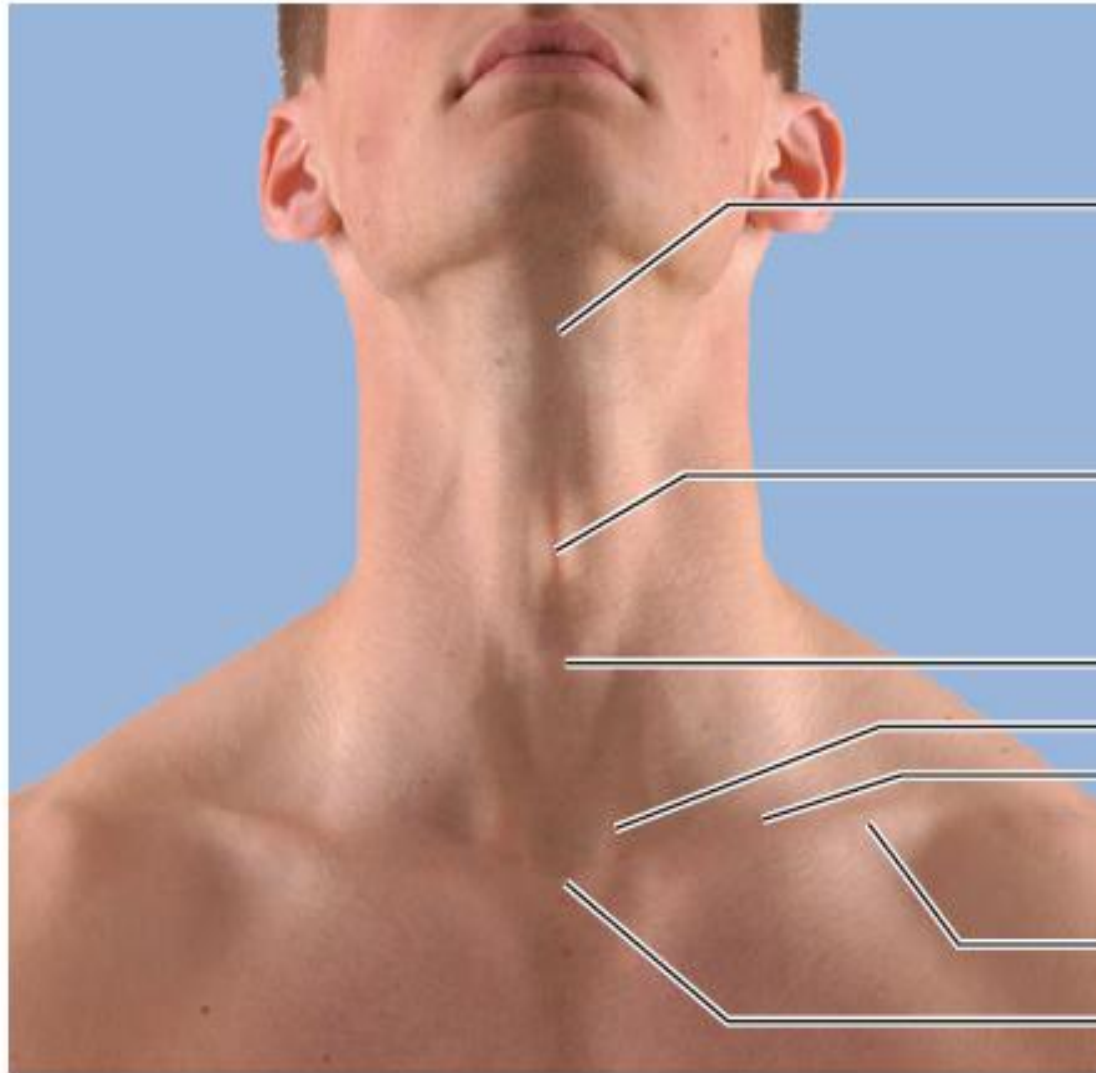
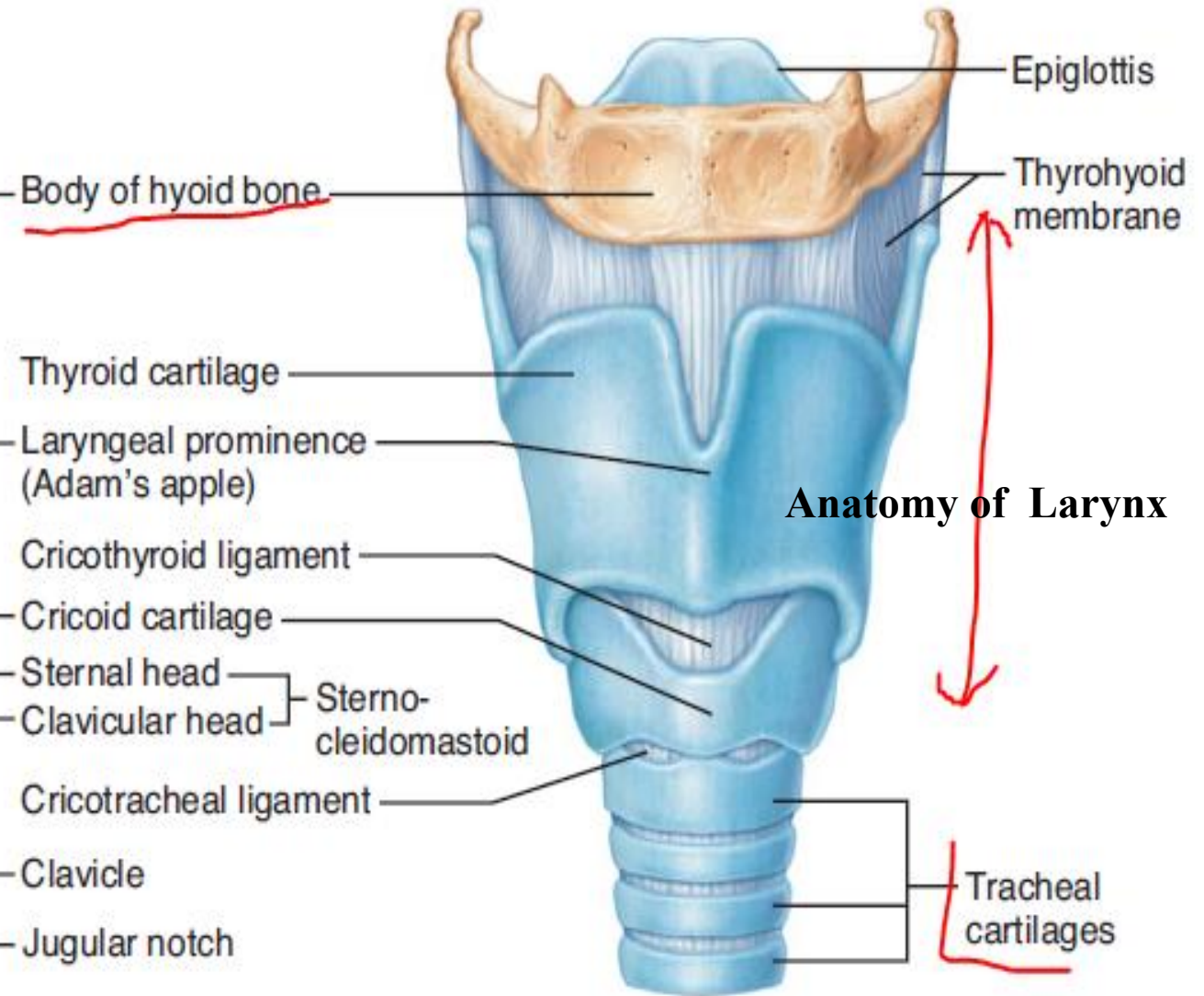


- *Gross structures and functions of the upper respiratory tract*

Anatomy of Larynx



(a) Surface view



(b) Anterior view

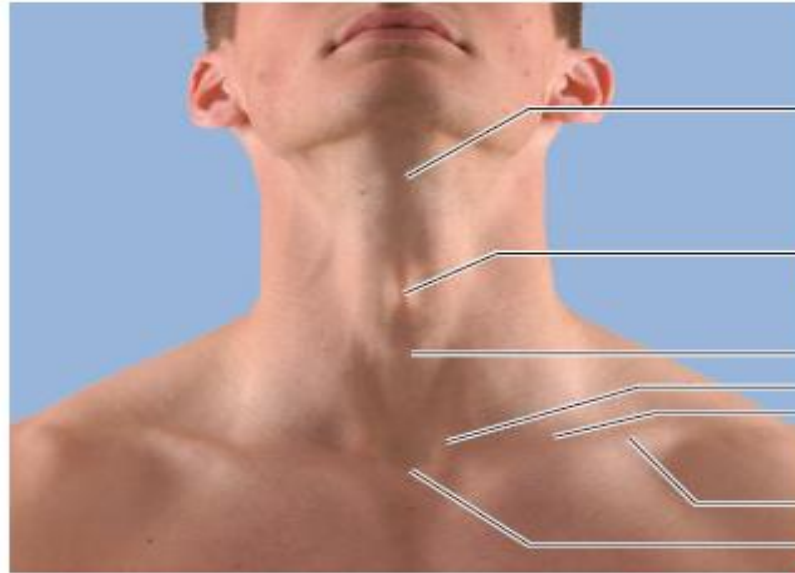
Anatomy of Larynx,,,

- The larynx or voice box, extends from the level of the **4th to the 6th** cervical vertebra.
- **Superiorly, it attaches to the hyoid bone** and opens into the **laryngopharynx; inferiorly**, it is continuous **with the trachea (windpipe).**

□ The larynx has three functions:

- I. producing vocalizations,
 - II. providing an open airway, and
 - III. acting as a switching mechanism to **route air and food** into the proper channels.
- ❖ **For the latter purposes, the inlet (superior opening) to the larynx is closed during swallowing and open during breathing.**

Anatomy of Larynx,,



(a) Surface view

Body of hyoid bone

Thyroid cartilage

Laryngeal prominence
(Adam's apple)

Cricothyroid ligament

Cricoid cartilage

Sternal head

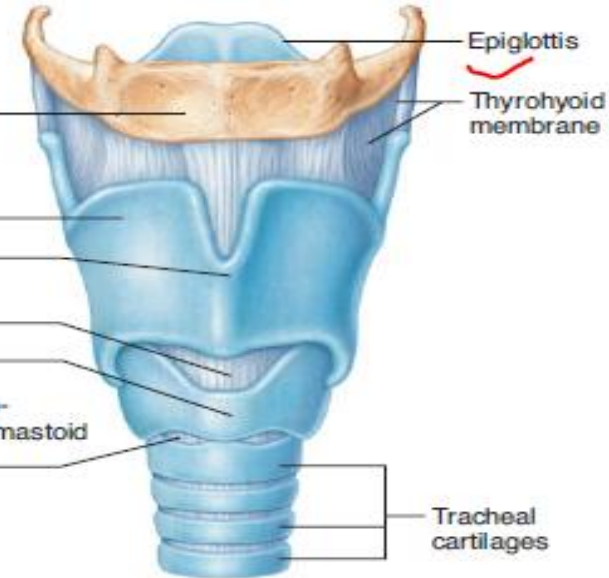
Clavicular head

Stero-
cleidomastoid

Cricotracheal ligament

Clavicle

Jugular notch

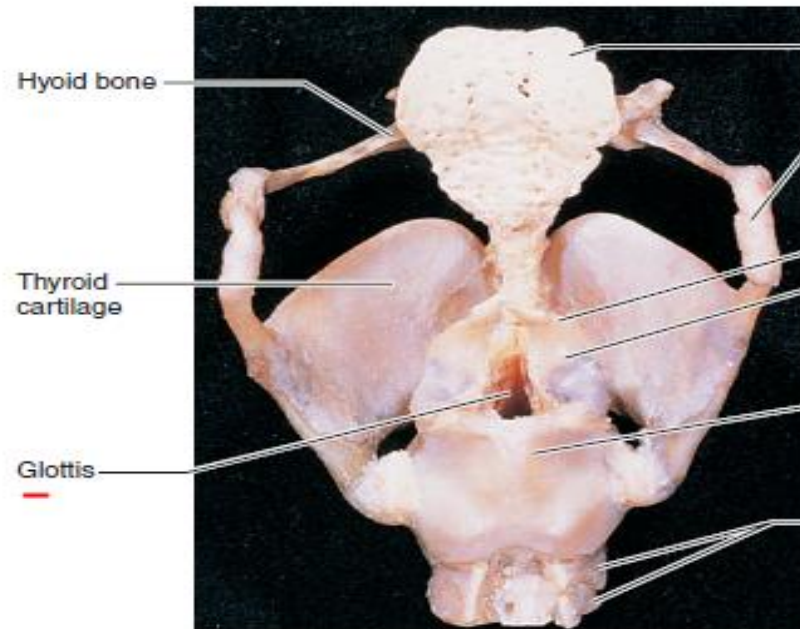


Epiglottis

Thyrohyoid
membrane

Tracheal
cartilages

(b) Anterior view



Hyoid bone

Thyroid
cartilage

Glottis

Epiglottis

Thyrohyoid
membrane

Cuneiform cartilage

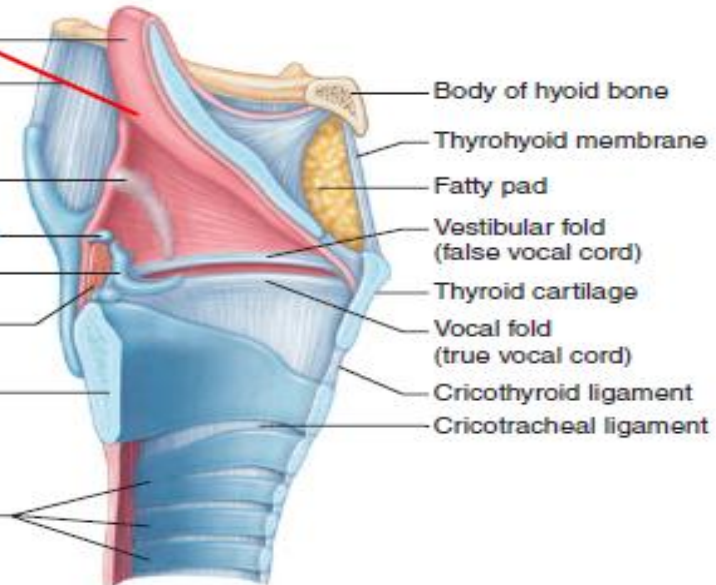
Corniculate cartilage

Arytenoid cartilage

Arytenoid muscle

Cricoid cartilage

Tracheal cartilages



Body of hyoid bone

Thyrohyoid membrane

Fatty pad

Vestibular fold
(false vocal cord)

Thyroid cartilage

Vocal fold
(true vocal cord)

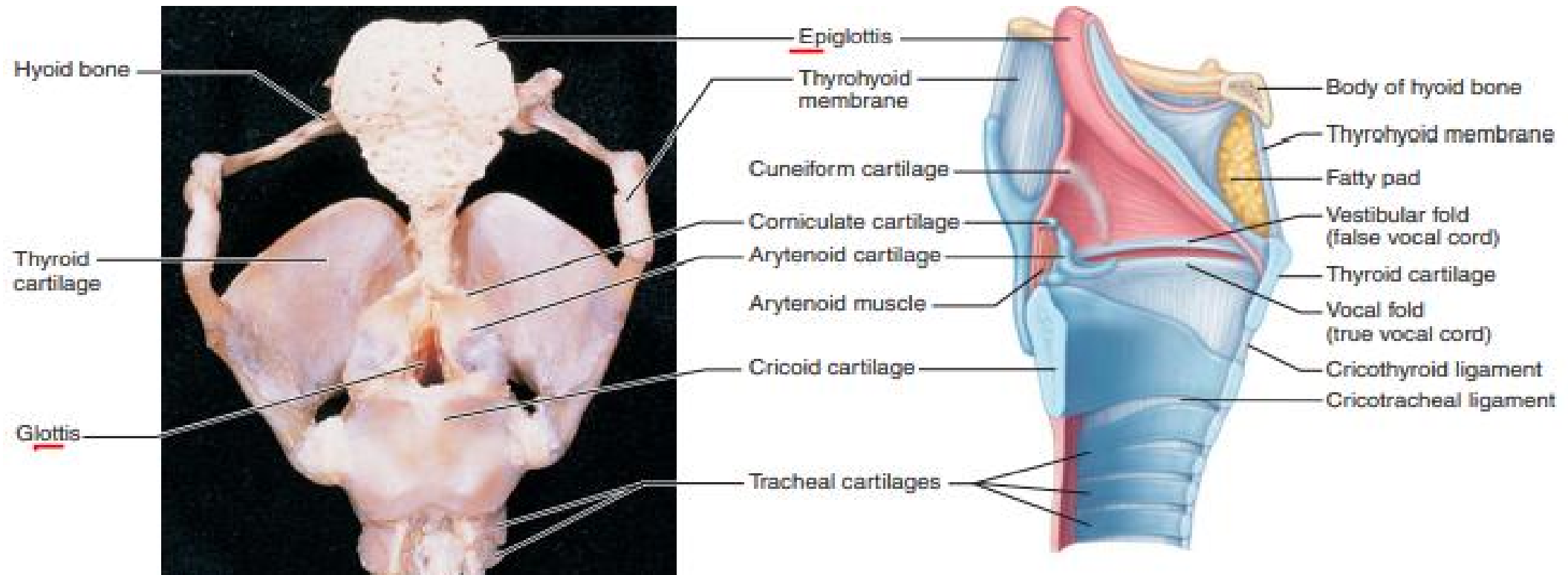
Cricothyroid ligament

Cricotracheal ligament

Anatomy of Larynx,,

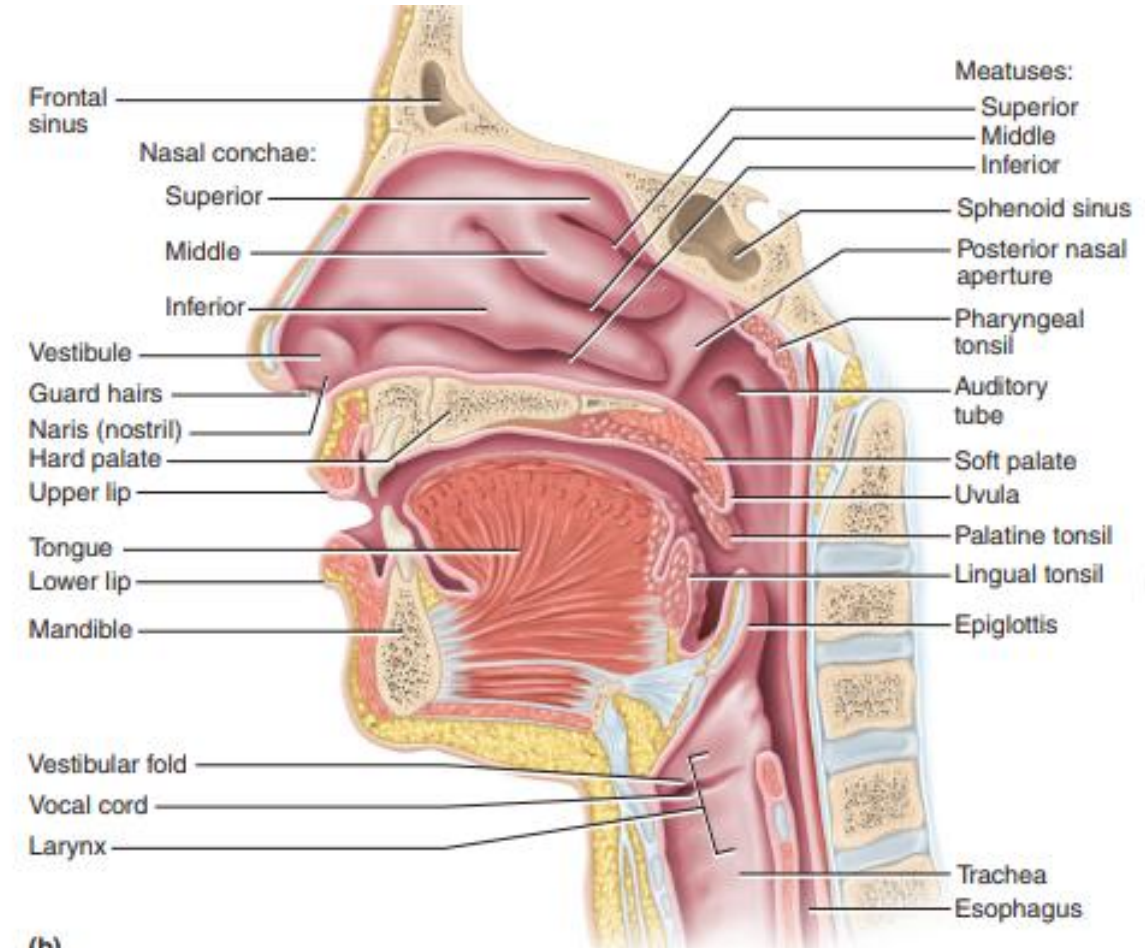
❖ The superior opening of the larynx is guarded by a flap of tissue called the **epiglottis**

- Superiorly continuous with laryngopharynx through the inlet of larynx
- Inferiorly continuous with trachea



Anatomy of Larynx,,

- ❑ At rest, the epiglottis stands almost vertically.
- ❑ During swallowing, however, **extrinsic muscles of the larynx pull the larynx upward toward the epiglottis, the tongue pushes the epiglottis downward to meet it, and the epiglottis closes the airway and directs food and drink into the esophagus behind it.**

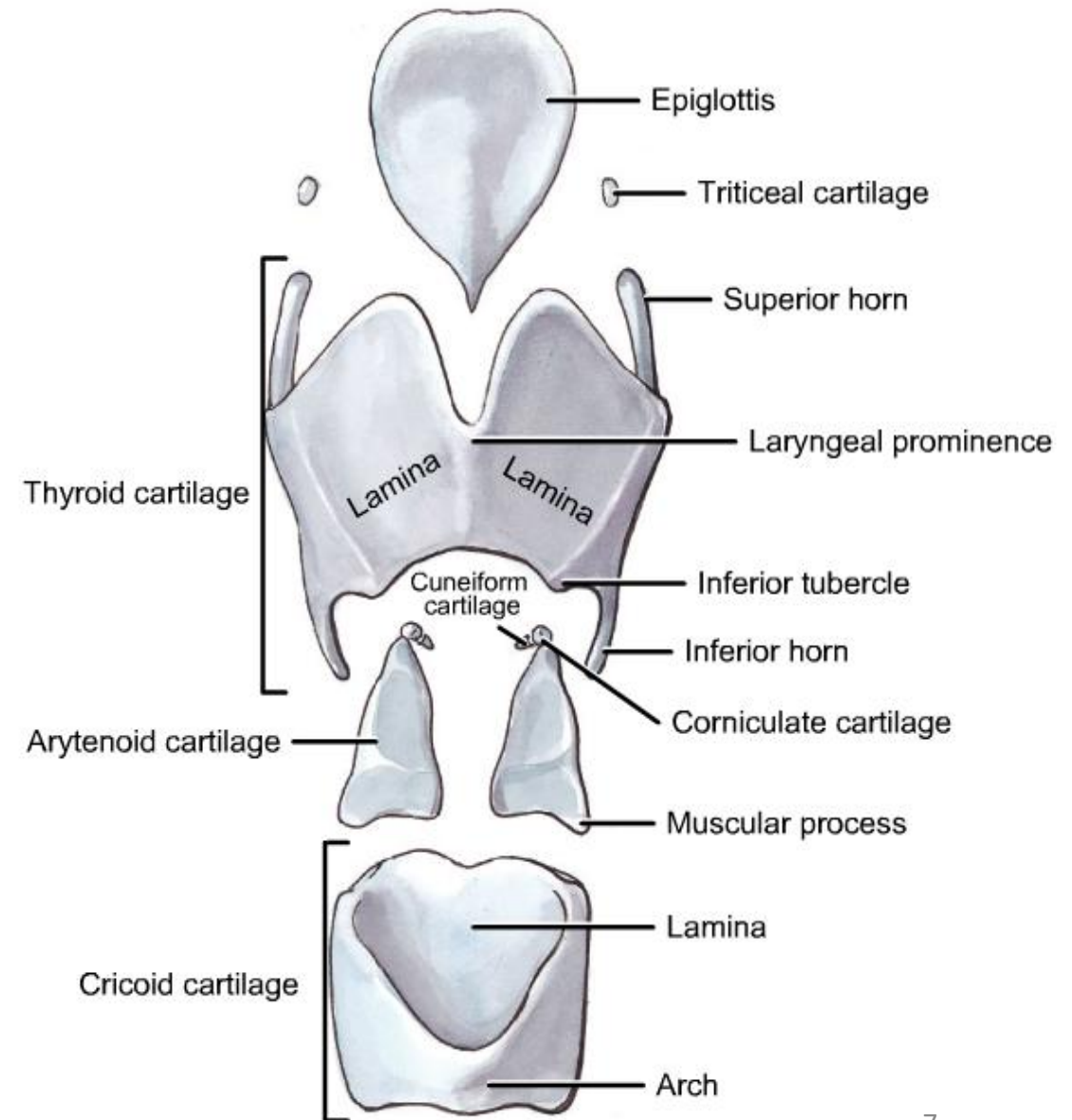


Laryngeal Skeleton

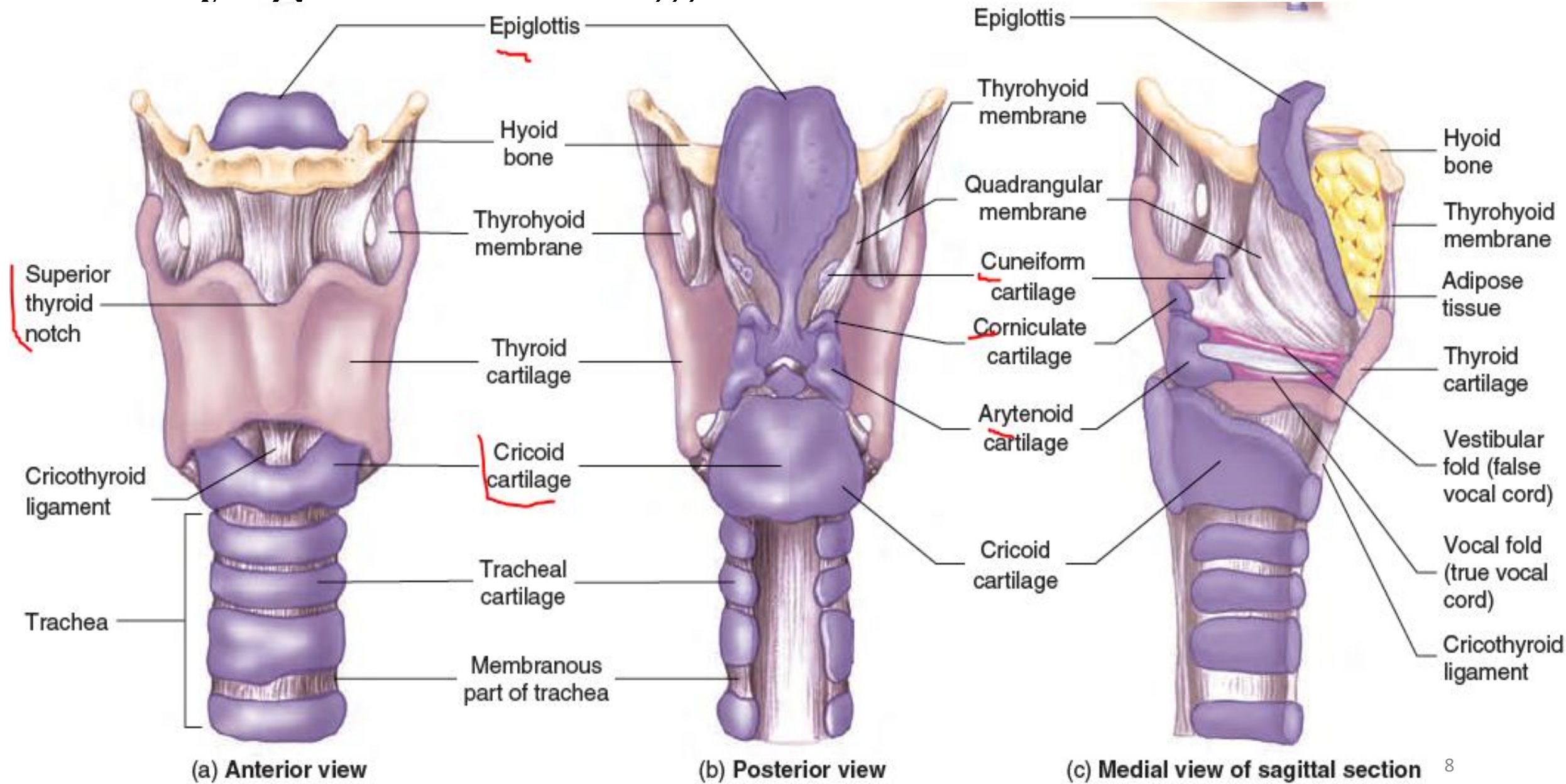
- The larynx is the complex organ of voice production composed of **nine cartilages** **connected by** membranes and ligaments and containing the vocal folds

□ The laryngeal skeleton consists of 9 cartilages:

- **Three are single** (thyroid, cricoid, and epiglottic)
- **Three are paired** (arytenoid, corniculate, and cuneiform)



Laryngeal Skeleton,,,



Laryngeal Skeleton,,

- The cartilages connected to one another by **muscles and ligaments**.

□ **The largest cartilage** is the **unpaired thyroid cartilage**, or Adam's apple.

- The thyroid cartilage is attached superiorly **to the hyoid bone**.

□ **The most inferior** cartilage of the larynx is **the unpaired cricoid(ring-shaped)** cartilage, which forms the base of the larynx on which the other cartilages rest.

- The **thyroid and cricoid cartilages** maintain an open passageway for air movement

Laryngeal Skeleton,,,

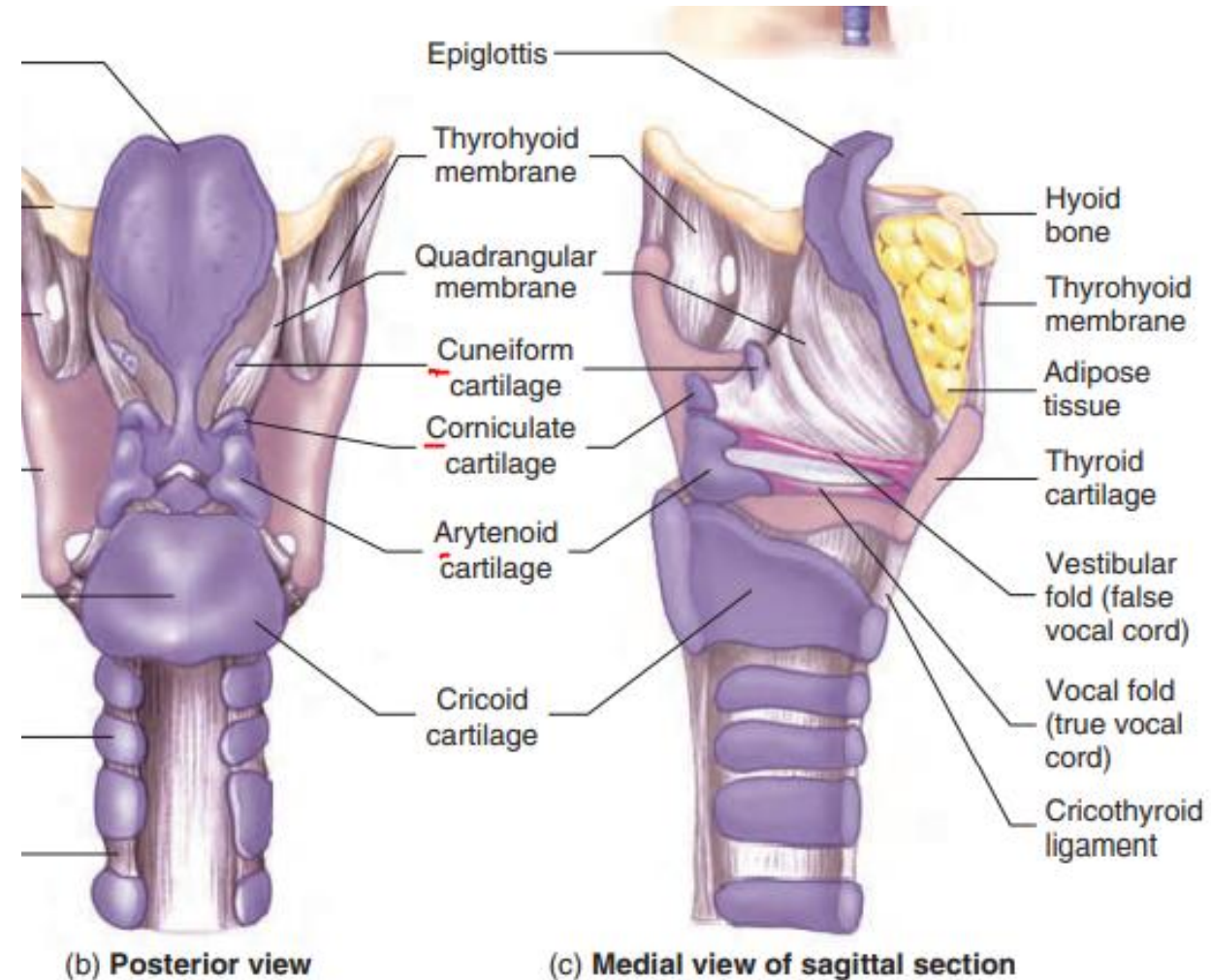
□ **The third unpaired cartilage** is the **epiglottis**.

- It differs from the other cartilages in that it consists of **elastic cartilage rather than hyaline cartilage**.
- Its inferior margin is attached to the thyroid cartilage anteriorly, and the superior part of the epiglottis projects superiorly as a free flap toward the tongue
- **The epiglottis helps prevent swallowed materials from entering the larynx.**
- As the larynx elevates during swallowing, the epiglottis tips posteriorly to cover the opening of the larynx.

Laryngeal Skeleton,,,

The six paired cartilages consist of three cartilages on each side of the posterior part of the larynx.

- The top cartilage on each side is the **cuneiform cartilage**, the middle cartilage is the **corniculate cartilage**, and the bottom cartilage is **the arytenoid cartilage**.
- The arytenoid cartilages articulate with the cricoid cartilage inferiorly.
- **The paired cartilages form an attachment site for the vocal folds.**

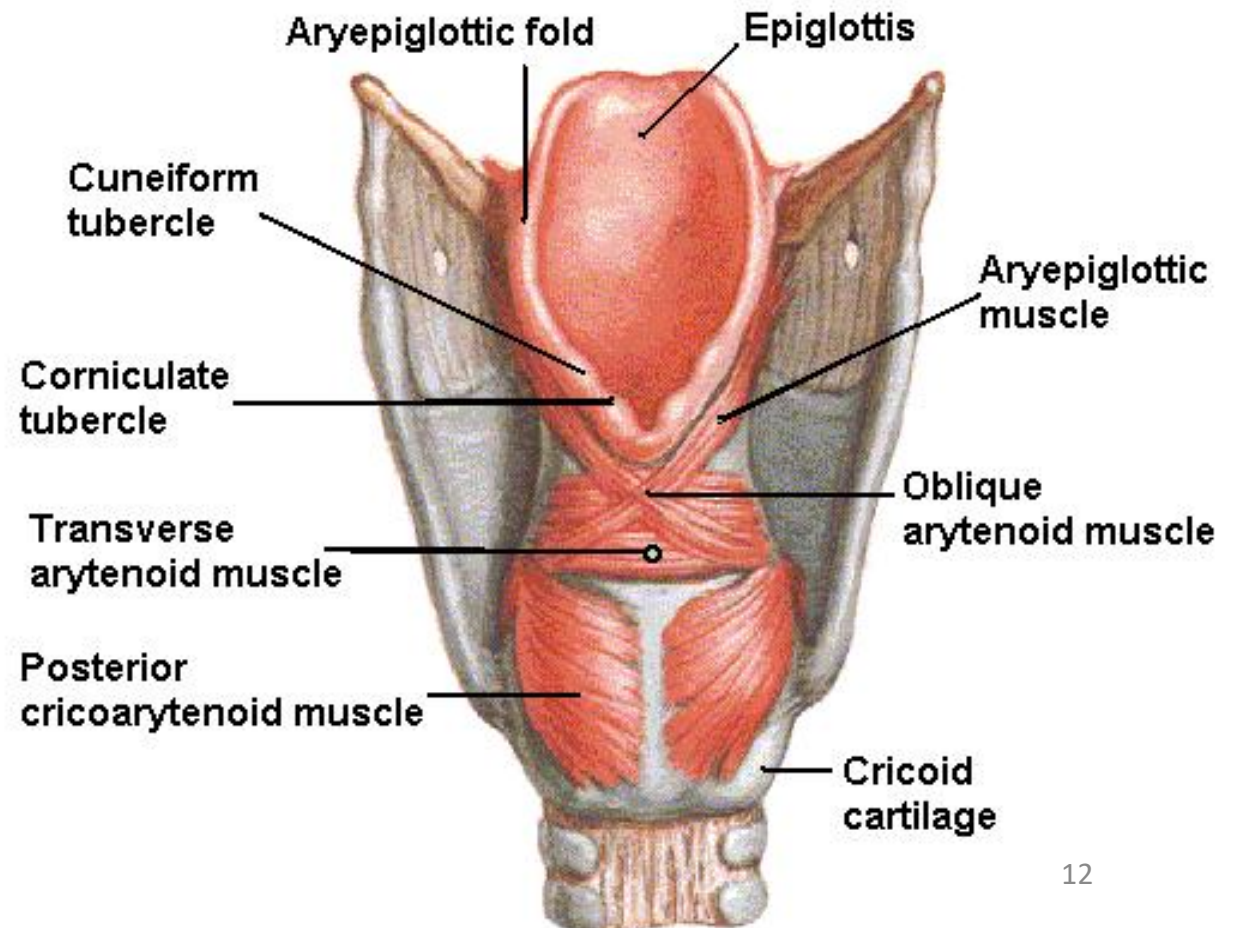


Cavity of the larynx,,,

- **Laryngeal Inlet faces** backward and upward and opens into the laryngeal part of the pharynx

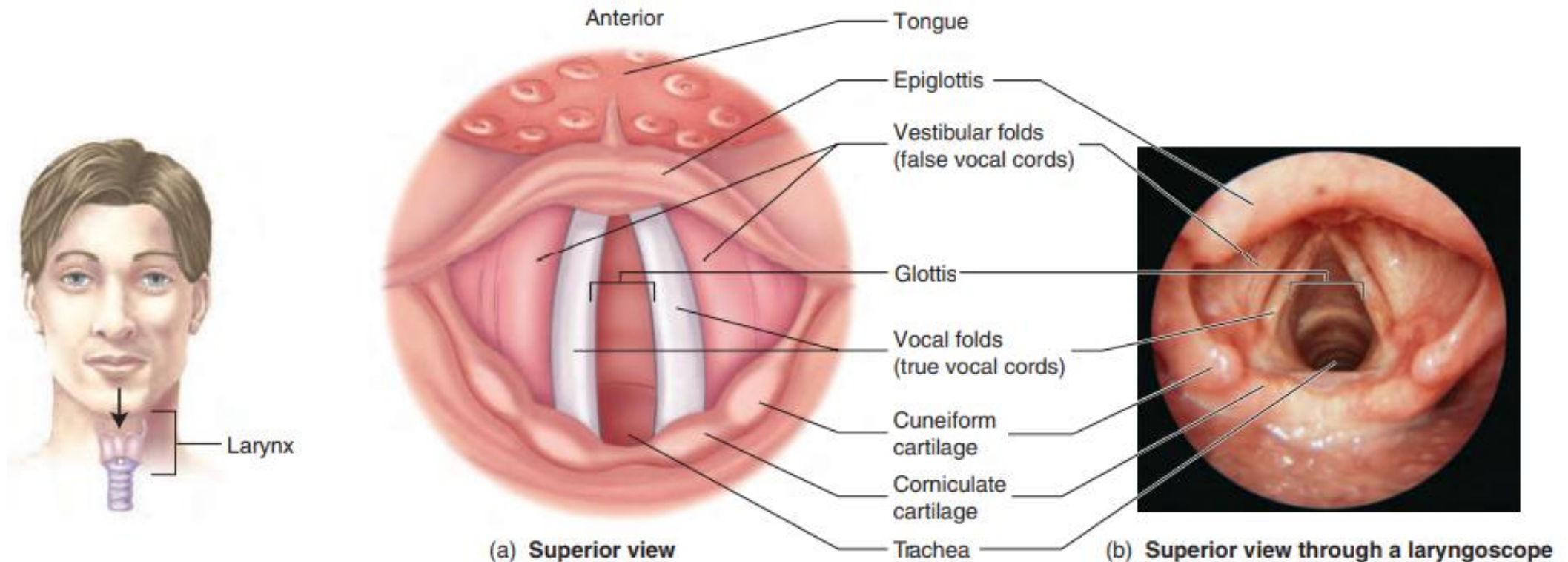
❖ *The opening is bounded:*

- ✓ Anteriorly: by the upper margin of epiglottis
- ✓ Posteriorly & below by arytenoid cartilages
- ✓ Laterally by aryepiglottic folds



The Structures of Voice Production

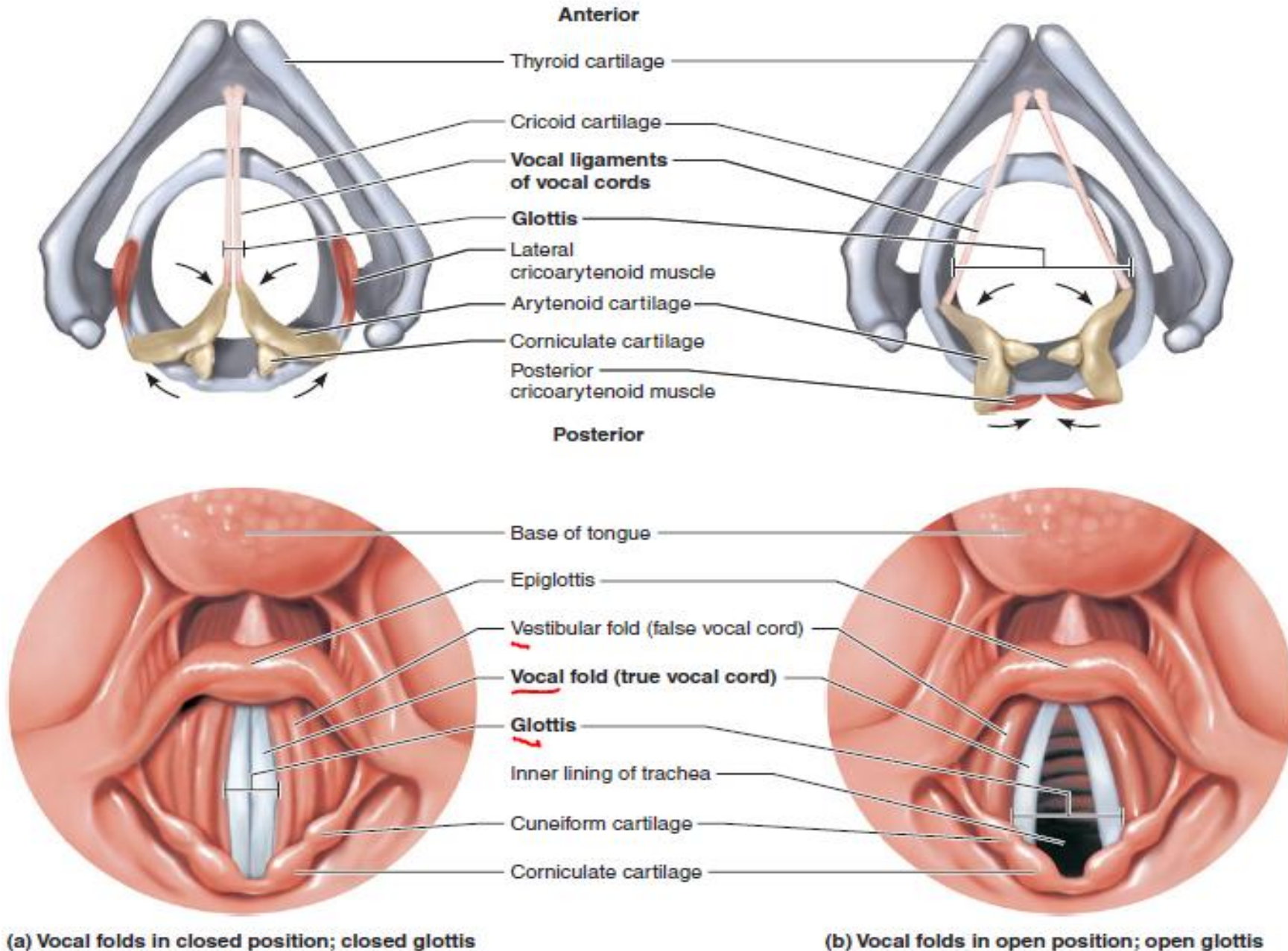
- The mucous membrane of the larynx forms two pairs of folds a superior pair **called the ventricular folds (false vocal cords)** and an inferior pair called simply the **vocal folds (true vocal cords)**.
- The space between the ventricular folds is known as the **rima vestibule**(Rima glottides)



The Structures of Voice Production,,

- *The vocal folds are the primary source of voice production.*
- Air moving past the vocal folds causes them to vibrate, producing sound
- In considering the process of speech, sounds are produced by blowing air past the vocal cords.
- **The air produces vibrations in the vocal cords**, forming the sounds that come out of the mouth.
- The air is pushed out of the lungs by relaxing the **diaphragm** and contracting **muscles of the abdominal wall**.

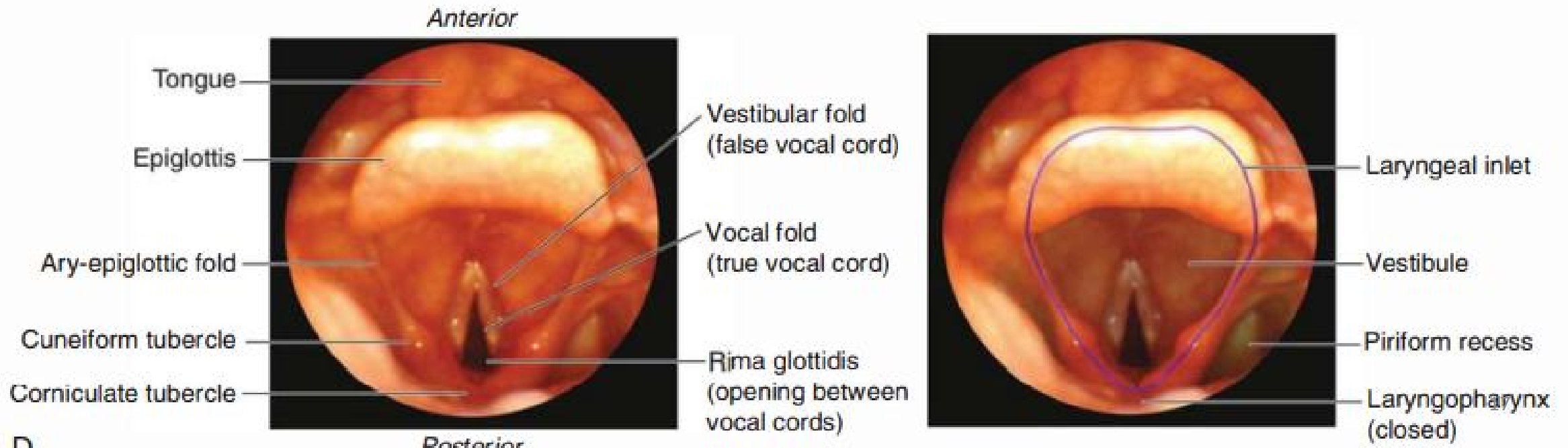
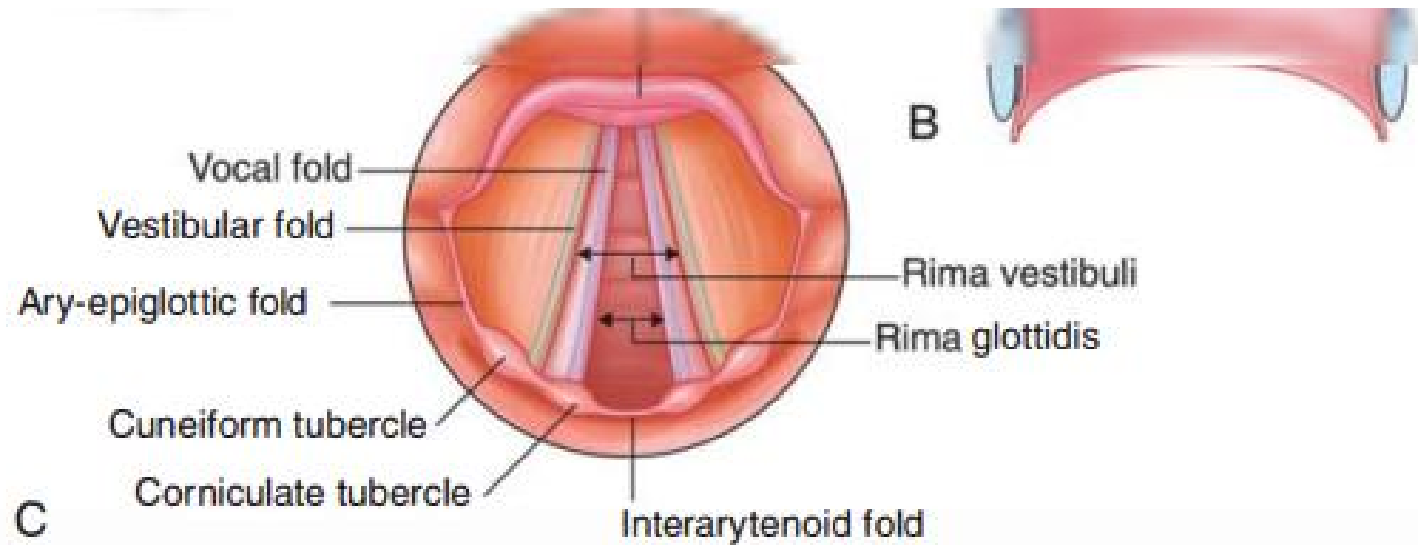
The Structures of Voice Production...



Rima vestibuli and rima glottidis

- **When viewed from above**, there is a triangular opening (**the rima vestibuli**) between the **two adjacent vestibular folds** at the entrance to the middle chamber of the laryngeal cavity.
 - **Inferior to the vestibular folds**, the vocal folds (true vocal cords) and adjacent mucosa-covered parts of the arytenoid cartilages form the lateral walls of a similar, but narrower triangular opening (**the rima glottidis between the two adjacent vocal folds**).
- **During quiet respiration**, the laryngeal inlet, vestibule, **rima vestibuli**, and **rima glottidis** are open.

Rima vestibuli and rima glottidis,,,



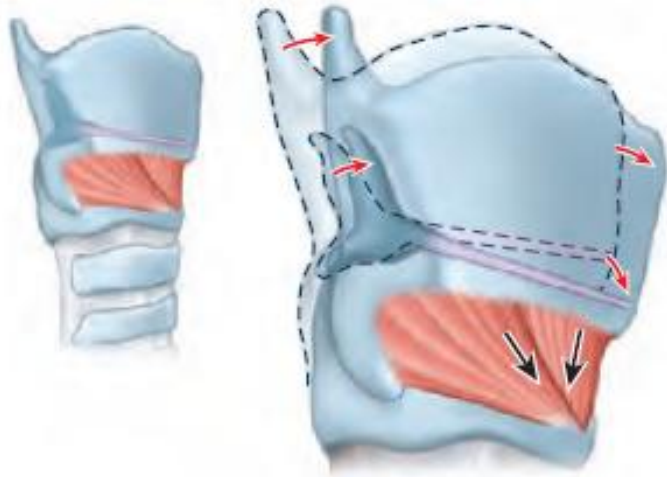
Rima vestibuli and rima glottidis,,,

- The arytenoid cartilages are **abducted and the rima glottidis** is triangular shaped.
- **During forced inspiration**, the arytenoid cartilages are rotated laterally, mainly by the action of the **posterior crico-arytenoid muscles**.
- **The walls of the larynx are quite muscular.**
- The superficial extrinsic muscles connect the larynx to the hyoid bone and elevate the larynx during swallowing.
- **The laryngeal muscles** are divided into **extrinsic and intrinsic groups**.
- The deeper intrinsic muscles control the vocal cords by pulling on the corniculate and arytenoid cartilages, causing the cartilages to pivot.

Rima vestibuli and rima glottidis,,,

- Depending on their action on the vocal cord , **the intrinsic laryngeal muscles are grouped as :**
 - ✓ adductors
 - ✓ abductors,
 - ✓ sphincters,
 - ✓ tensors and
 - ✓ relaxers.
- Air forced between the adducted vocal cords vibrates them, **producing a high-pitched sound.**
- **The crude sounds from the larynx are formed into words by actions of the pharynx, oral cavity, tongue, and lips.**

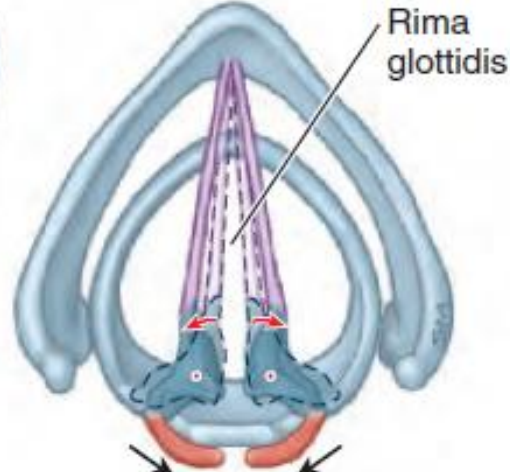
Rima vestibuli and rima glottidis,,,



(A) Lateral view
Cricothyroid



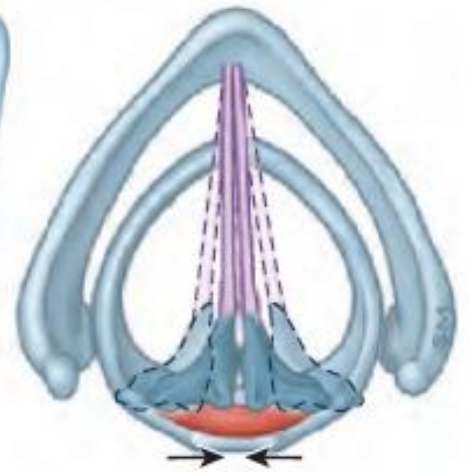
(B) Superlor view
Thyro-arytenoid



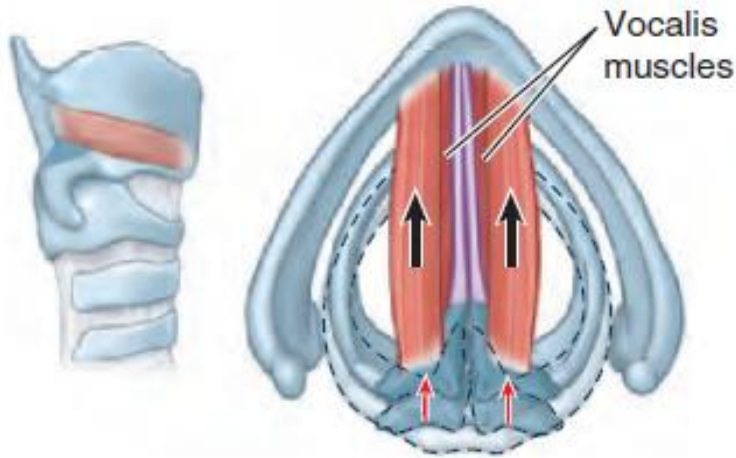
(C) Superlor view
Posterior crico-arytenoid



(D) Superlor view
Lateral crico-arytenoid



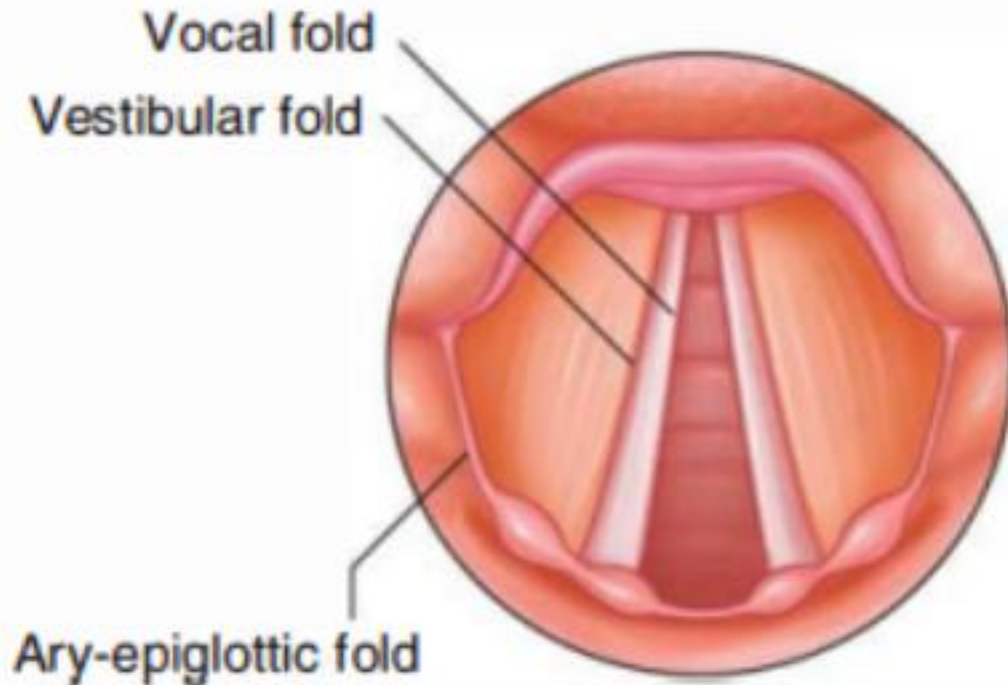
(E) Superlor view
Transverse arytenoid



(F) Superlor view
Oblique arytenoid

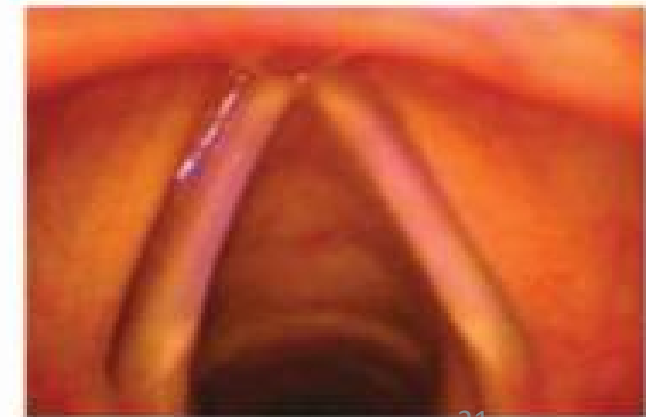
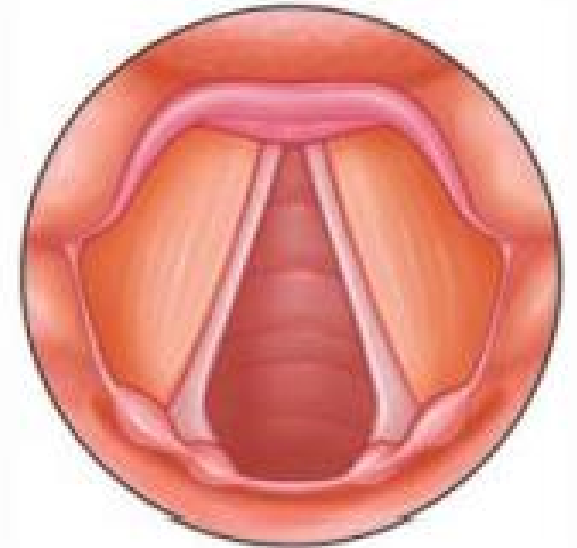
Rima vestibuli and rima glottidis,,,

Quiet respiration

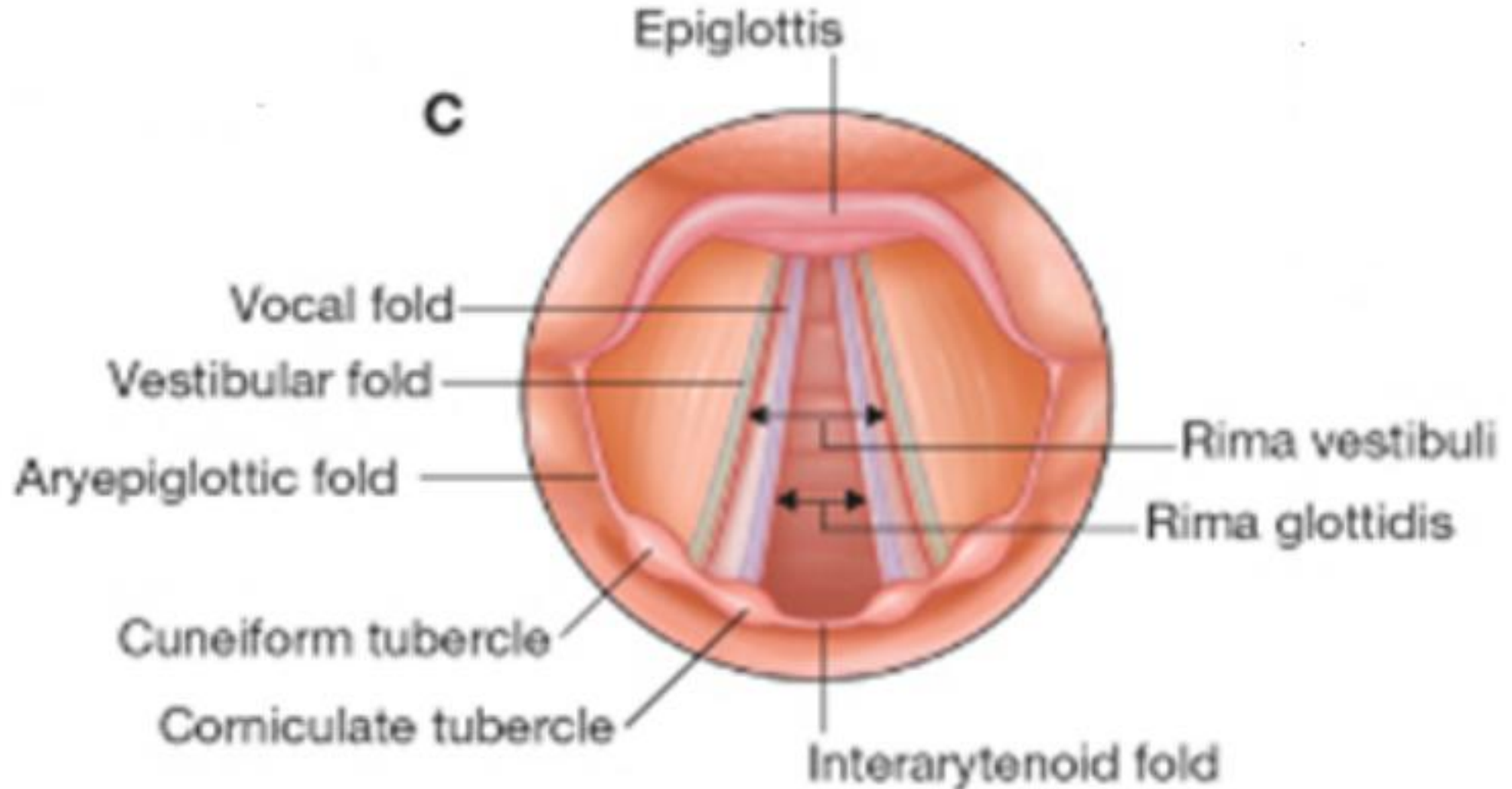


Forced inspiration

- Vocal folds abducted and rima glottidis wide open
- Vestibule open



Rima vestibuli and rima glottidis,,

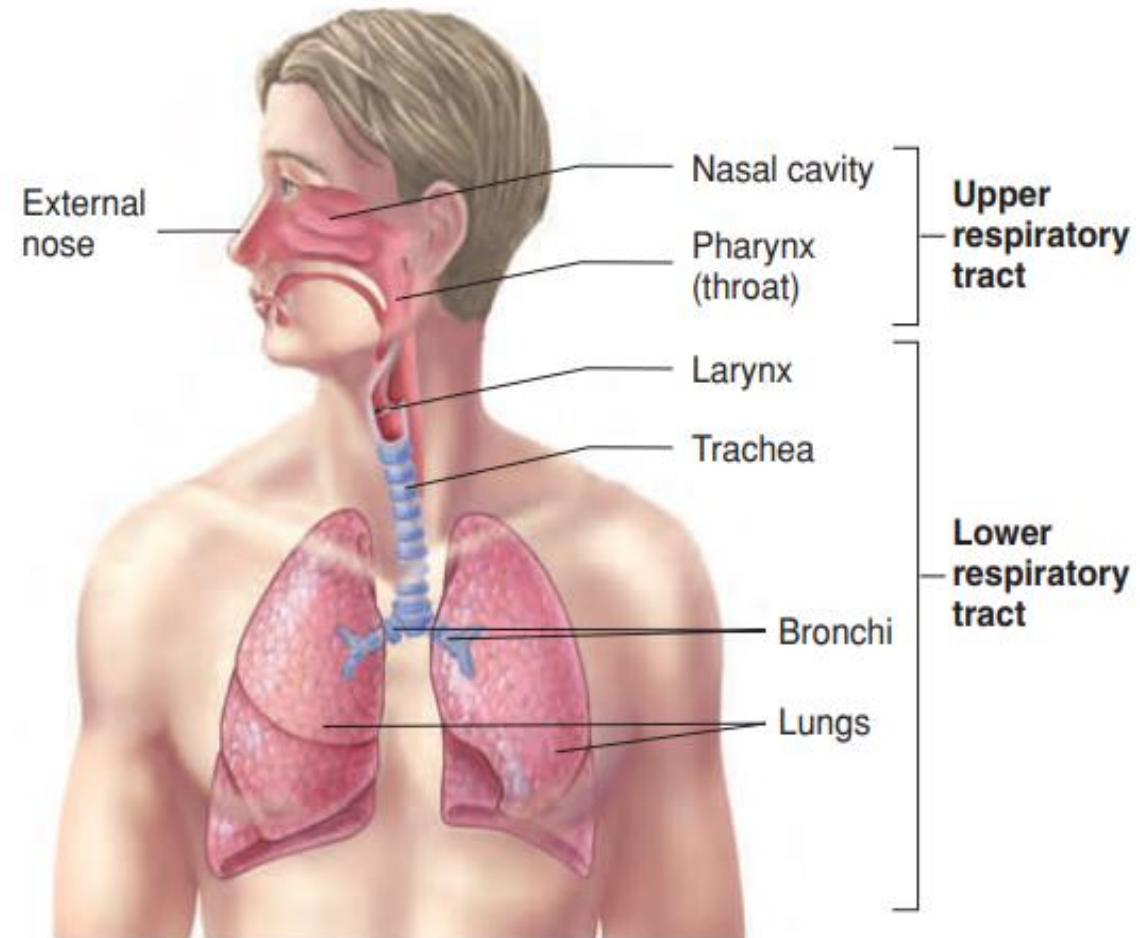


Gross structures and functions of the lower respiratory tract

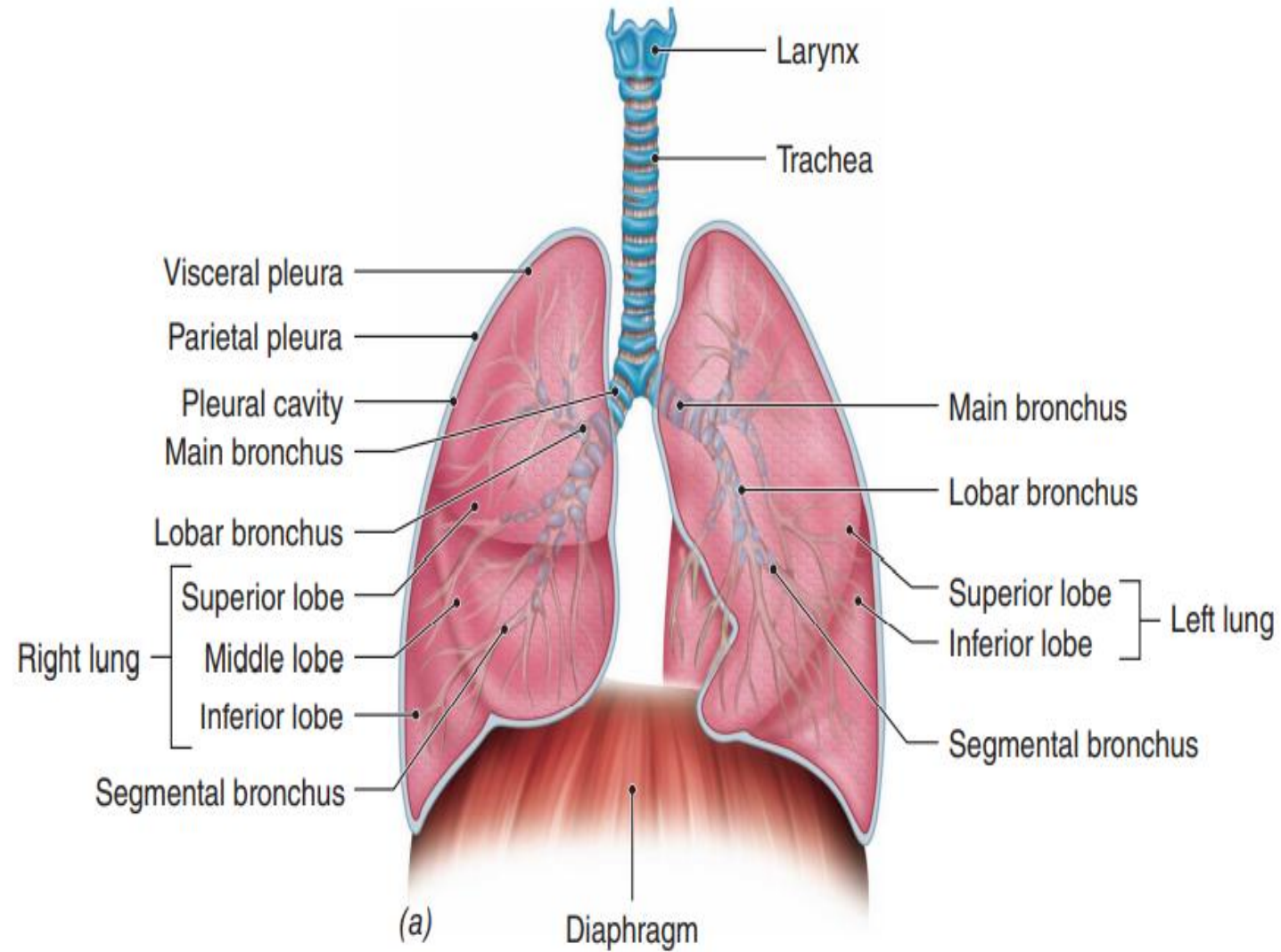
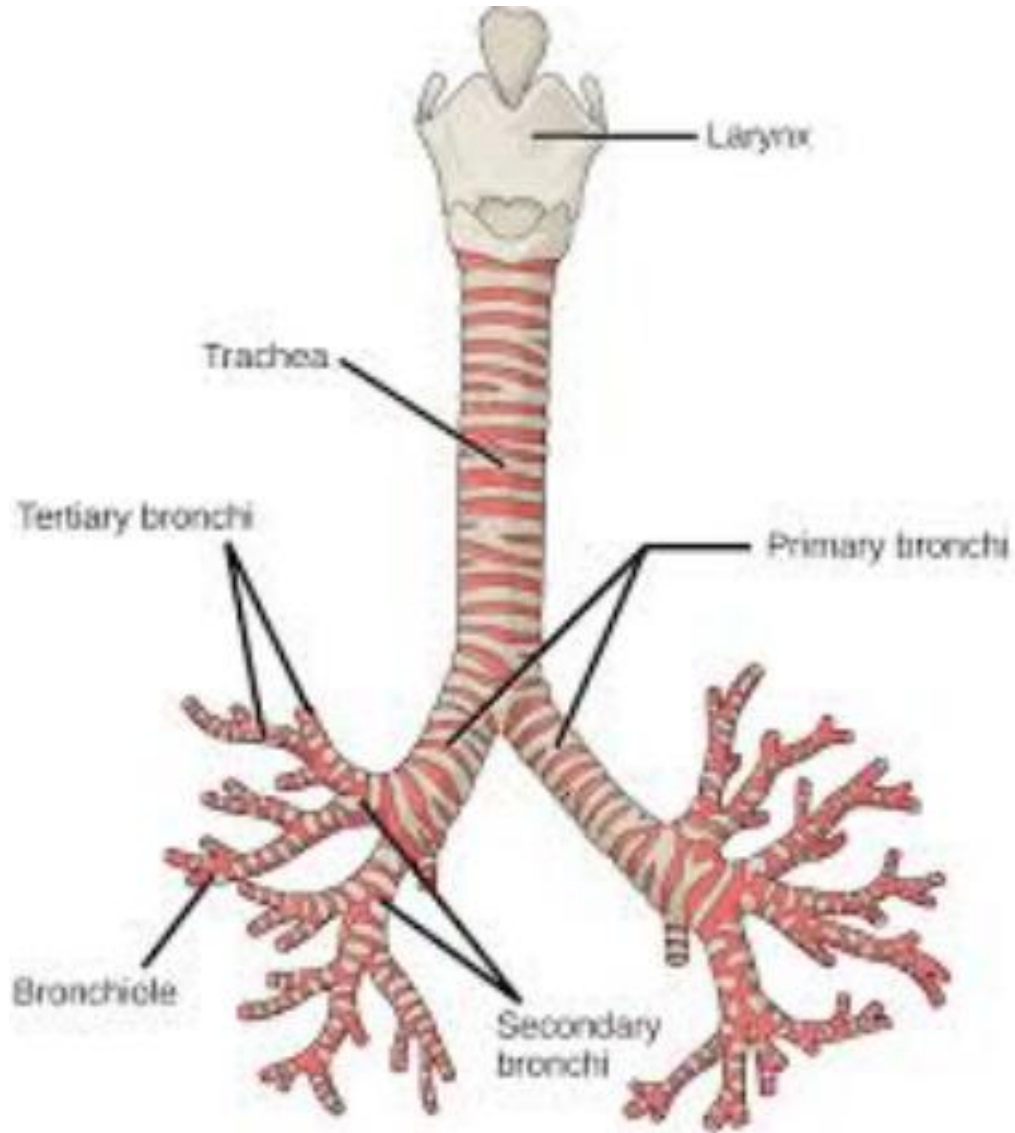
Gross structures and functions of the lower respiratory tract

- Structurally, the respiratory system consists of two parts:

- I. the upper respiratory system includes the nose, nasal cavity, pharynx, **larynx**, and associated structures.
- II. the lower respiratory system includes the, trachea, bronchi, and lungs

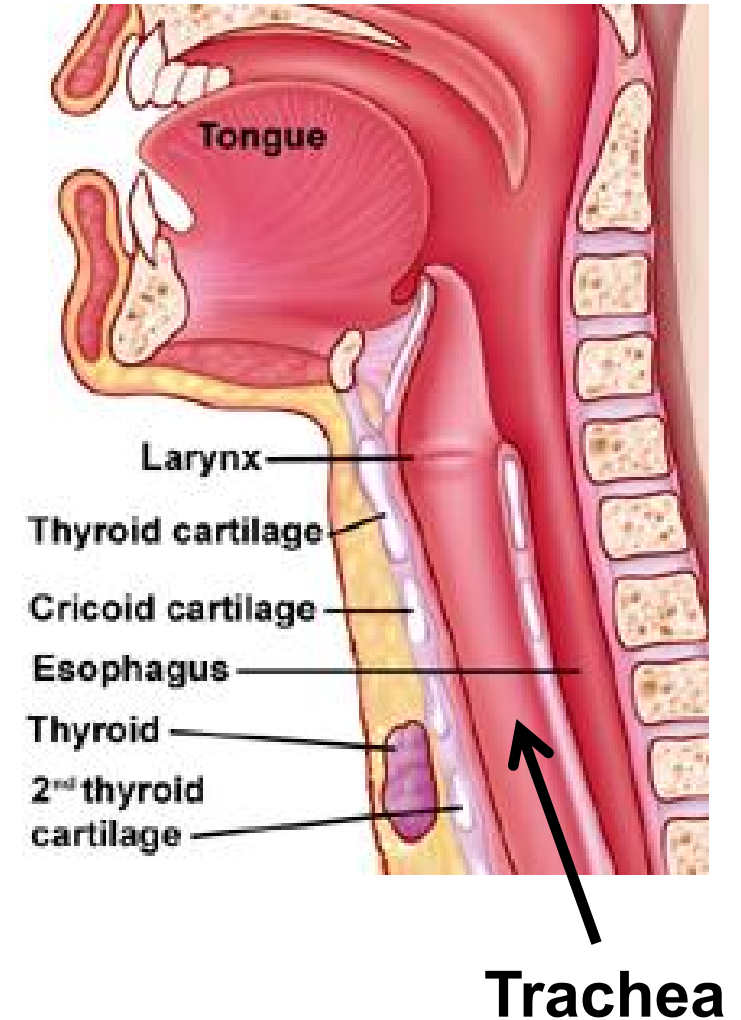


The Trachea

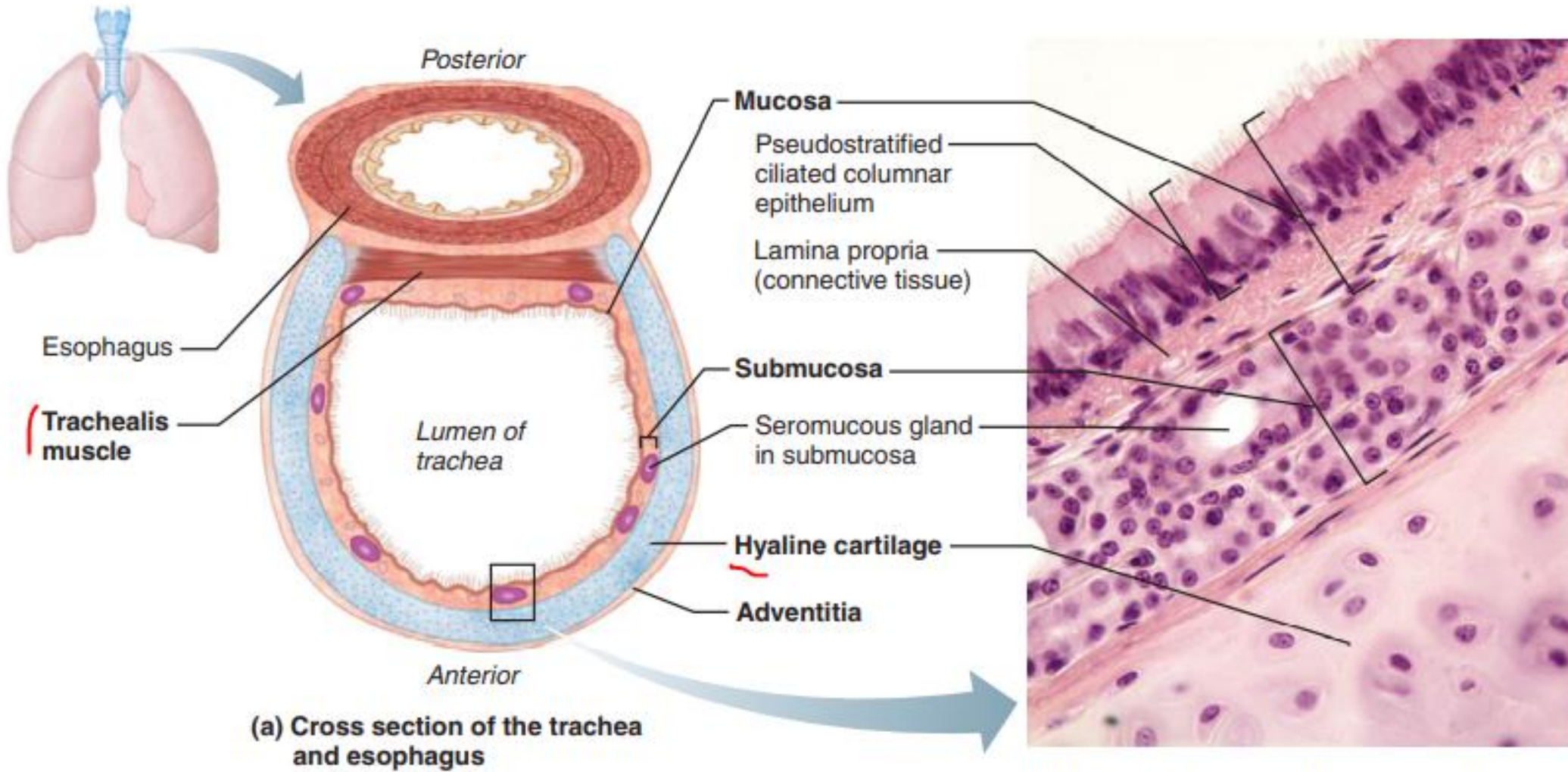


The Trachea

- The trachea or windpipe, is a tubular passageway for air that is about 12 cm long and 2.5 cm in diameter.
- It is located anterior to the **esophagus** and extends from the **larynx** to the superior border of the fifth thoracic vertebra (T5), where it divides into **the right and left primary bronchi**.
- It transports air to and from the lungs,



The Trachea,,,



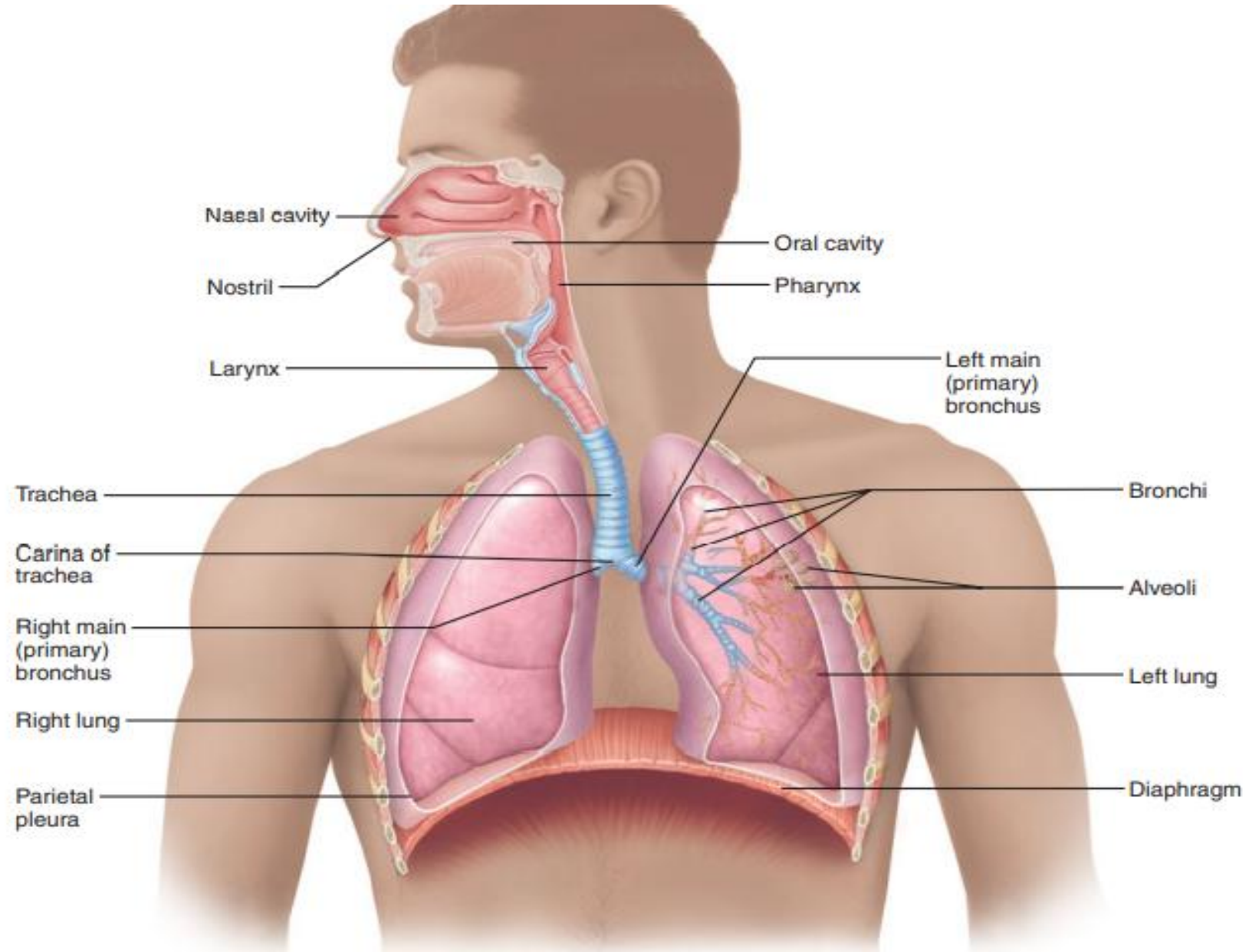
The Trachea,,,

- The walls of the trachea are supported by 15-20 incomplete C-shaped tracheal cartilages that hold the passageway open in spite of the air pressure changes that occur during breathing
- The open portion of the tracheal cartilages is oriented posteriorly against the esophagus
- This orientation allows the esophagus to expand slightly as food passes down to the stomach.

The Trachea,,,

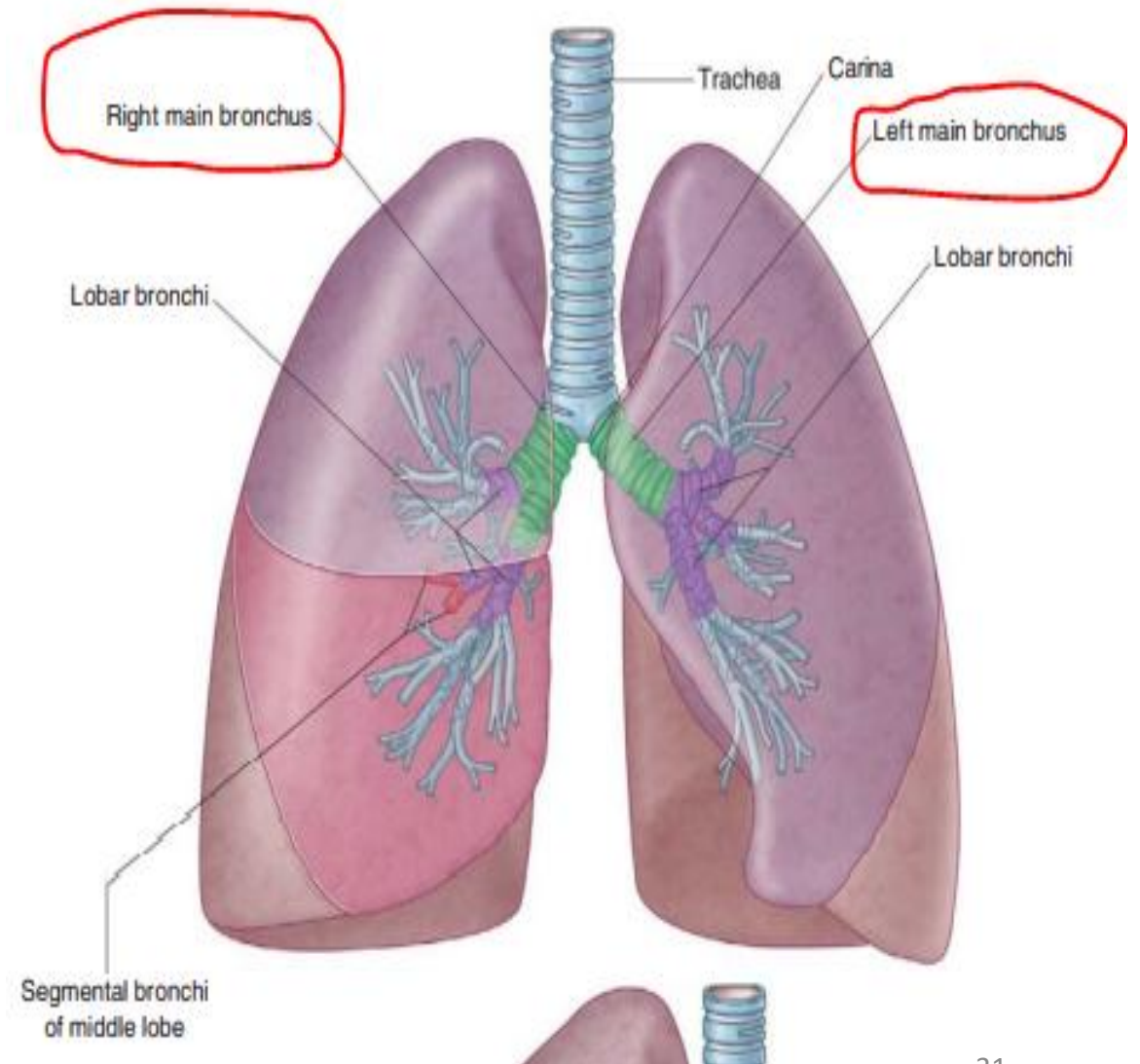
- This incomplete, “C” shaped cartilaginous tracheal rings (the posterior gap) is closed by **the involuntary trachealis muscle**, connecting the ends of the tracheal rings.
 - Due to this the posterior wall of the trachea is flat
- A ridge on the internal aspect of the last tracheal cartilage, called the carina, marks the point where the trachea branches into the two main (primary) bronchi.
- **The mucosa that lines the carina is highly sensitive to irritants, and the cough reflex often originates here**

The Trachea,,,



The Bronchial Tree

- The collective term “bronchial tree” refers to the bronchi and all of their subsequent branches
- The bronchi are the airways of the lower respiratory tract.
- At the level of the 3rd or 4th thoracic vertebra, the trachea bifurcates into the left and right main bronchi.

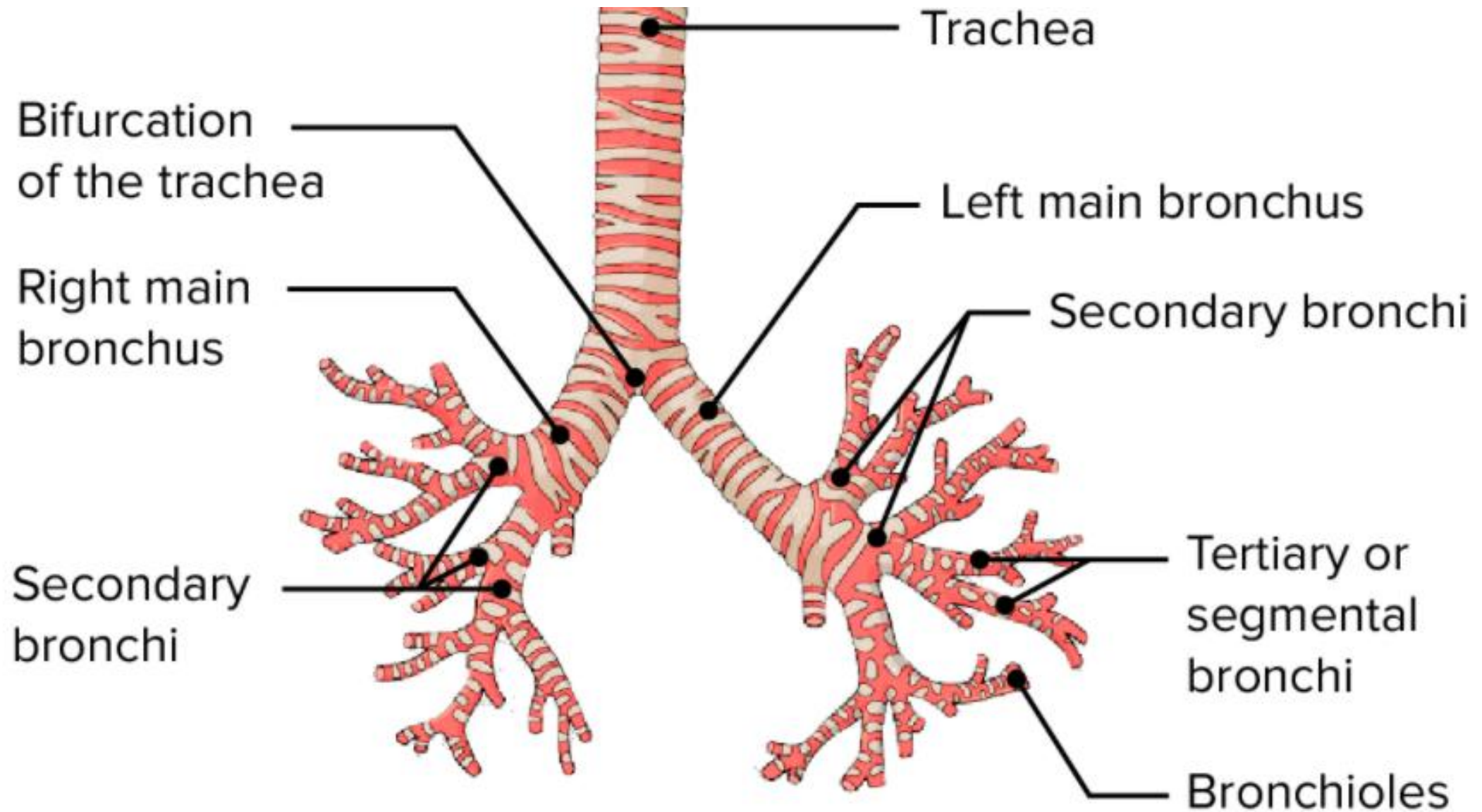


The Bronchial Tree,,,

❑ *Main branching structure and components:*

- ❖ Trachea → carina → main bronchi → lobar bronchi → segmental bronchi → terminal bronchioles → respiratory bronchioles → alveolar duct → alveolar sac → alveoli
- ✓ The walls of the trachea & bronchi are supported by **C-shaped of cartilage**
- ✓ The right main bronchi has different angle of inclination.
- ✓ The **left primary bronchus** diverges at a greater angle than the **right primary bronchus** to reach the left lung.

The Bronchial Tree,,,



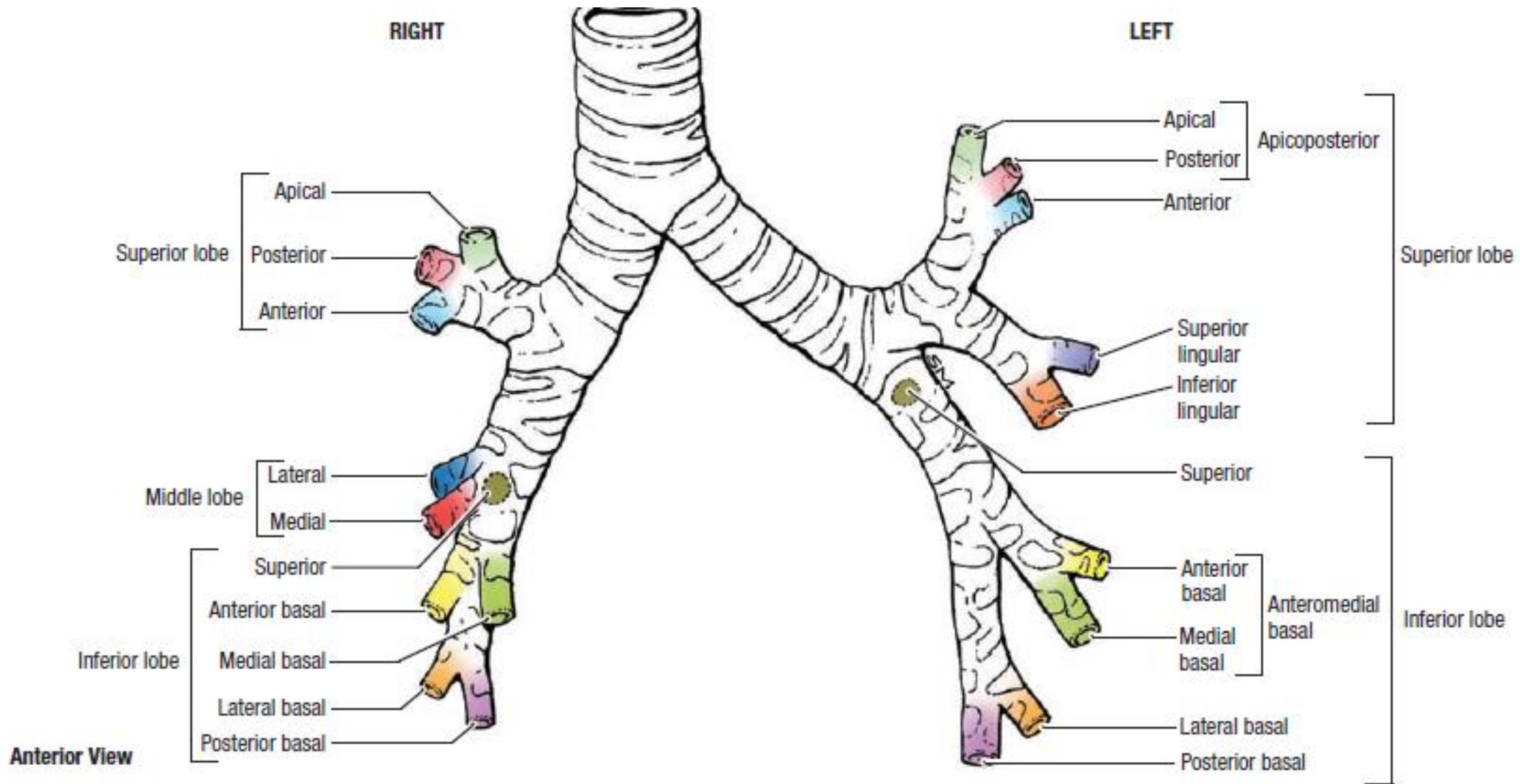
The Bronchial Tree,,,

- The **left main bronchus** passes inferolaterally, inferior to the arch of the aorta & anterior to the esophagus & thoracic aorta, to reach the hilum of the lung.
- Each **primary bronchus** divides into secondary **lobar bronchi 2 on the left and 3 on the right**, each of which supplies each **lobe** of the lung.
- Each **lobar bronchus** divides into several tertiary **segmental bronchi** that supply the **bronchopulmonary segments**.

The Bronchial Tree,,,

- **In both lungs**, the lobar bronchi branch into segmental (tertiary) bronchi.
- **There are 10 of these in the right lung and 8 in the left.**
- Each one ventilates a functionally independent unit of lung tissue **called a bronchopulmonary segment.**
- **The main bronchi are supported**, like the trachea, by C-shaped rings of hyaline cartilage, whereas the lobar and segmental bronchi are supported by overlapping crescentshaped cartilaginous plates.
- All of the bronchi are lined with **ciliated pseudostratified columnar epithelium**, but the cells grow shorter and the epithelium thinner as we progress distally.

The Bronchial Tree,,,



The bronchopulmonary segments are

- ✓ The largest subdivisions of a lobe.
- ✓ Pyramidal-shaped segments of the lung.
- ✓ Separated from adjacent segments by connective tissue septa.
- ✓ Supplied independently by a **segmental bronchus** and a **tertiary branch of the pulmonary artery**.
- ✓ Named according to the segmental bronchi supplying them.
- ✓ Drained by **intersegmental parts of the pulmonary veins**.
- ✓ Usually 18–20 in number (10 in Rt; 8–10 in Lt).
- ✓ ***Surgically resectable.***

Tracheobronchial tree and bronchopulmonary segments,,,

Lobes of right lung:

Superior lobe

- Apical
- Posterior
- Anterior

Middle lobe

- Lateral
- Medial

Inferior lobe

- Superior
- Anterior basal
- Medial basal
- Lateral basal
- Posterior basal

Lobes of left lung:

Superior lobe

- Apical
- Posterior
- Anterior

Middle lobe

- Superior lingular
- Inferior lingular

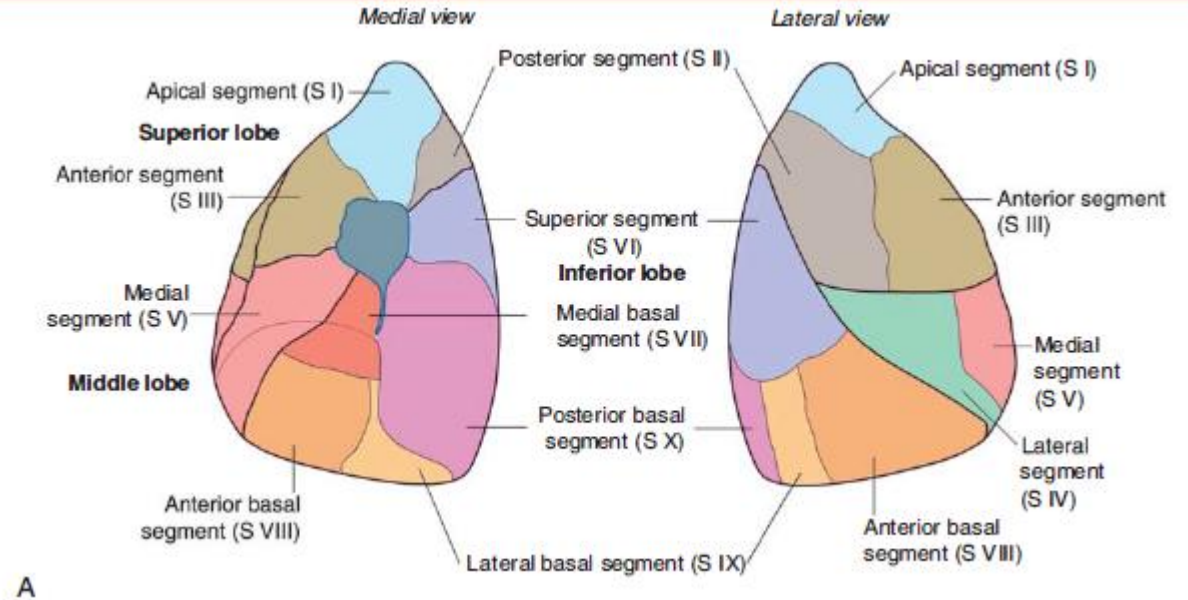
Inferior lobe

- Superior
- Anterior basal
- Medial basal
- Lateral basal
- Posterior basal

**

*

** Typically combine into apicoposterior segment
 * Often combined into anteriomedial basal segment



A

B

Tracheobronchial tree and bronchopulmonary segments,,,

- Beyond the tertiary segmental bronchi, there are 20 to 25 generations that eventually end as **terminal bronchioles**.
- **Bronchioles** lack cartilage in their walls.
- **Conducting bronchioles** transport air but **lack glands or alveoli**.
- Each terminal bronchiole gives rise to several generations of **respiratory bronchioles**.
- *The pulmonary alveolus is the basic structural unit of gas exchange in the lung.*
- Due to the presence of the alveoli, **the respiratory bronchioles** are involved both in air transportation and gas exchange.
- Each respiratory bronchiole gives rise to **2–11 alveolar ducts then to 5–6 alveolar sacs**.

➤ **Gross structures and function of the lung and pleura**

Gross structures and function of the lung and pleura,,,

❖ *The thoracic cavity is divided into 3 compartments:*

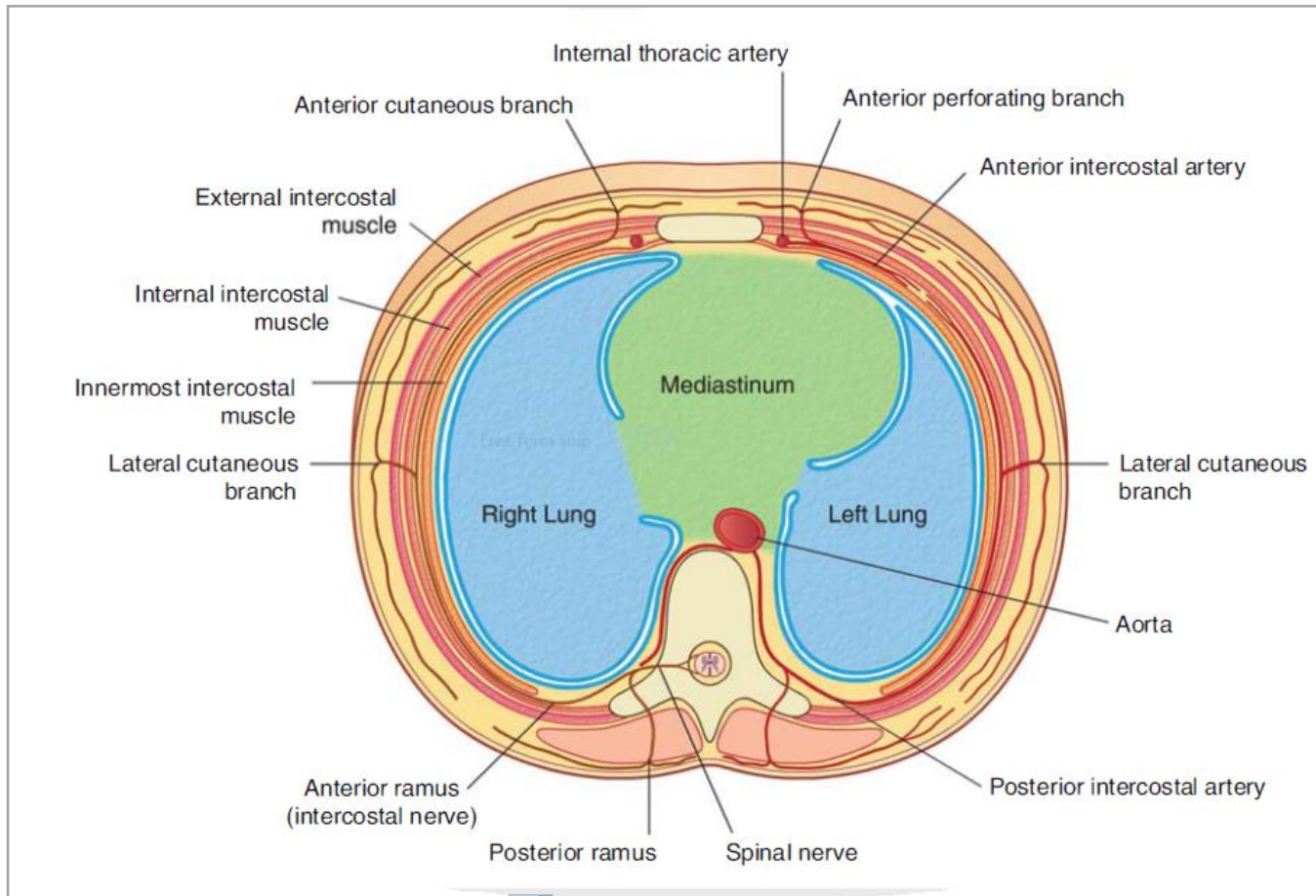
➤ **Right pulmonary cavity**

➤ **Left pulmonary cavity**

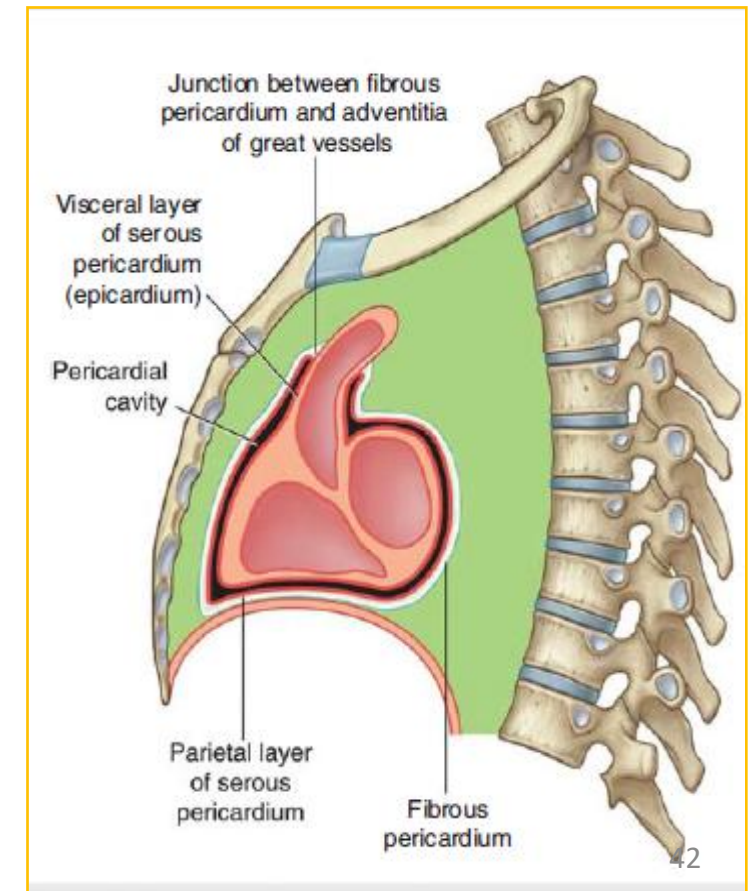
➤ **A central mediastinum**

Gross structures and function of the lung and pleura,,,

Transverse section of the thoracic cavity



Sagittal section of the thoracic cavity



The *pulmonary* cavities

- **Right and left pulmonary cavities:** contain the lungs & pleurae.
- **A central mediastinum:** contains all other thoracic structures ,the heart, thoracic parts of the great vessels, thoracic part of the trachea, esophagus, thymus, and lymph nodes.
- It extends vertically from the **superior thoracic aperture to the diaphragm** and **anteroposteriorly** from the **sternum** to the thoracic **vertebral bodies**.

Gross structures and function of the lungs,,,

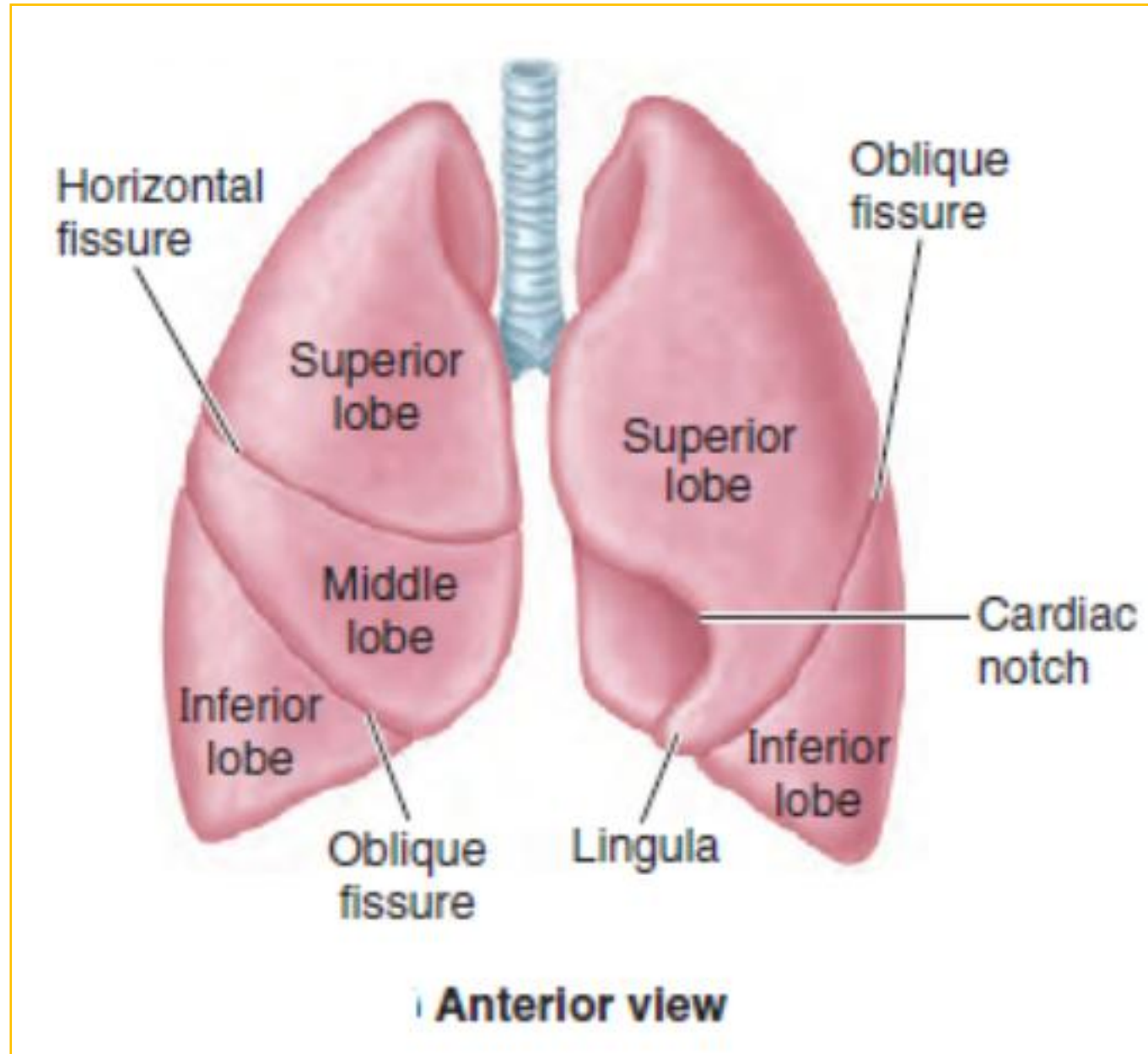
❖ **Lungs:** They are the principal organs of respiration.

- **The most important function of the lungs** is to take oxygen from the environment and transfer it to the bloodstream.
- The lungs also perform several important non-respiratory functions that are vital for normal physiology

• **Non-respiratory functions of the lungs, include:**

- Filtration
 - Immune defence
 - Blood reservoir
 - Metabolism etc
- Healthy lungs in living people are **normally light, soft, and spongy.**

Gross structures and function of the lungs,,



Gross structures and function of the lungs,,

- **Lungs are highly elastic structures.**
- Each lung lies free in its own pleural cavity
- **Its only attachment** is at its root.
- The lungs connect the with trachea by **bronchii** and heart by **pulmonary vessels**
- **The anterior, lateral, and posterior surfaces** of a lung contact the ribs and form a continuously curving costal surface

Gross structures and function of the lungs,,

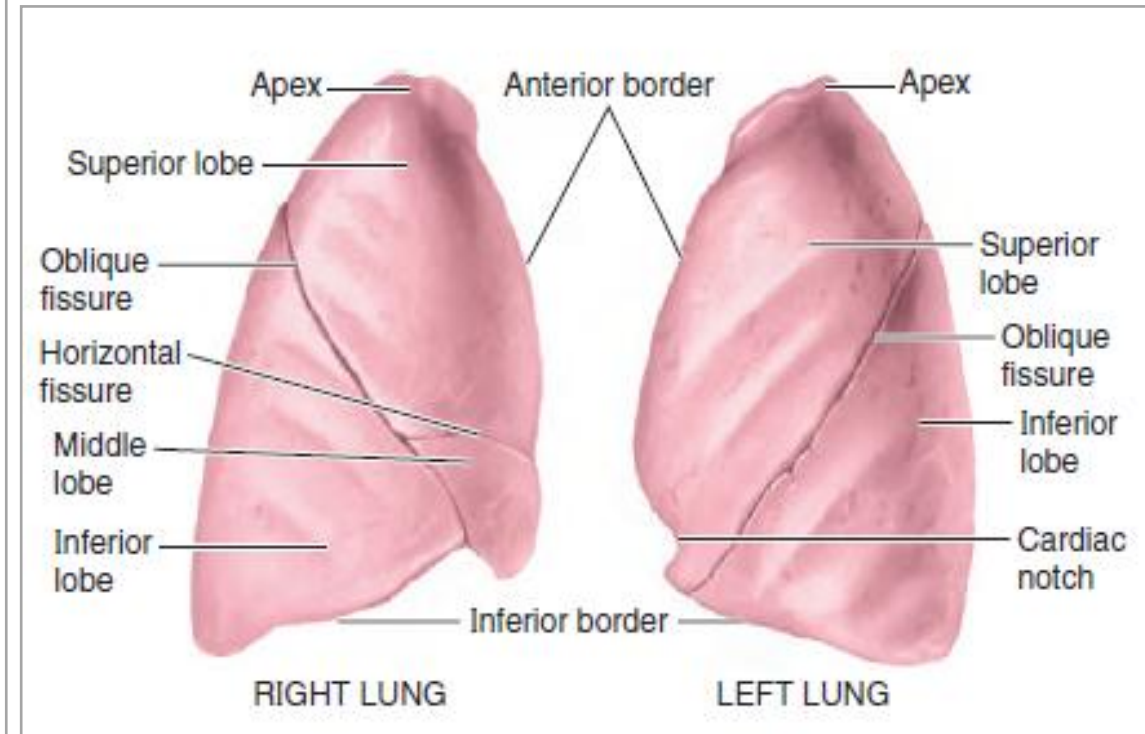
❖ Each lung has the following parts:

✓ An **apex**, superior end of the lung, a **base** is concave inferior surface of the lung.

✓ **2 or 3 lobes**, created by **1 or 2 fissures**.

✓ **3 surfaces**; costal, mediastinal & diaphragmatic.

✓ **3 borders**; anterior, inferior & posterior.



Gross structures and function of the lungs,,

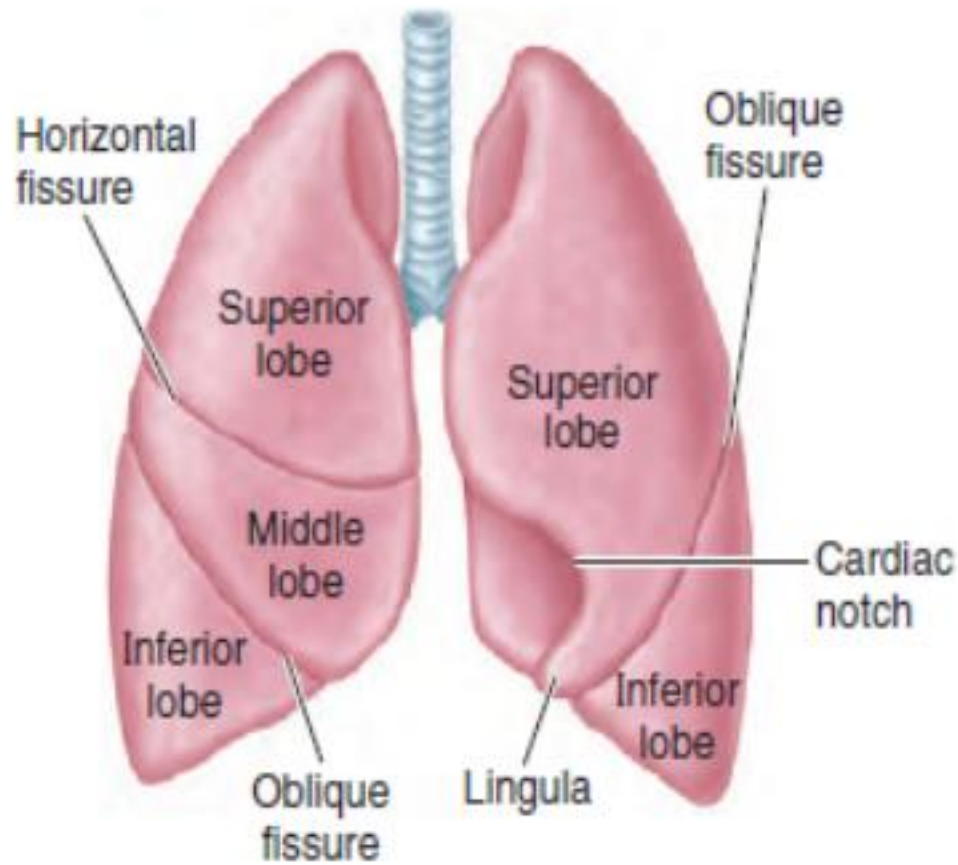
• The right lung

- Has 2 fissures oblique & horizontal that divide it into **3 right lobes: superior, middle & inferior.**
- **Is larger and heavier than the left.**
- The *anterior border of the right lung* is relatively straight.

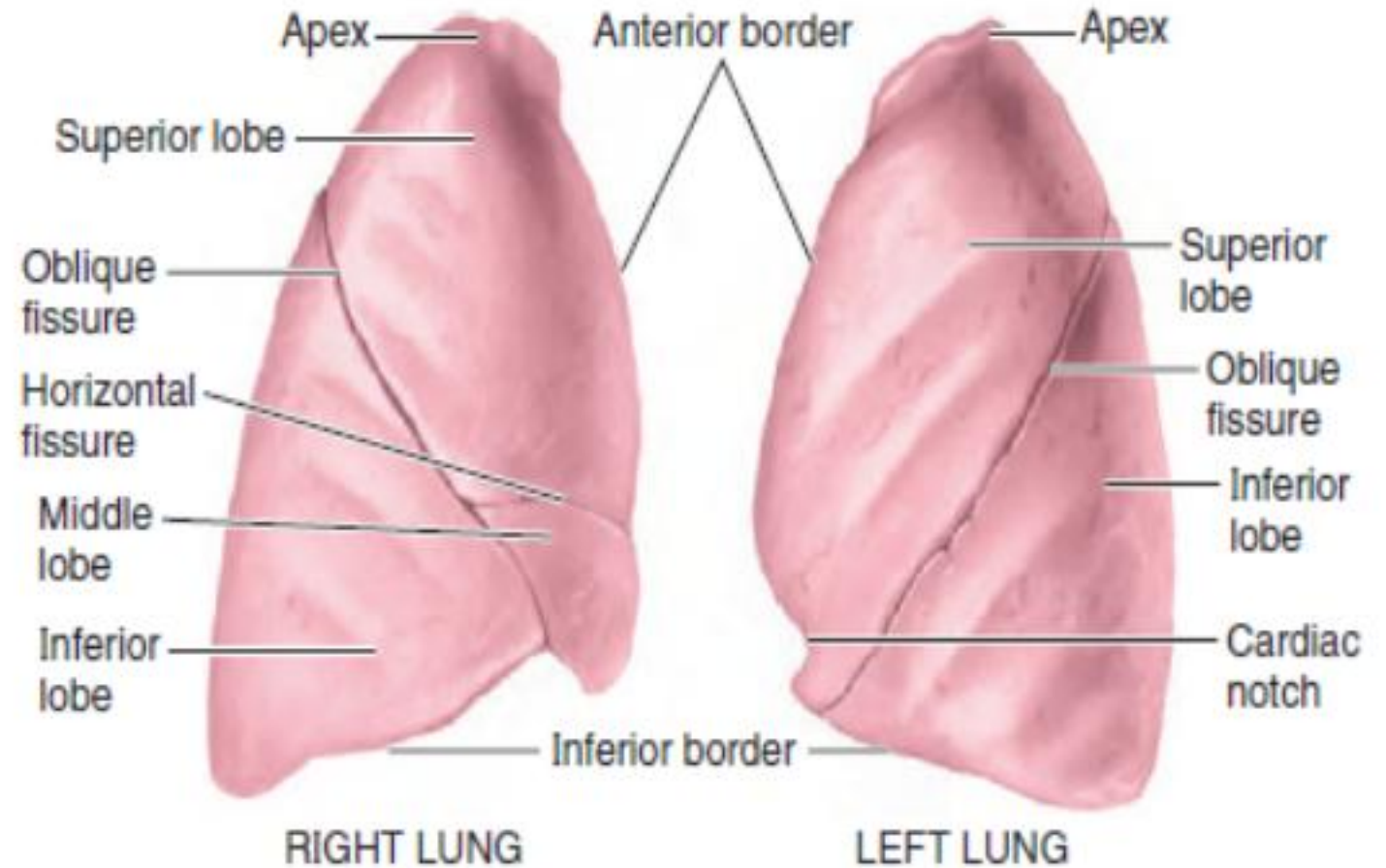
• Left lung

- Has 1 **oblique fissure** dividing it into 2 **lobes, superior and inferior.**
- The *anterior border of the left lung* has a deep **cardiac notch**
- **Has** tongue-like process, the **lingula** w/c extends below the cardiac notch & slides in & out of the **costomediastinal** recess during inspiration and expiration

Gross structures and function of the lungs,,,



(A) Anterior view

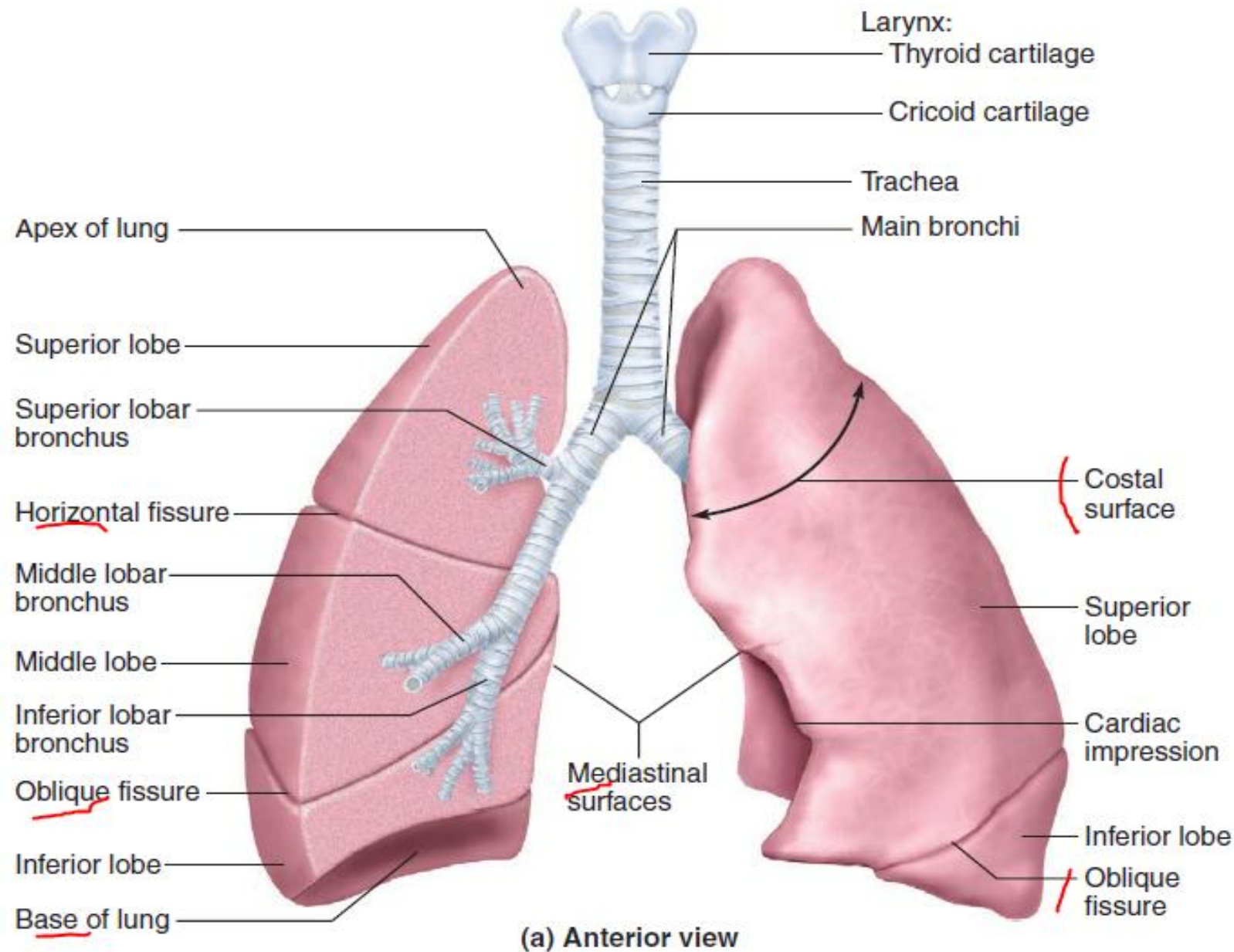


(B) Lateral views

Gross structures and function of the lungs,,,

- **Note: The costal surface of the lung is large, smooth, and convex.**
- The mediastinal surface of the lung is **concave**.
- The mediastinal surface includes the *hilum*, w/c receives the root of the lung.
- The **diaphragmatic surface of the lung** is concave & forms the **base of the lung**.
- **The concavity is deeper in the right lung b/c of the higher position of the right lobe of the liver.**

Gross structures and function of the lungs,,,



Root of the lungs

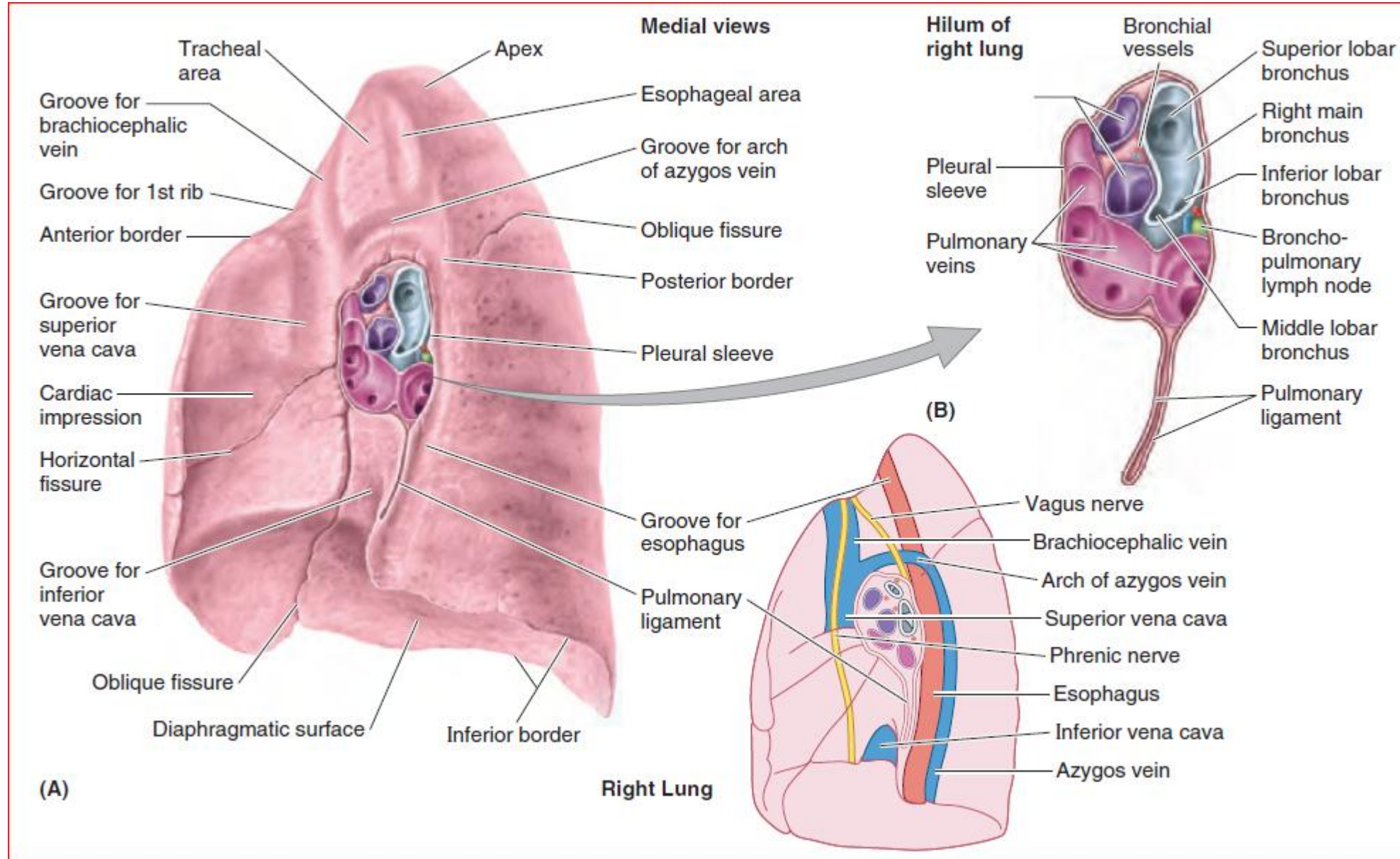
The **root** of each lung is a short tubular collection of structures that together attach the lung to structures in the mediastinum.

The lung's root has, **bronchi, pulmonary artery, 2 pulmonary veins, the pulmonary plexuses of nerves and lymphatic vessels..**

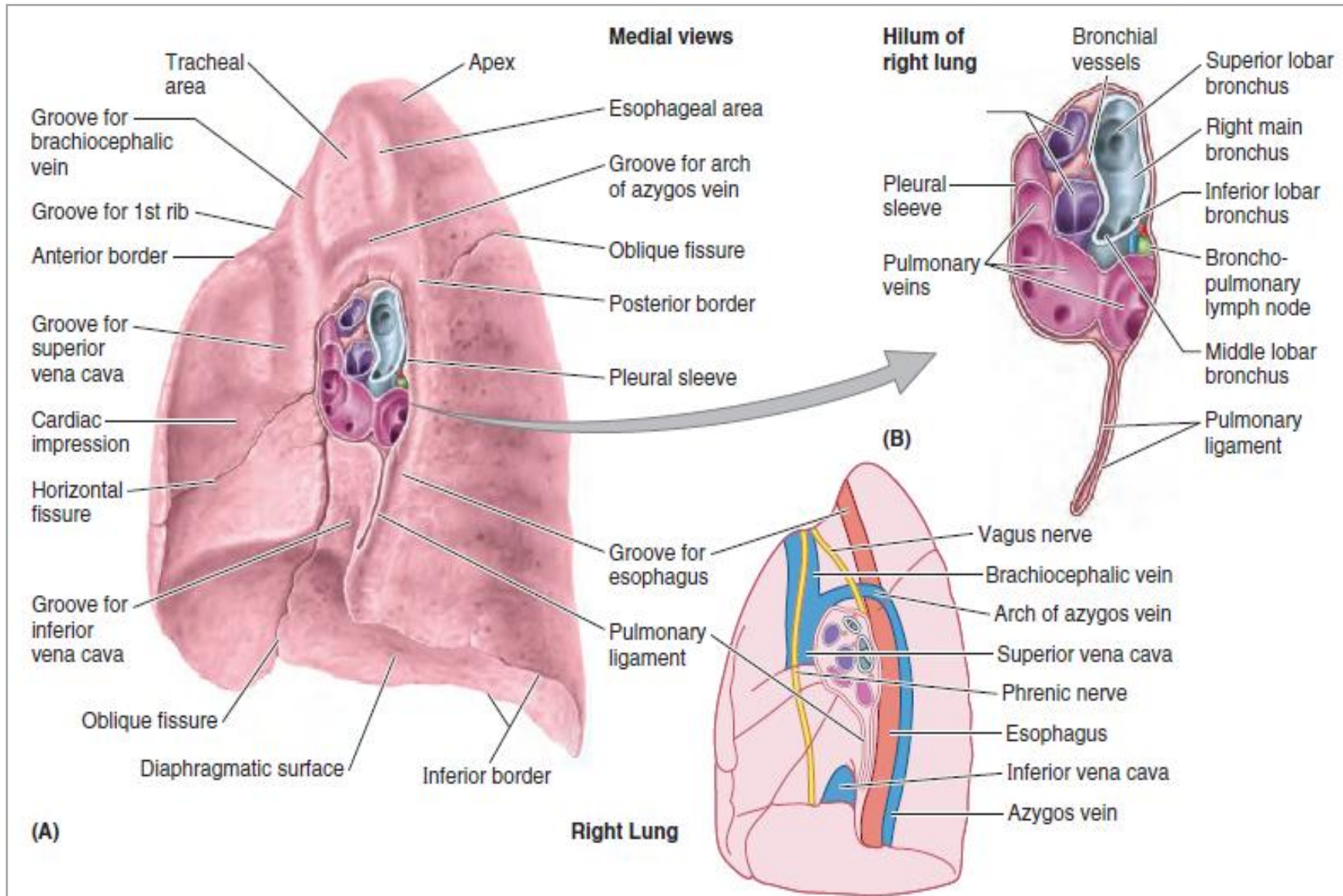
❖ If the root is sectioned before branching of the main (primary) bronchus and pulmonary artery, its general arrangement is different on both lungs.

- **Pulmonary artery: superiormost on left** (the superior lobar or “eparterial” **bronchus may be superiormost on the right**).
- Superior & inferior pulmonary veins: anteriormost & inferiormost, respectively.

Root of the lungs,,,



Root of the Right lungs,,,



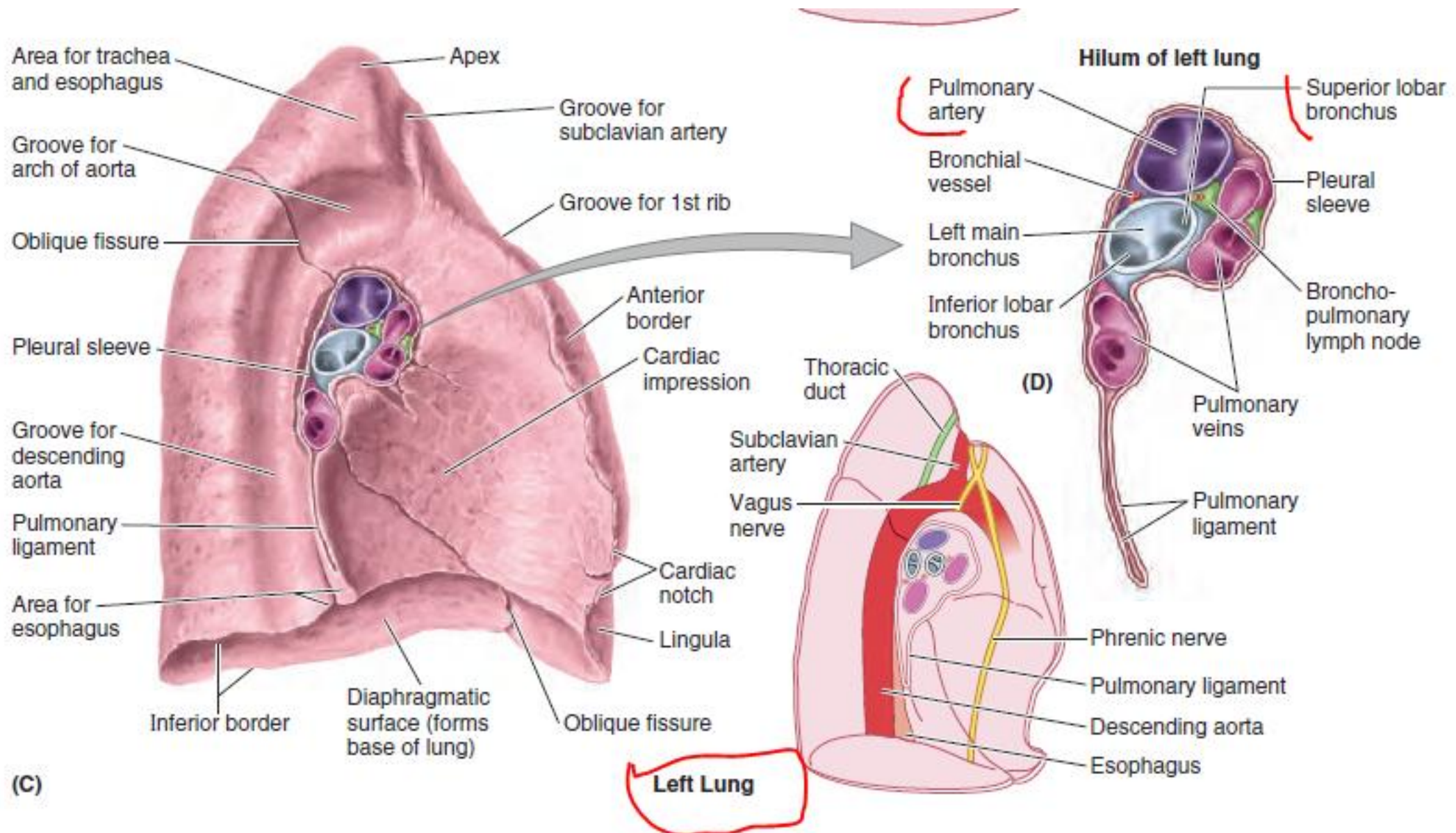
❖NOTE HEAR!!

✓ *pulmonary arteries are called arteries because they are leaving the heart.*

✓ *In this case the pulmonary veins are oxygen-rich.*

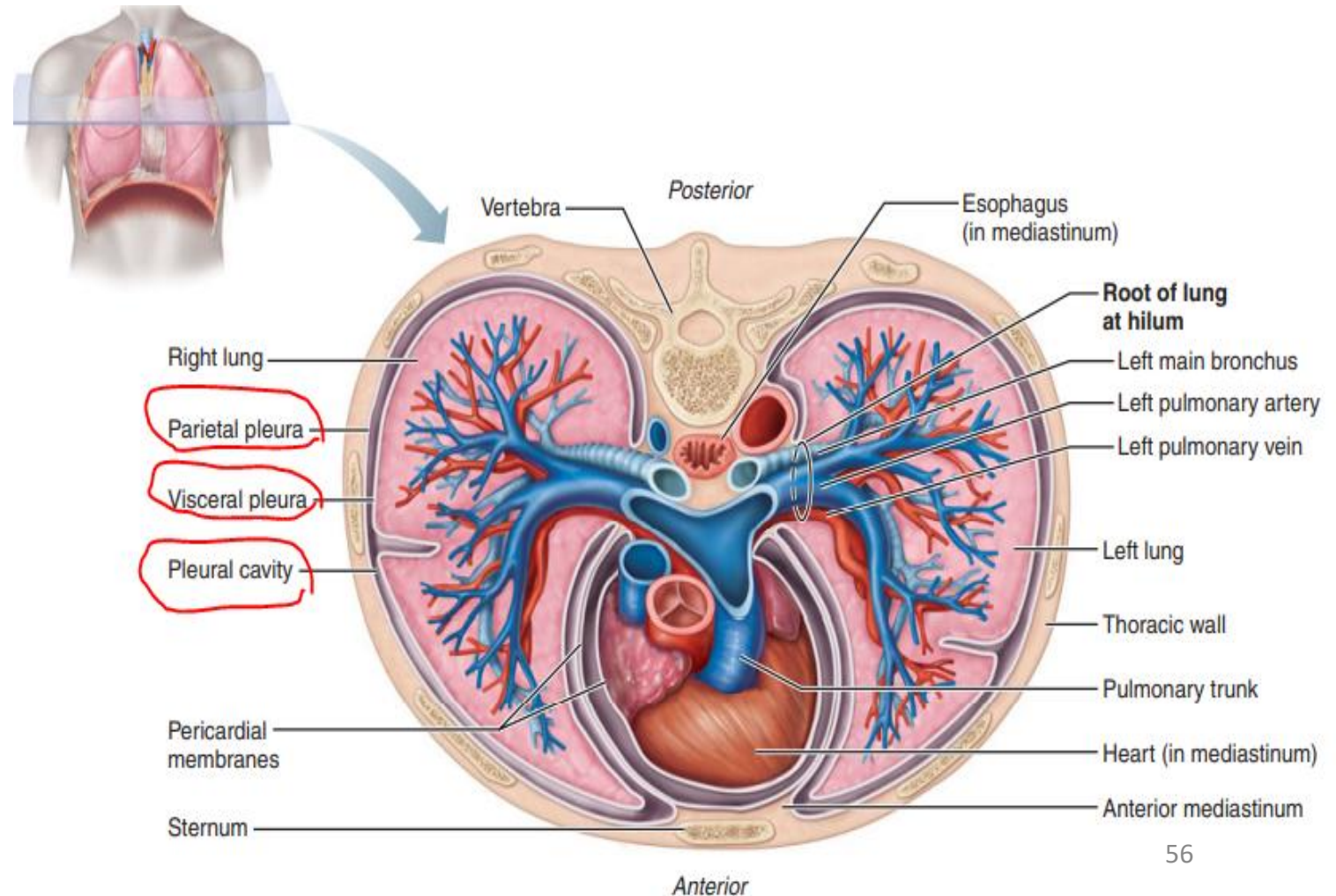
✓ *Veins also return blood to the heart.*

Root of the Left lungs,,,



The Pleurae

- Around each lung is a flattened sac whose walls consist of a serous membrane called **pleura**.
- The outer layer of this sac is **the parietal pleura**, whereas the inner layer, directly on the lung, is **the visceral pleura**.



The Pleurae,,,

The **parietal pleura** lines the pulmonary cavities, adhering to the **thoracic wall, the mediastinum, and the diaphragm.**

- **The pleural cavity** is a potential space b/n **visceral and parietal pleurae** that contains **serous pleural fluid**.
- This fluid lubricates the pleural surfaces and allows the layers of pleura to slide smoothly over each other during respiration.
- In the area where these vessels enter the lung, the parietal pleura is continuous with the visceral pleura, which covers the external lung surface

The visceral pleura

- The **visceral pleura** is continuous with the parietal pleura at the **hilum of the lung**.
- The **visceral pleura** closely covers the lung & adheres to all its surfaces, including those within **the horizontal and oblique fissures**
- In cadaver dissection, the visceral **pleura cannot** usually be dissected from the surface of the lung.
- The visceral pleura is **innervated by visceral afferent nerves** that accompany bronchial vessels.

The parietal pleura

❖ **The parietal pleura** consists of 4

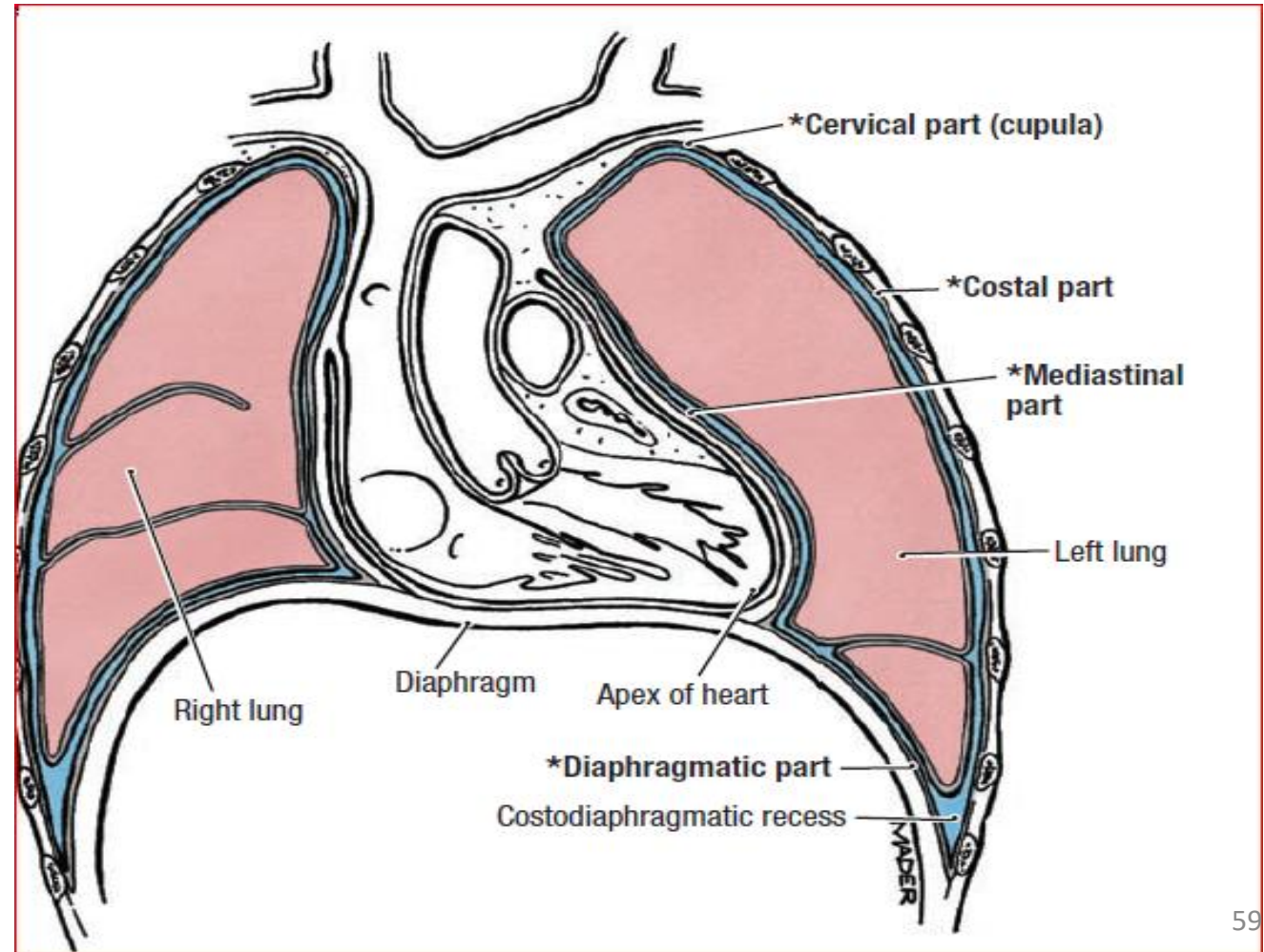
parts:

✓ *Costal,*

✓ *Mediastinal*

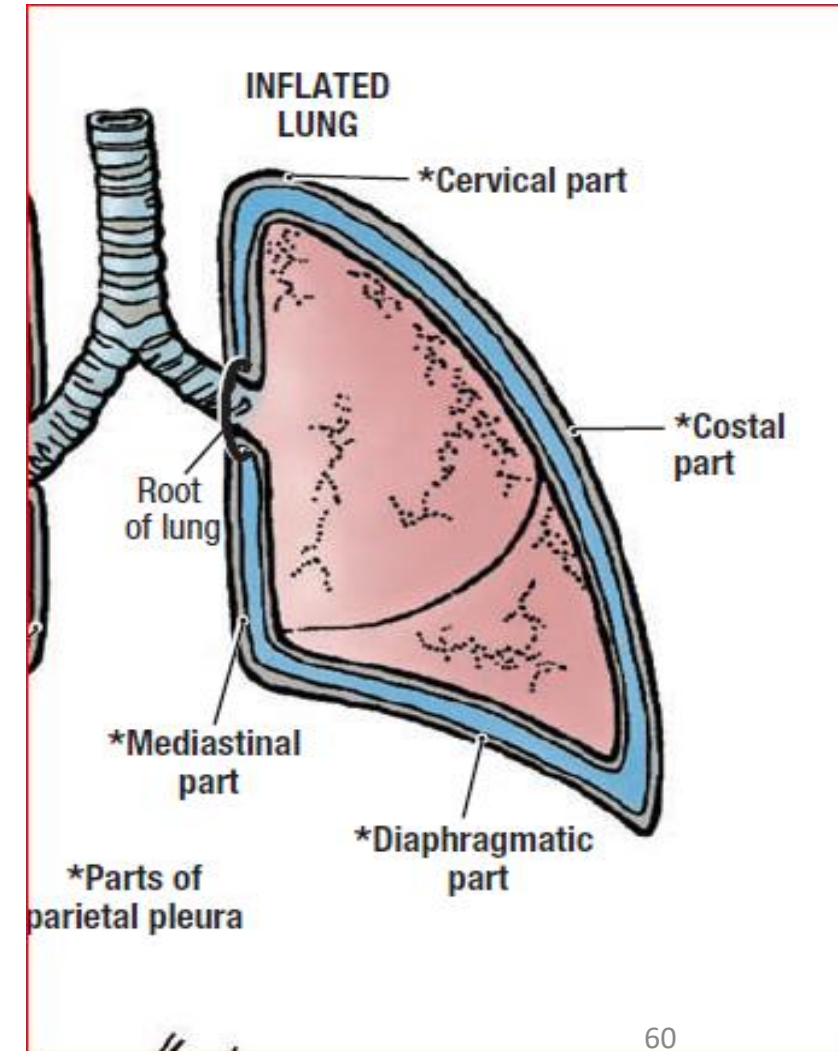
✓ *Diaphragmatic and*

✓ *Cervical pleura*



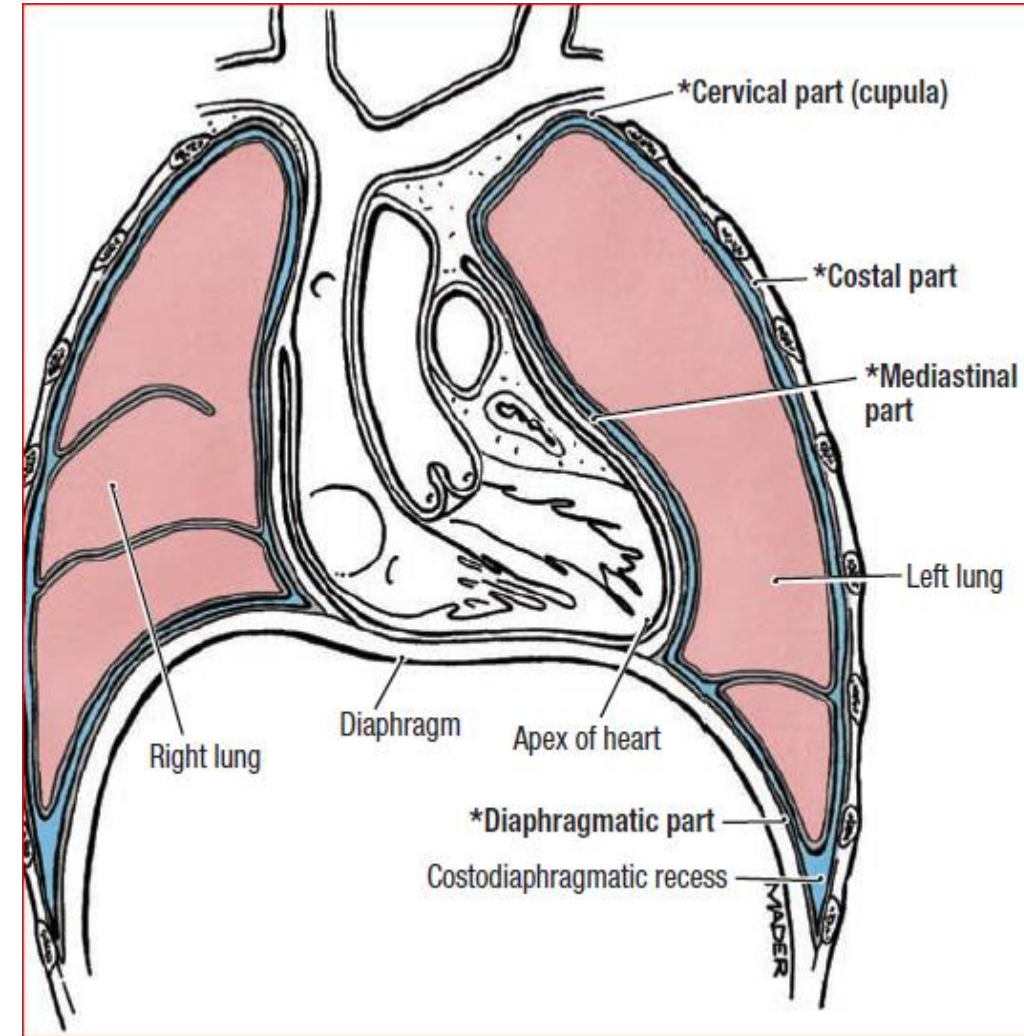
The parietal pleura,,,

- The **costal pleura** lines the inner surfaces of the ribs, the costal cartilages, the intercostal spaces, the **vertebral bodies & the back of the sternum**.
- The **mediastinal part** covers the lateral aspects of the mediastinum.
- It is continuous with costal pleura **anteriorly and posteriorly**, and with the diaphragmatic pleura **inferiorly**.



The parietal pleurae,,,

- At the *hilum of the lung*, it is the **mediastinal pleura** that reflects laterally onto the root of the lung.
- The **diaphragmatic part of the parietal pleura** covers the superior surface of the diaphragm, except along its **costal attachments** and where the diaphragm is fused to the *pericardium*.

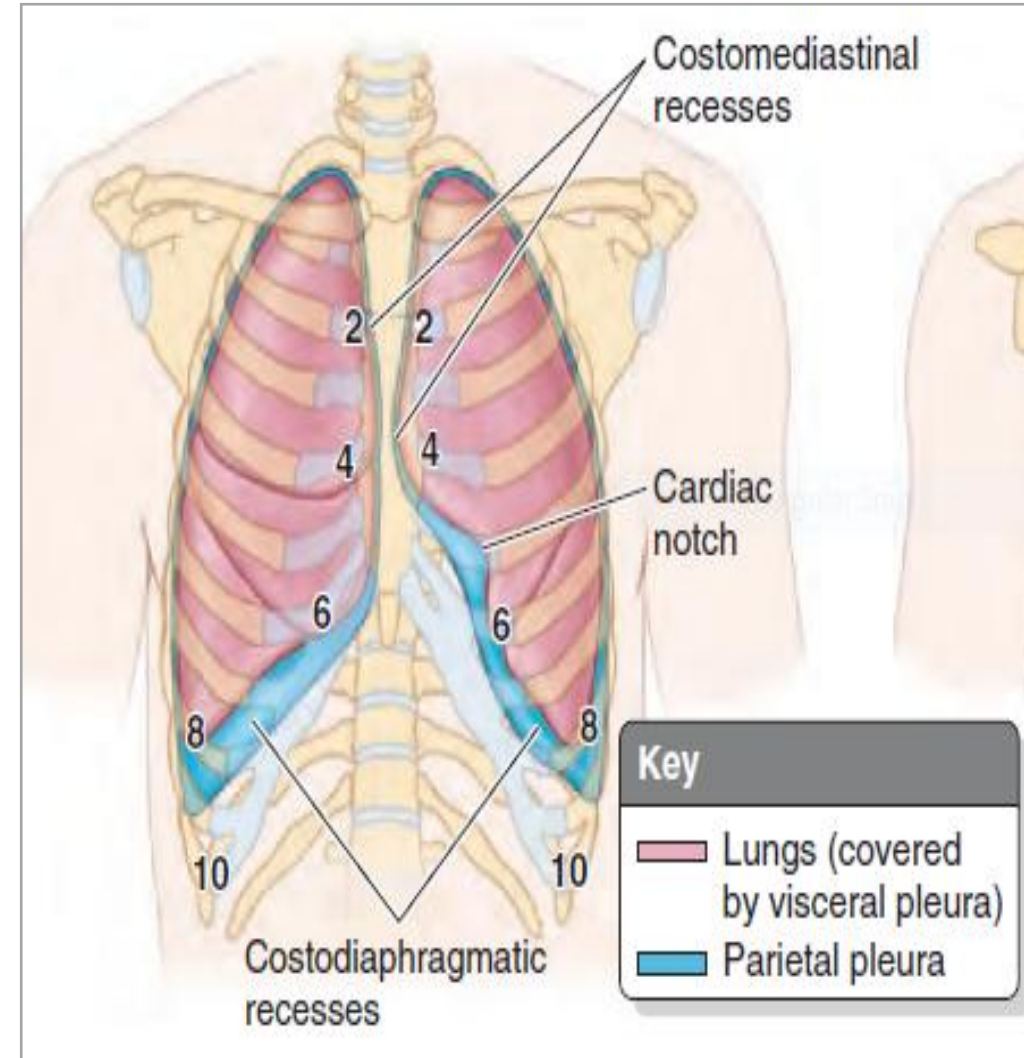


The parietal pleurae,,,

- A thin, more elastic layer of **endothoracic fascia**, the **phrenicopleural fascia**, connects the **diaphragmatic pleura** with the **muscular fibers** of the diaphragm
- The **cervical pleura** covers the apex of the lung.
- It is a superior continuation of the **costal & mediastinal** parts of the parietal pleura.
- The cervical pleura forms **pleural cupula**, cover the apex that reaches 2–3 cm superior to the level of the **medial third of the clavicle** at the level of the neck of the 1st rib.

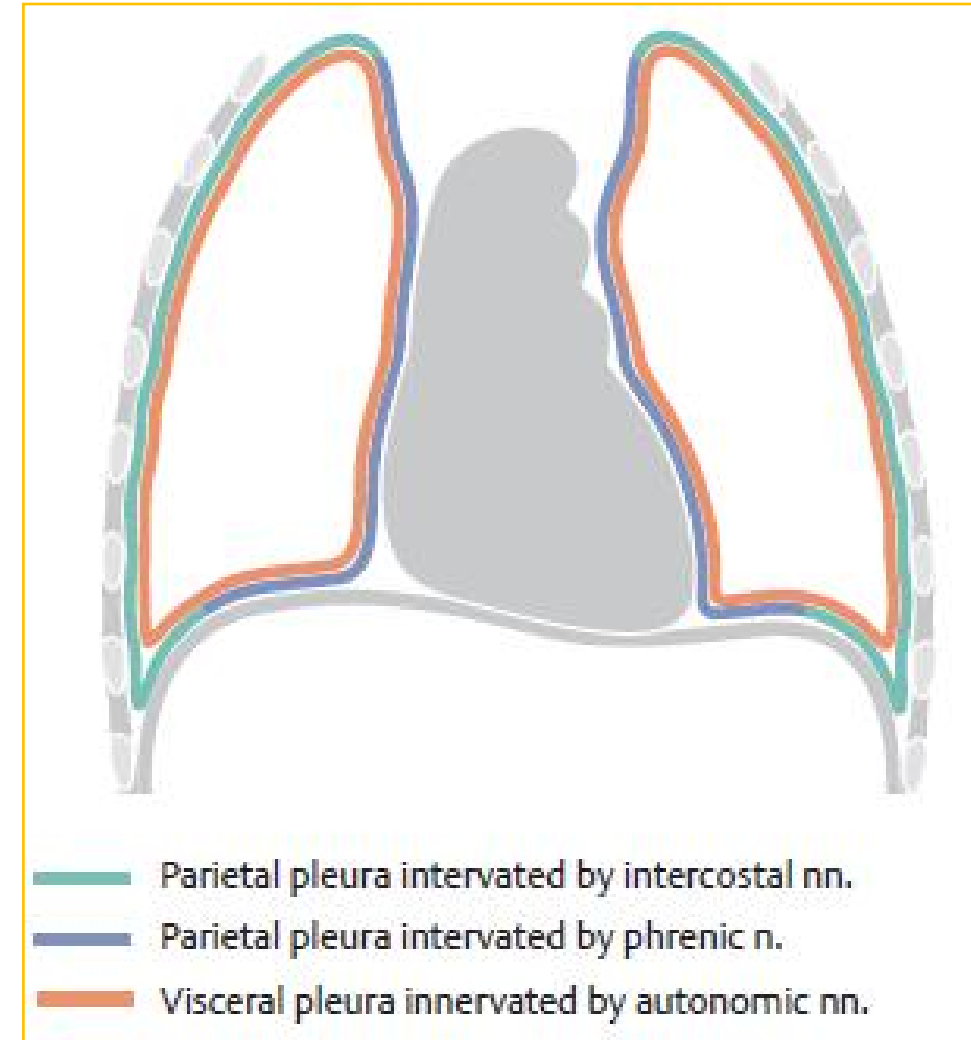
Pleural *recesses*

- The lungs do not completely occupy the pleural cavities especially **during expiration.**
- The potential pleural spaces are the **costodiaphragmatic recesses.**
- **Similar but smaller** pleural recesses are the **costomediastinal recesses.**
- The potential **costomediastinal** recesses; the left recess is potentially larger (less occupied) because of the **cardiac notch in the left lung.**
- The inferior borders of the lungs move farther into the pleural recesses during deep **inspiration.**



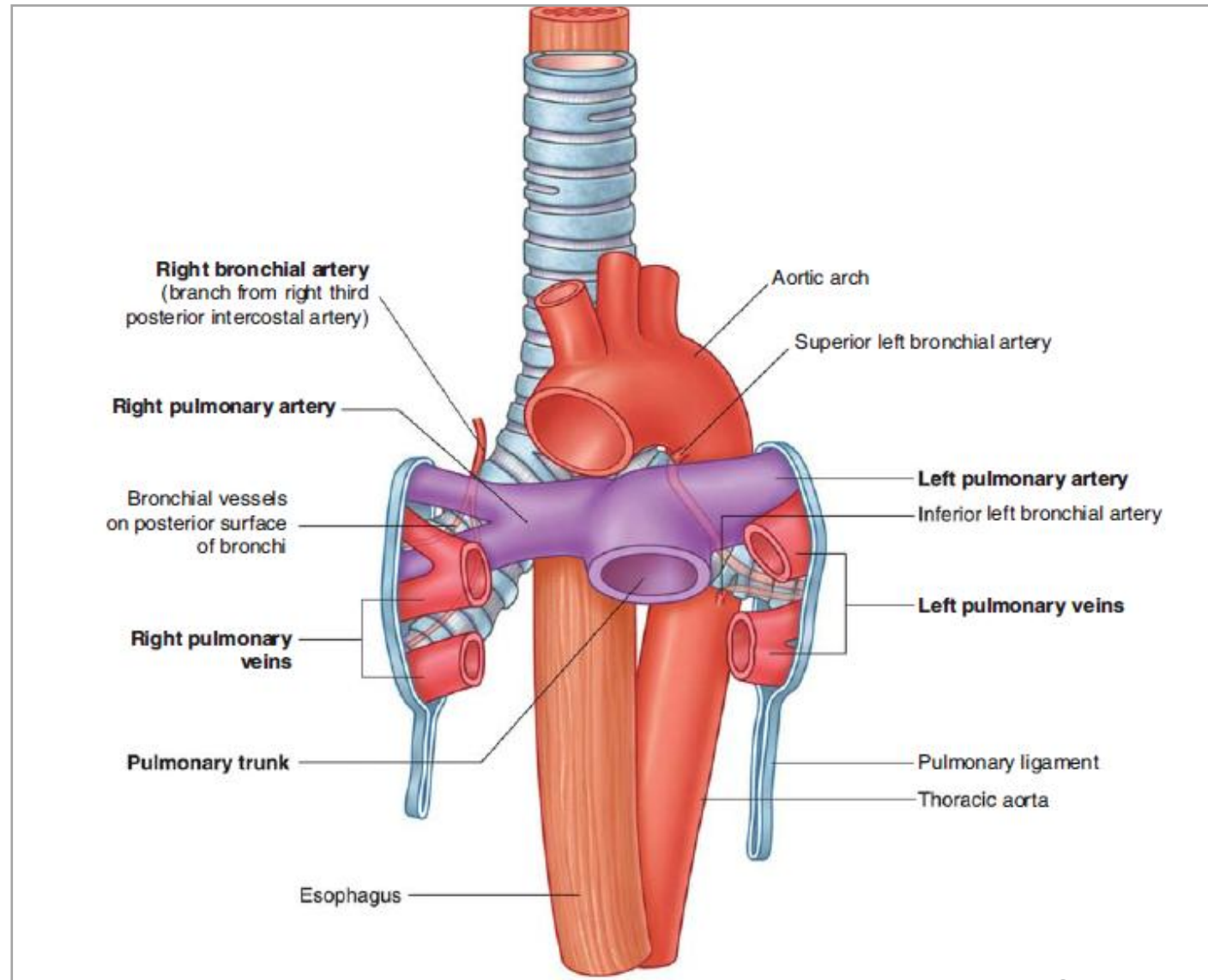
Innervation of the pleurae

- **The parietal pleura:** The **costal, cervical & the periphery of the diaphragmatic** portion are innervated by the **intercostal nerves**.
- ✓ The **mediastinal & central** part of the diaphragmatic pleura are innervated by the **phrenic nerves**.
- **The visceral pleura & the lung itself** receives innervation from the **autonomic NS**.

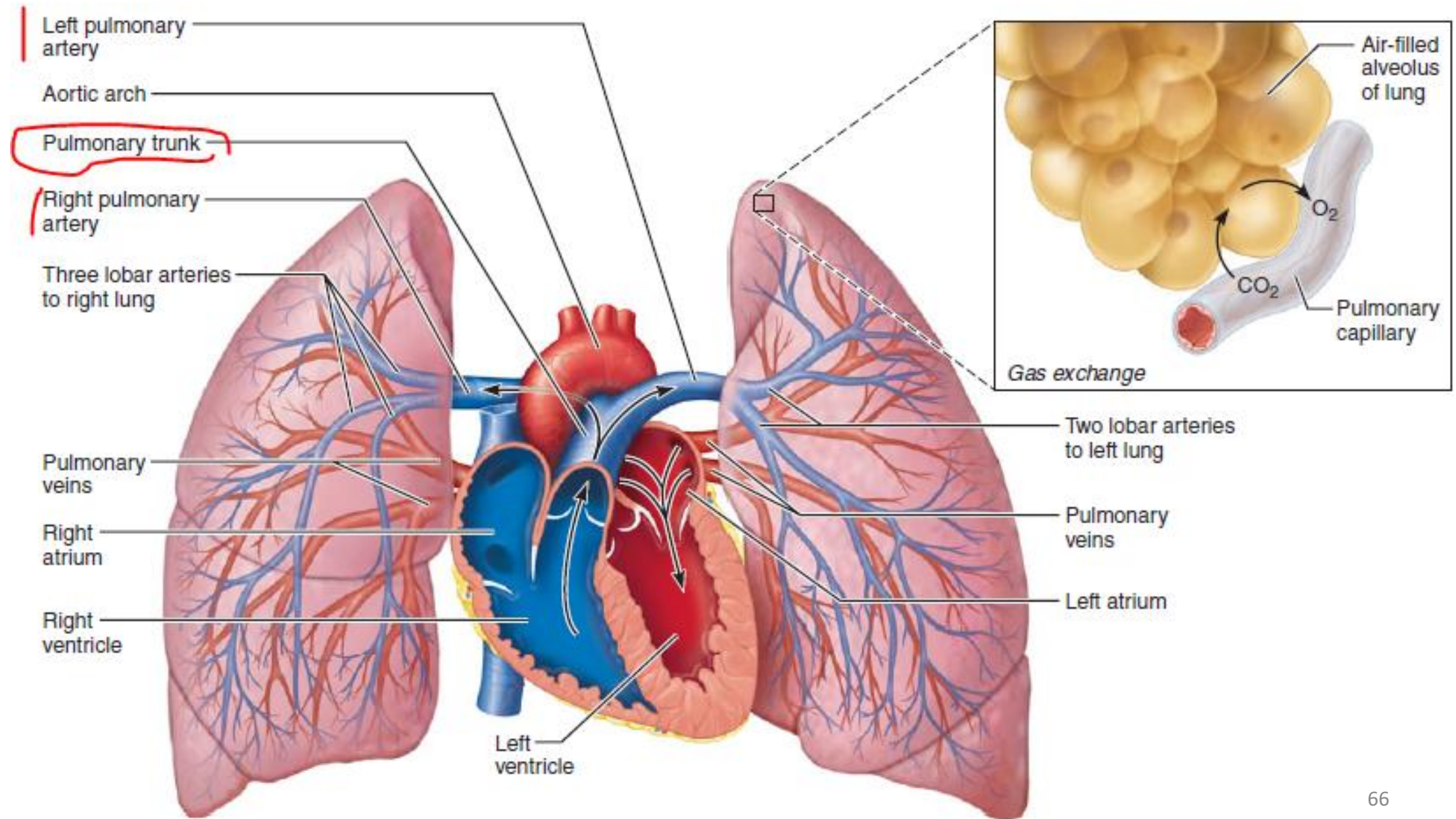


Vasculature of lungs and pleurae

- Each lung has **a pulmonary artery** supplying blood to it & **2 pulmonary** veins draining blood from it .
- The Rt & Left pulmonary arteries arise from the pulmonary trunk at the level of the sternal angle & **carry poorly oxygenated (venous) blood to the lungs for oxygenation**



Vasculature of lungs and pleurae,,



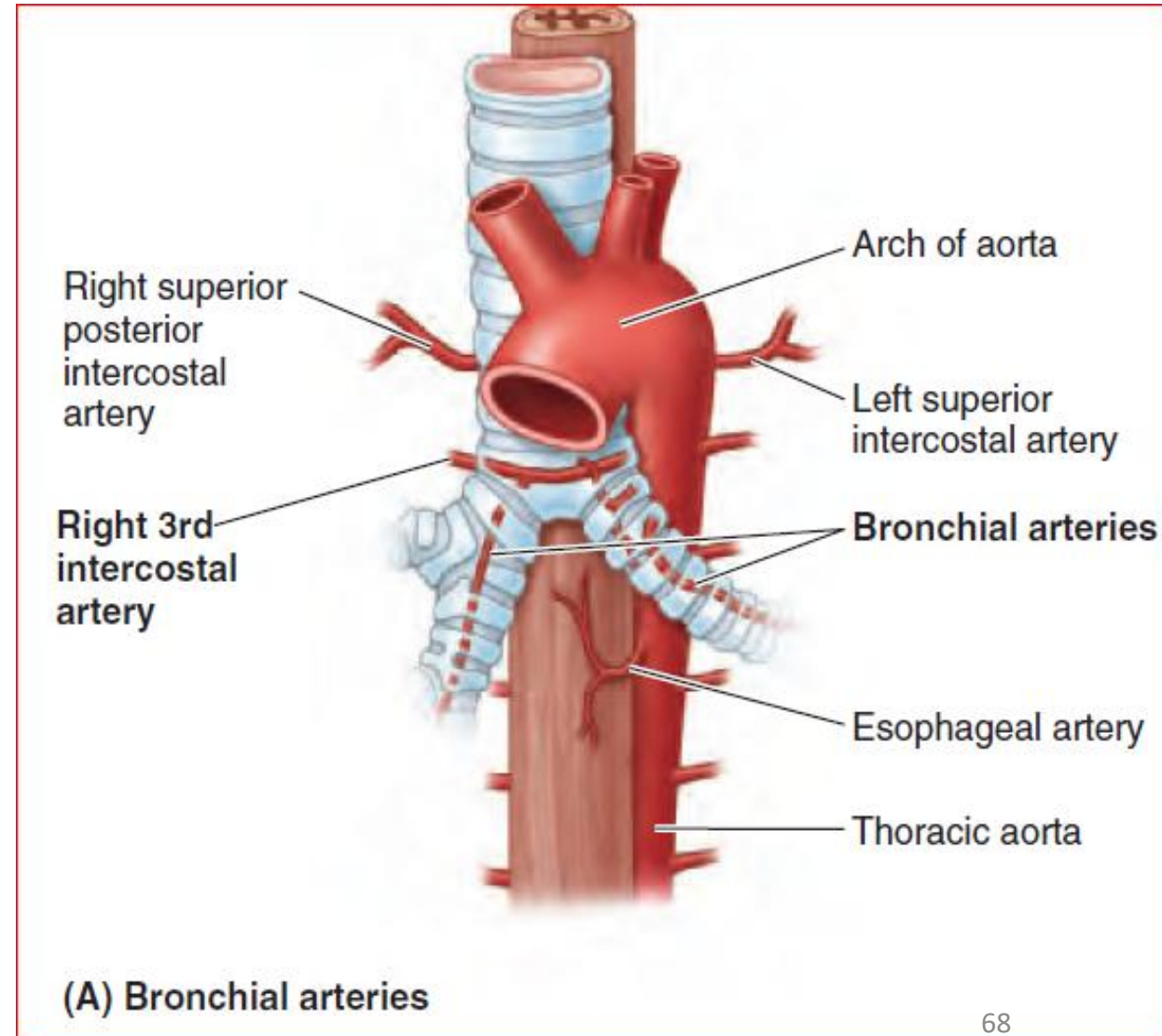
Vasculature of lungs and pleurae,,,

- **Superior & inferior pulmonary vein** on each side, carry oxygen-rich blood from corresponding lobes of each lung to the left atrium of the heart.
- The **middle lobe vein** is a tributary of the right superior pulmonary vein.
- The pulmonary veins run independently of the arteries and bronchi in the lung.

□ **Bronchial arteries** supply blood for nutrition of the structures making up the root of the lungs, the supporting tissues of the lungs, and the visceral pleura.

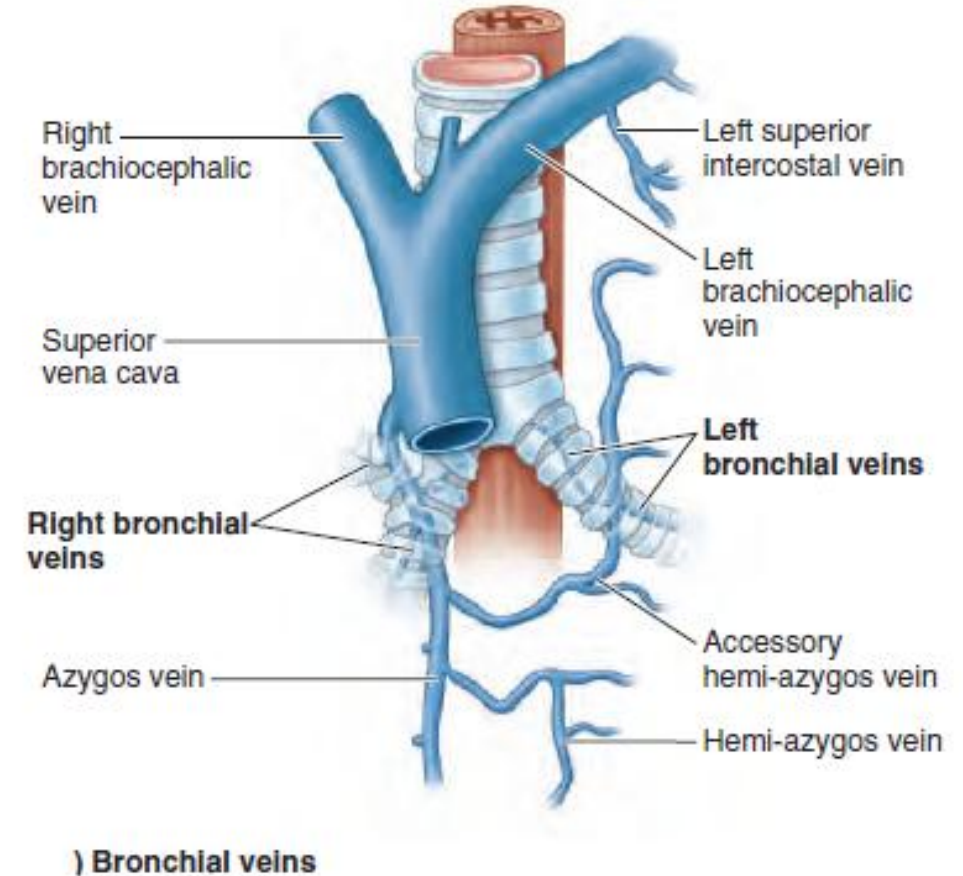
Vasculature of lungs and pleurae,,,

- **Left bronchial arteries arise from the superior thoracic aorta.**
- The origin of the right bronchial artery is variable; it may arise:
 1. from the **right 3rd posterior intercostal artery.**
 2. from a **common trunk shared** with the **left superior bronchial artery.**
 3. **directly from the aorta**



Vasculature of lungs and pleurae,,

- **The bronchial veins drain** the more proximal capillary beds supplied by the **bronchial arteries**.
- The *right bronchial vein* drains into the *azygos vein*, and
- The *left bronchial vein* drains into the *accessory hemi-azygos vein* or the *left superior intercostal vein*.
- **Bronchial veins also receive some blood from esophageal veins.**



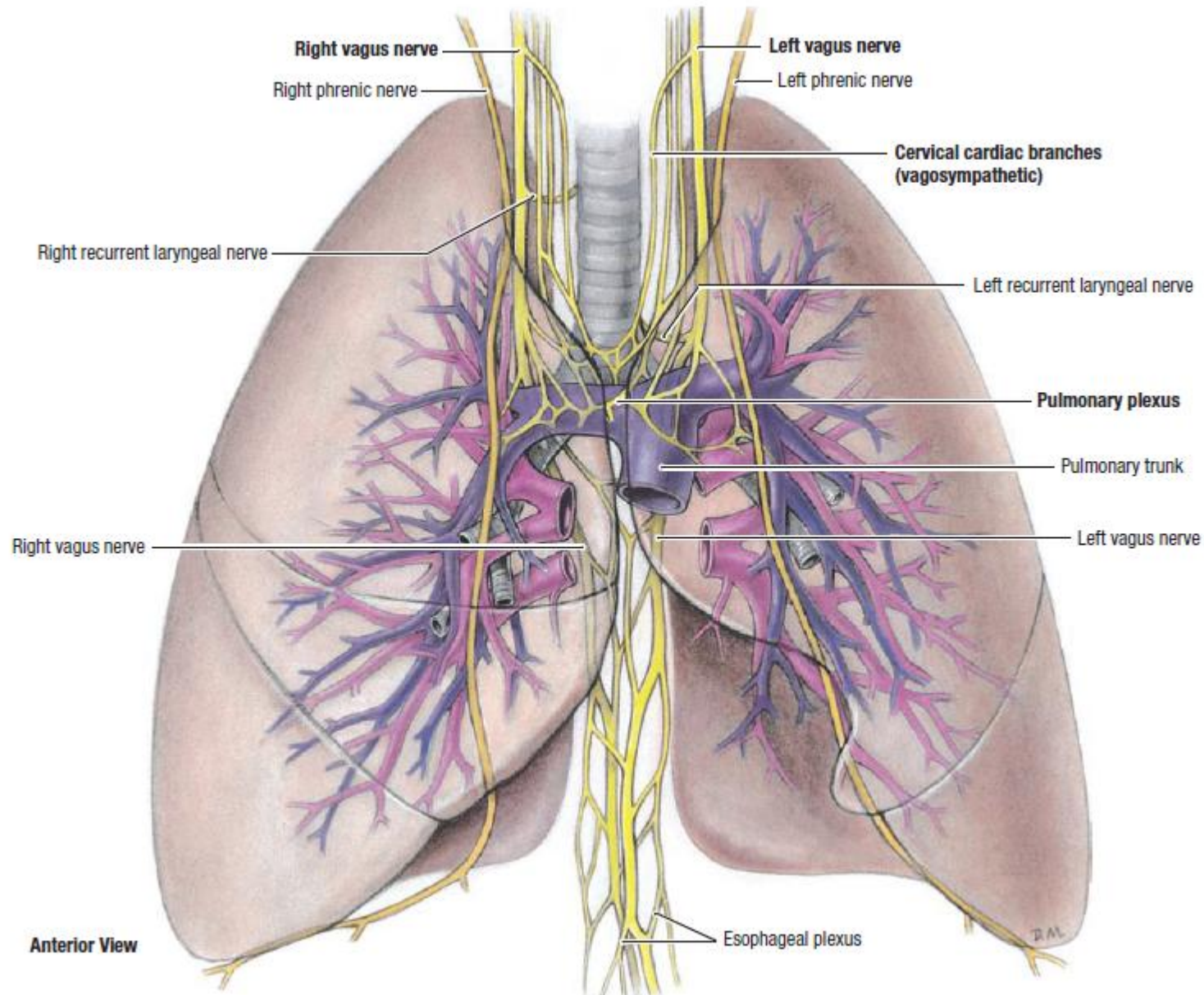
Nerves of the lungs

- The nerve supply of the lungs is derived from *anterior & posterior pulmonary branches* of the **vagus** which are joined by rami of *2nd, 3^r & 4th thoracic sympathetic nerves* to form the **pulmonary plexus**.
- These **plexus** contain **parasympathetic, sympathetic & visceral afferent fibers**.
 - The **parasympathetic fibers** are:
 - ✓ **Motor** to the smooth muscle of the **bronchial tree** (*bronchoconstrictor*),
 - ✓ **Inhibitory to the pulmonary vessels** (*vasodilator*), and
 - ✓ **Secretory** to the glands of the bronchial tree (*secretomotor*).

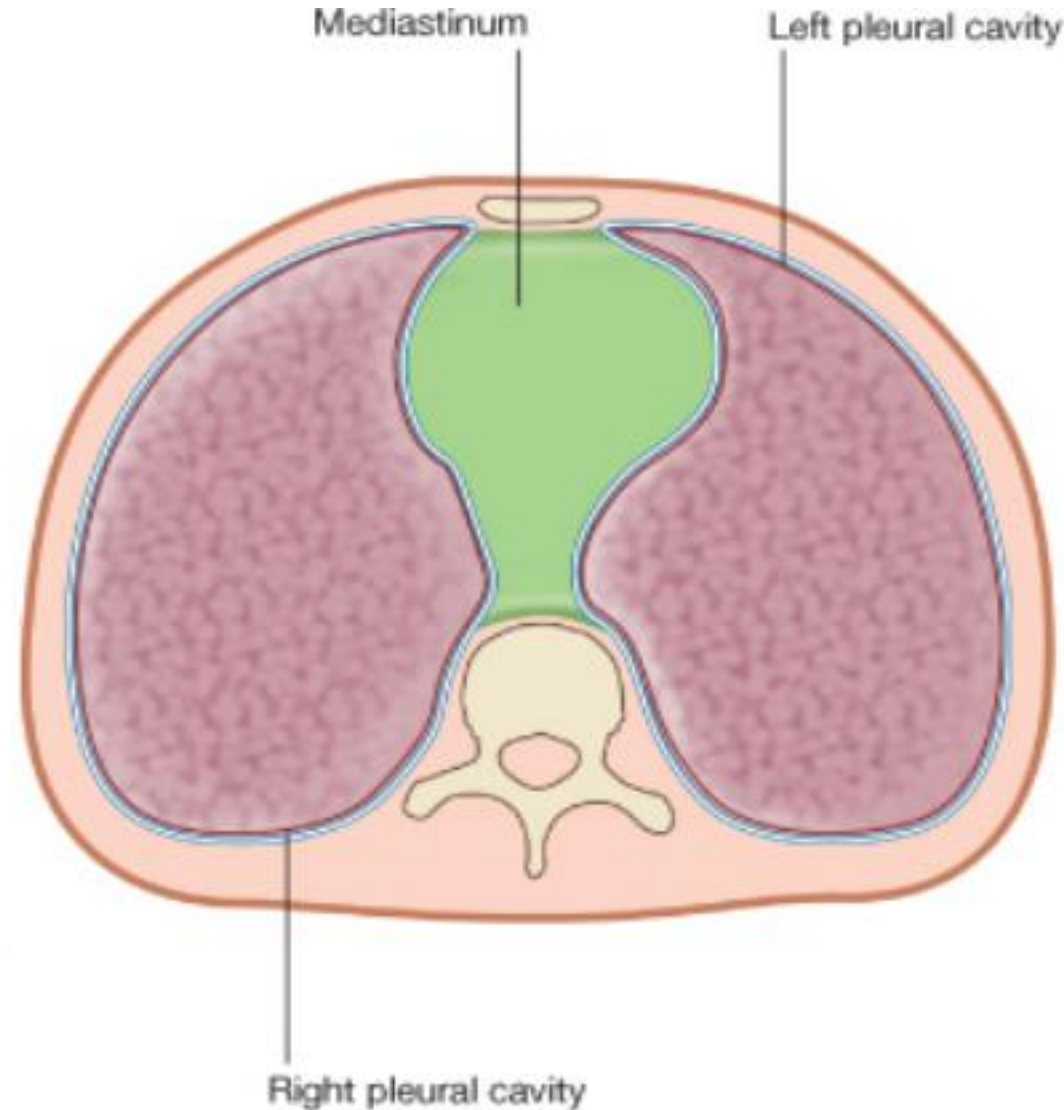
Nerves of the lungs,,

- **The sympathetic fibers** are:
 - ✓ **Inhibitory to the bronchial muscle** (*bronchodilator*),
 - ✓ **Motor to the pulmonary vessels** (*vasoconstrictor*) and
 - ✓ **Inhibitory to the alveolar glands of the bronchial tree.**
- **The *visceral afferent fibers* are either reflexive or nociceptive.**
- ***Reflexive visceral afferent fibers*** with cell bodies in the sensory ganglion of the vagus nerve (CN X) accompany the **parasympathetic** fibers.

Innervation of the lungs,,,



Introduction to *Posterior mediastinum*

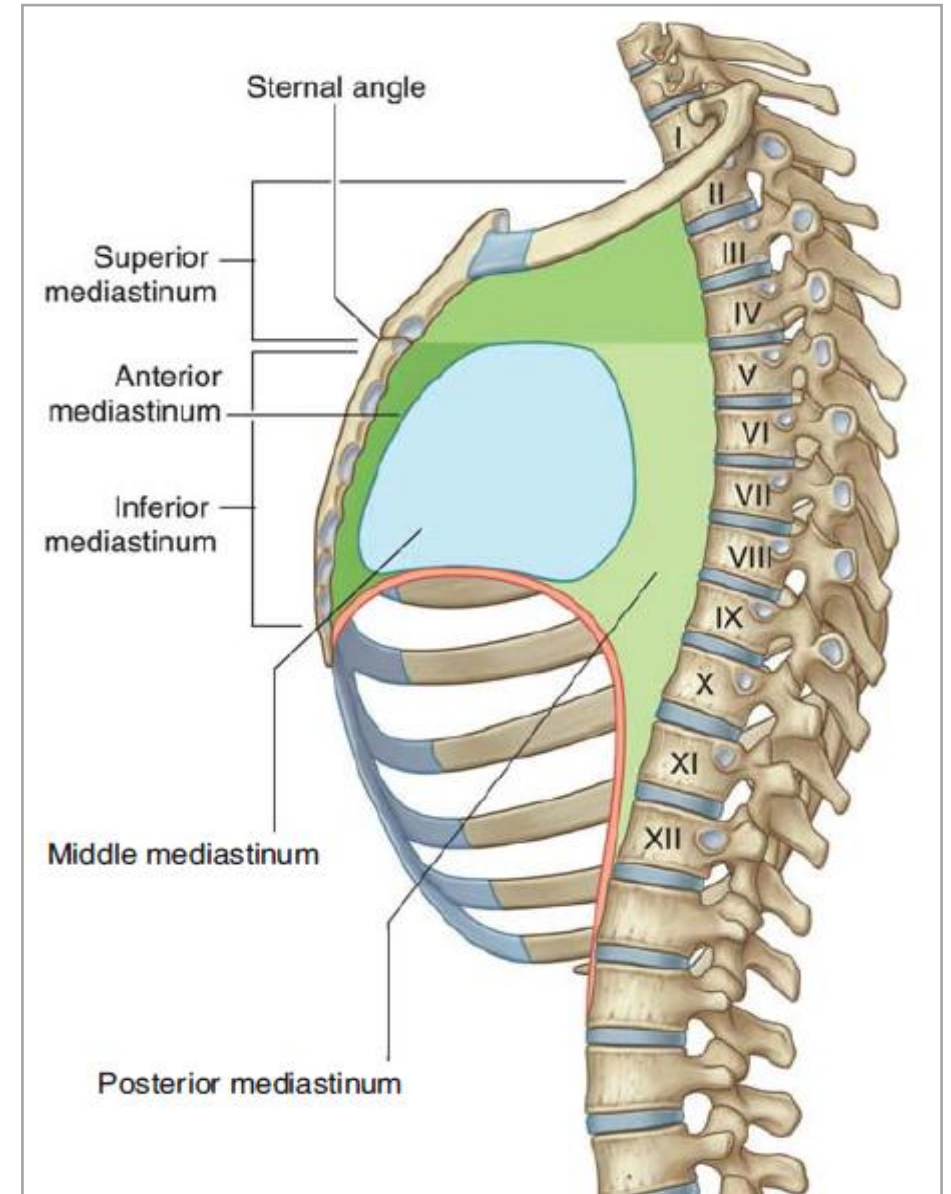


Introduction to *Posterior mediastinum*

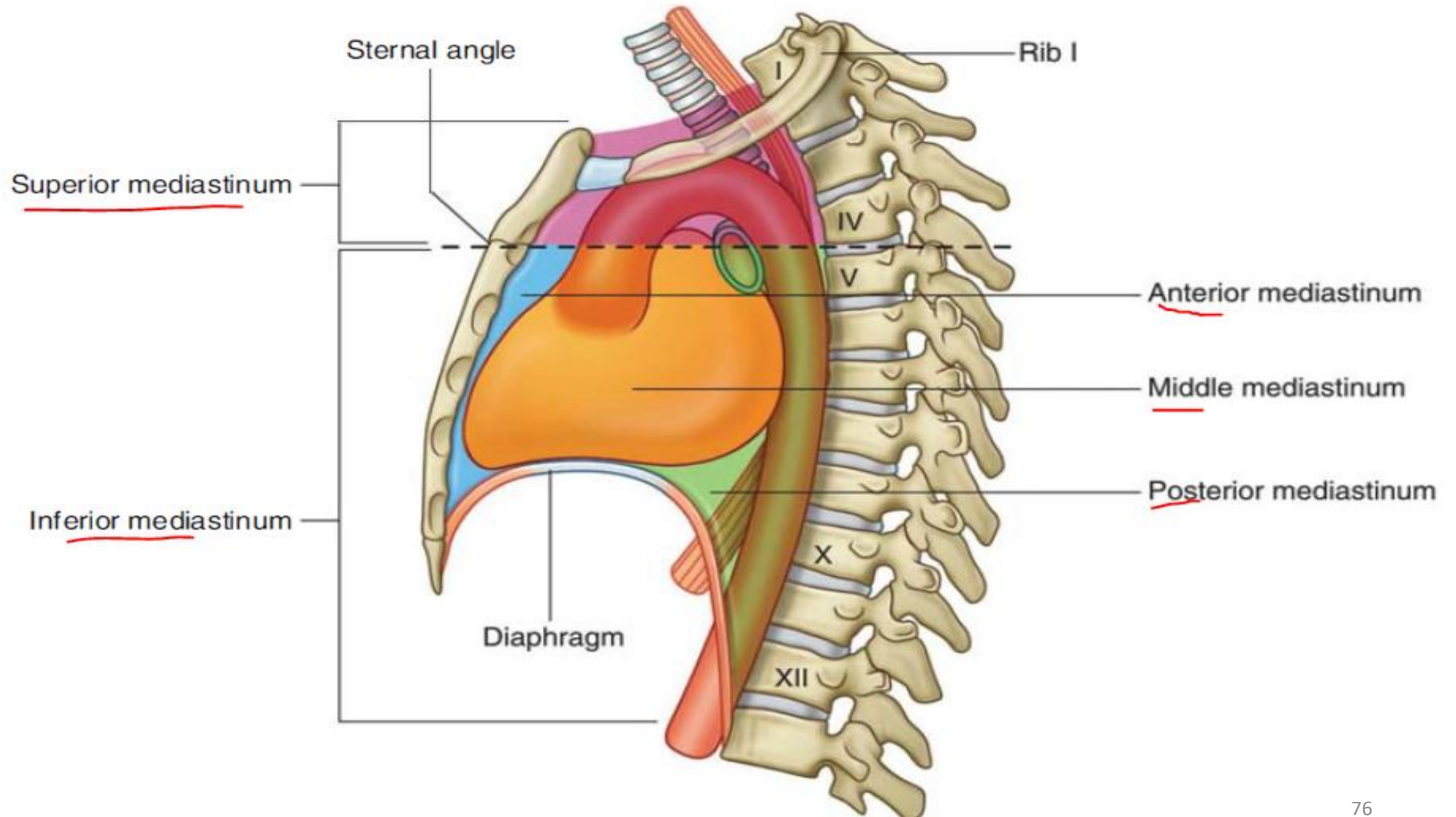
- ❖ The **mediastinum** (L. middle septum), is the central compartment of the thoracic cavity.
- Occupied by the mass of tissue.
- It extends from the **superior thoracic aperture to the diaphragm & from the sternum to the bodies of the T. vertebrae.**
- It serves as a passageway for structures from the neck & head to the abdomen.
- Is covered on each side by mediastinal pleura.
- **contains all the thoracic viscera except the lungs.**

Introduction to *Posterior mediastinum*,,,

- A horizontal plane passing through the sternal angle & to the intervertebral disc b/n vertebrae TIV and TV divides the mediastinum into **superior** and **inferior** parts.
- **Inferior mediastinum** further partitioned into the **anterior**, **middle**, and **posterior** mediastinum by the pericardial sac.

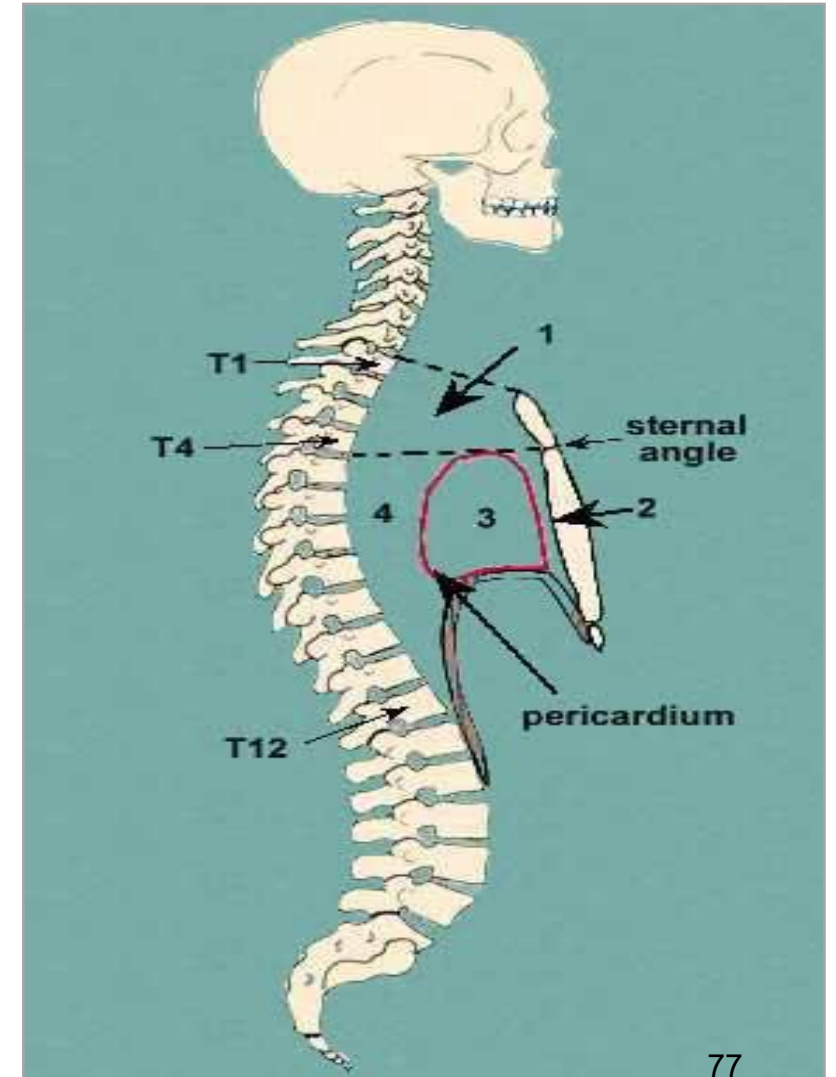


Introduction to *Posterior mediastinum*,,,



Introduction to *Posterior mediastinum*,,

- ❑ The inferior mediastinum, is divided by the pericardial sac into the:
 - anterior(2)
 - middle (3)
 - posterior (4) mediastinum

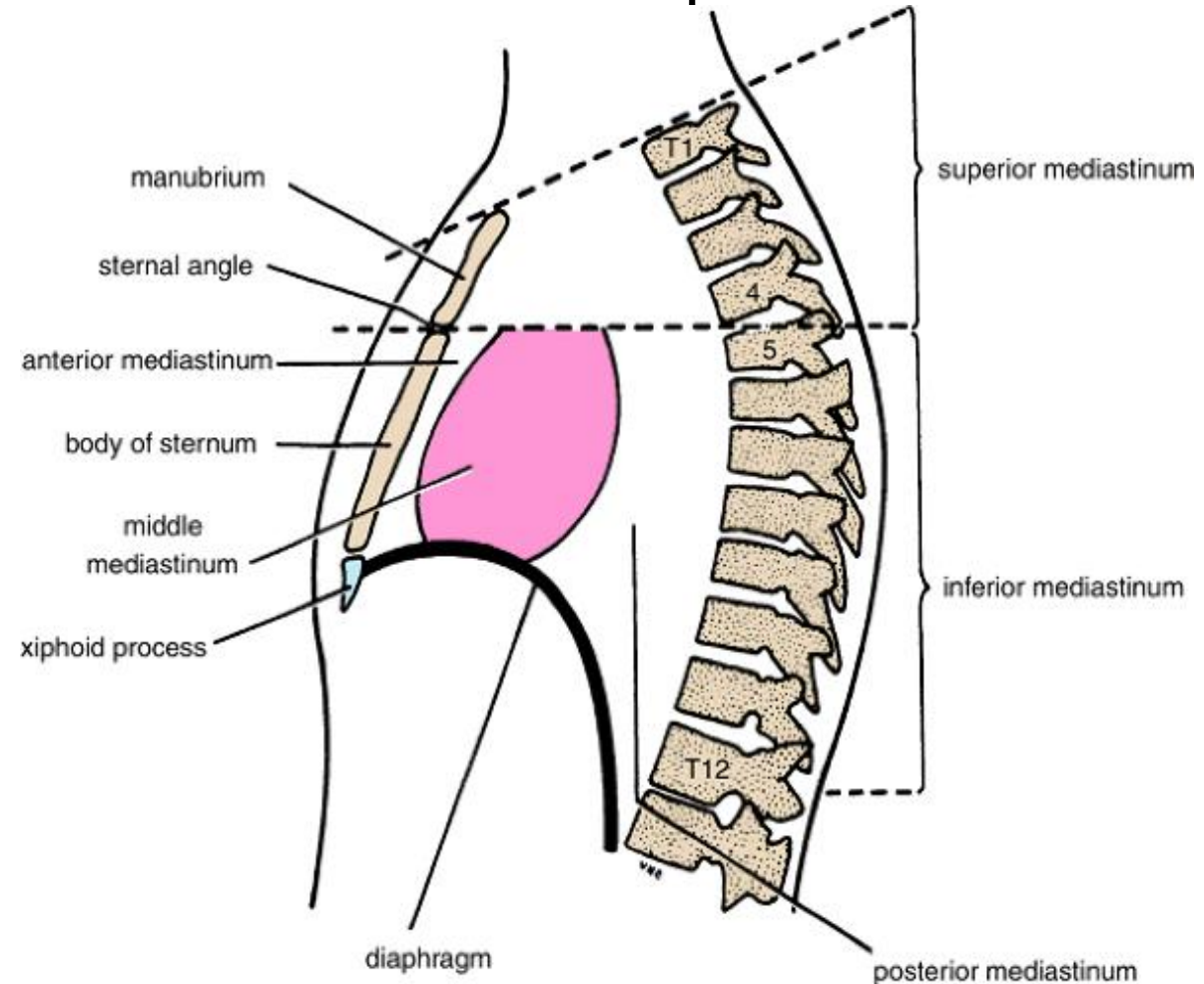


Posterior Mediastinum

The posterior mediastinum (the posterior part of the inferior mediastinum) is located inferior to the transverse thoracic plane

□ Boundaries of posterior mediastinum:

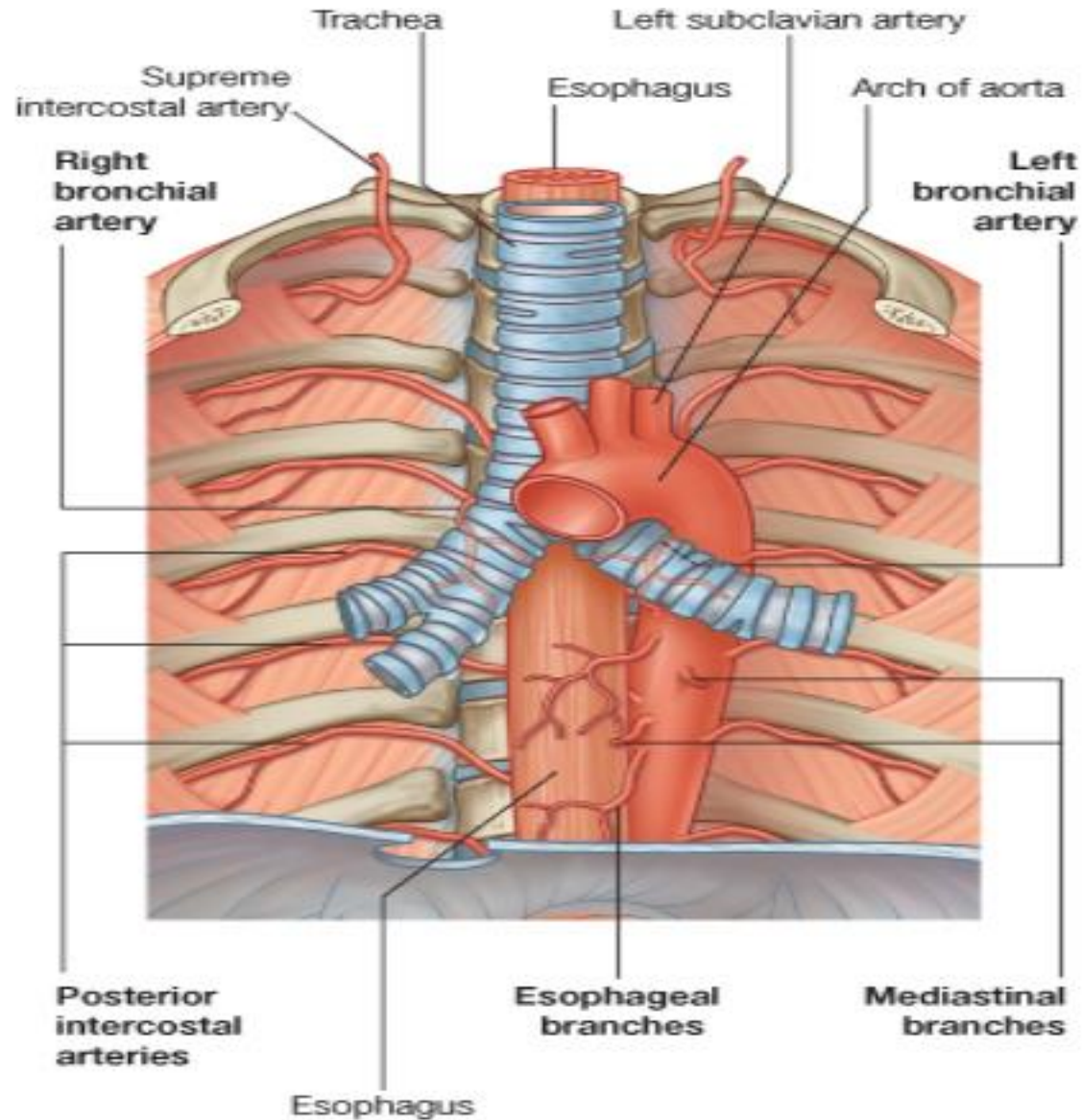
- ✓ **Posteriorly** – thoracic vertebrae (T5- T12)
- ✓ **Superiorly**: boundary is a transverse plane passing from the **sternal angle** to the vertebra T4/T5
- ✓ **Inferiorly**: diaphragm;
- ✓ **Laterally**: mediastinal part of **parietal pleura** on either side;



Contents of Posterior Mediastinum

- The posterior mediastinum contains the **thoracic aorta, thoracic duct and lymphatic trunks, posterior mediastinal lymph nodes, azygos and hemiazygos veins, and esophagus and esophageal nerve plexus**
- Some authors also include the thoracic sympathetic trunks and thoracic splanchnic nerves; however, these structures lie lateral to the vertebral bodies and are not within the posterior mediastinal compartment or space per se.

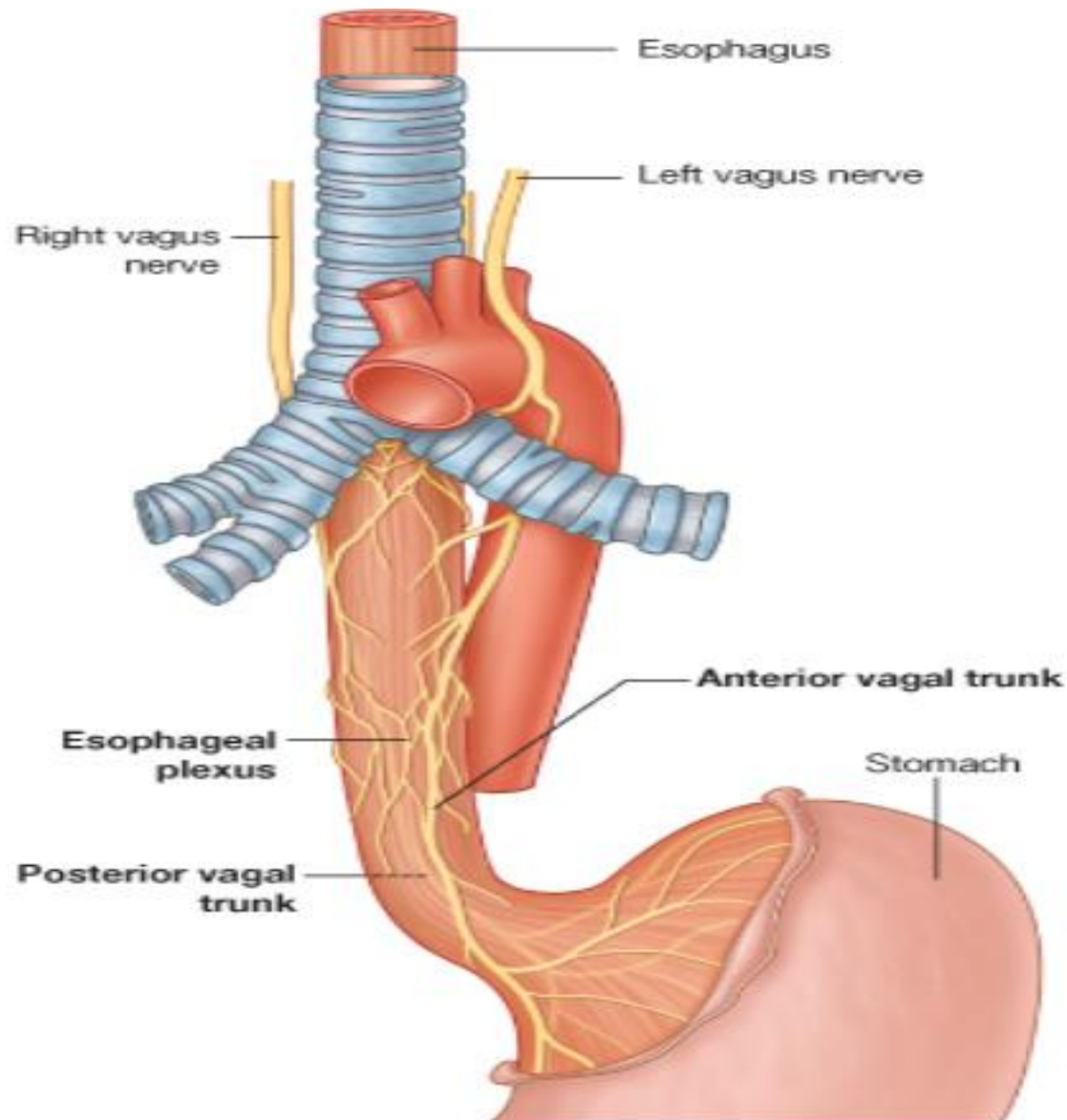
Contents of Posterior Mediastinum,,,



Contents of Posterior Mediastinum,,

- ❖ **The esophagus** descends into the posterior mediastinum from the superior mediastinum, passing posterior and to the right of the arch of the aorta and posterior to the pericardium and left atrium
- The esophagus constitutes the primary posterior relationship of the base of the heart
- It then deviates to the left and passes through the esophageal hiatus in the diaphragm at the level of the T10 vertebra, anterior to the **aorta**

Contents of Posterior Mediastinum,,,

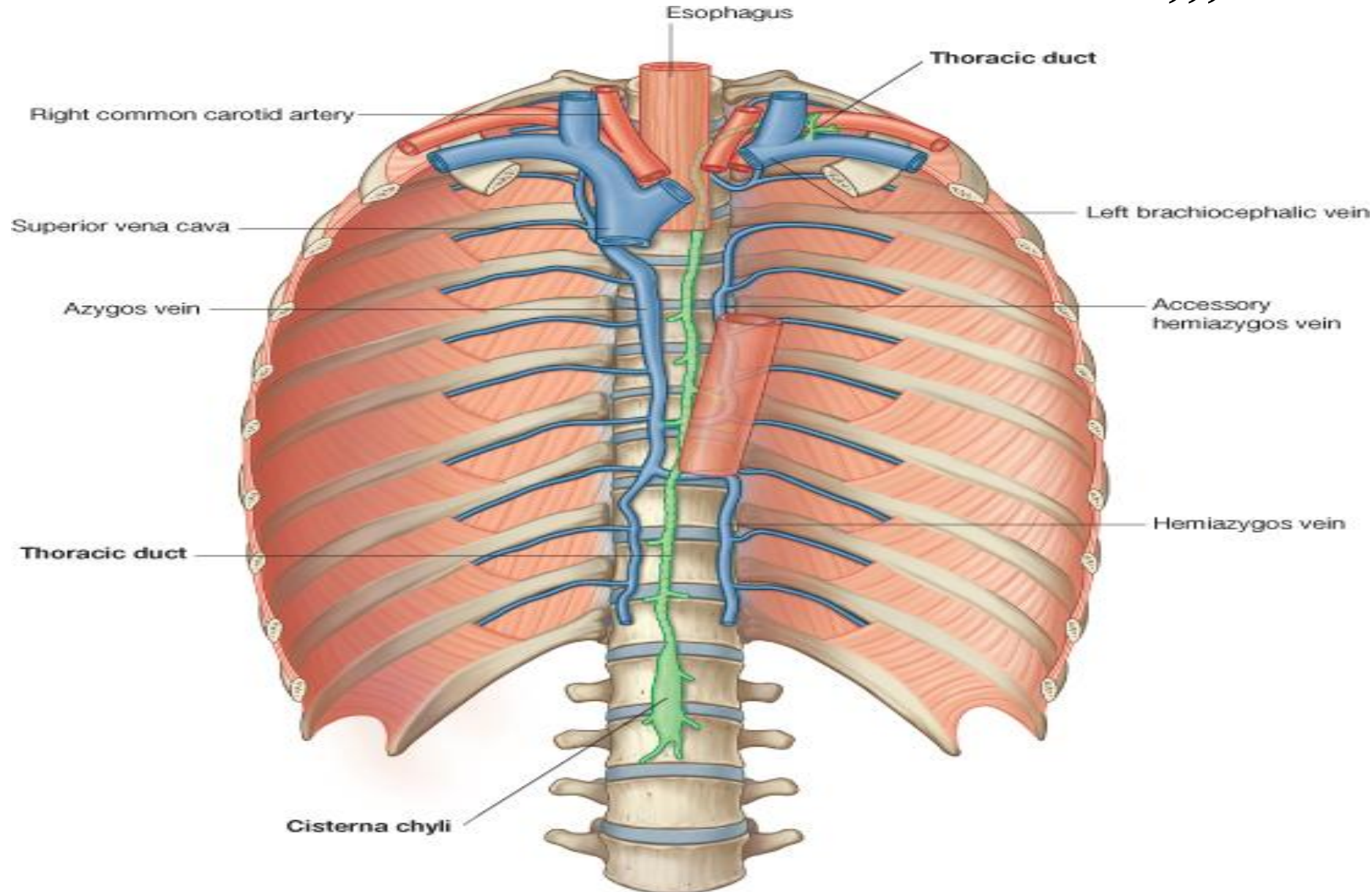


Contents of Posterior Mediastinum,,,

❖ Thoracic Duct and Lymphatic Trunks

- the posterior mediastinum, **the thoracic duct lies** on the anterior aspect of the bodies of the inferior 7 thoracic vertebrae
- **It is the largest lymphatic channel in the body**
- The thoracic duct conveys most lymph of the body to the venous system, that from the lower limbs; pelvic cavity; abdominal cavity; left upper limb; and left side of the thorax, head, and neck, that is, all but the right superior quadrant

Contents of Posterior Mediastinum,,,

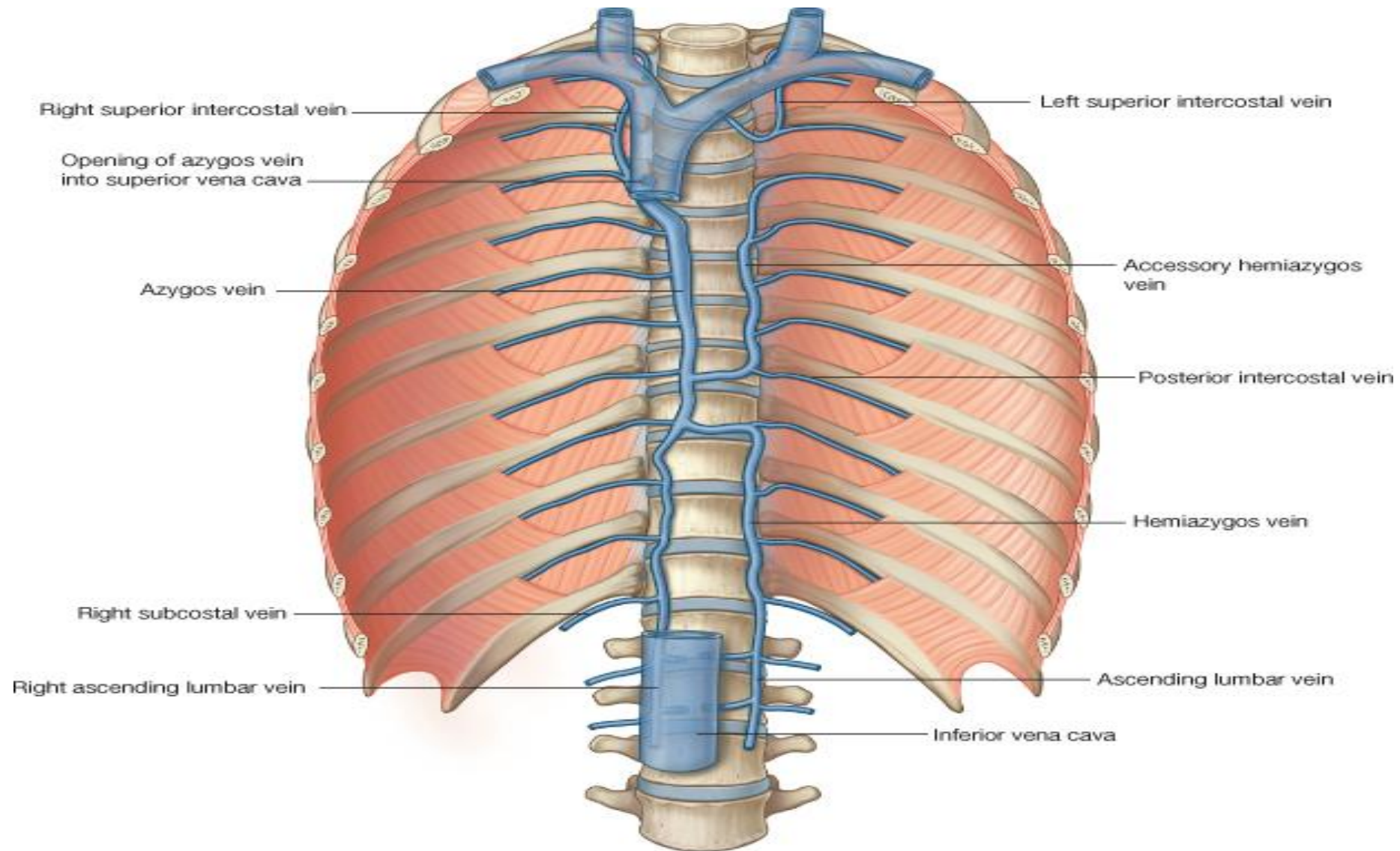


Contents of Posterior Mediastinum,,,

❖ Veins of the posterior mediastinum

- The azygos system of veins, on each side of the vertebral column, drains the back and thoracoabdominal walls as well as the mediastinal viscera
- The azygos system exhibits much variation in its origin, course, tributaries, and anastomoses

Contents of Posterior Mediastinum,,,



Contents of Posterior Mediastinum,,,

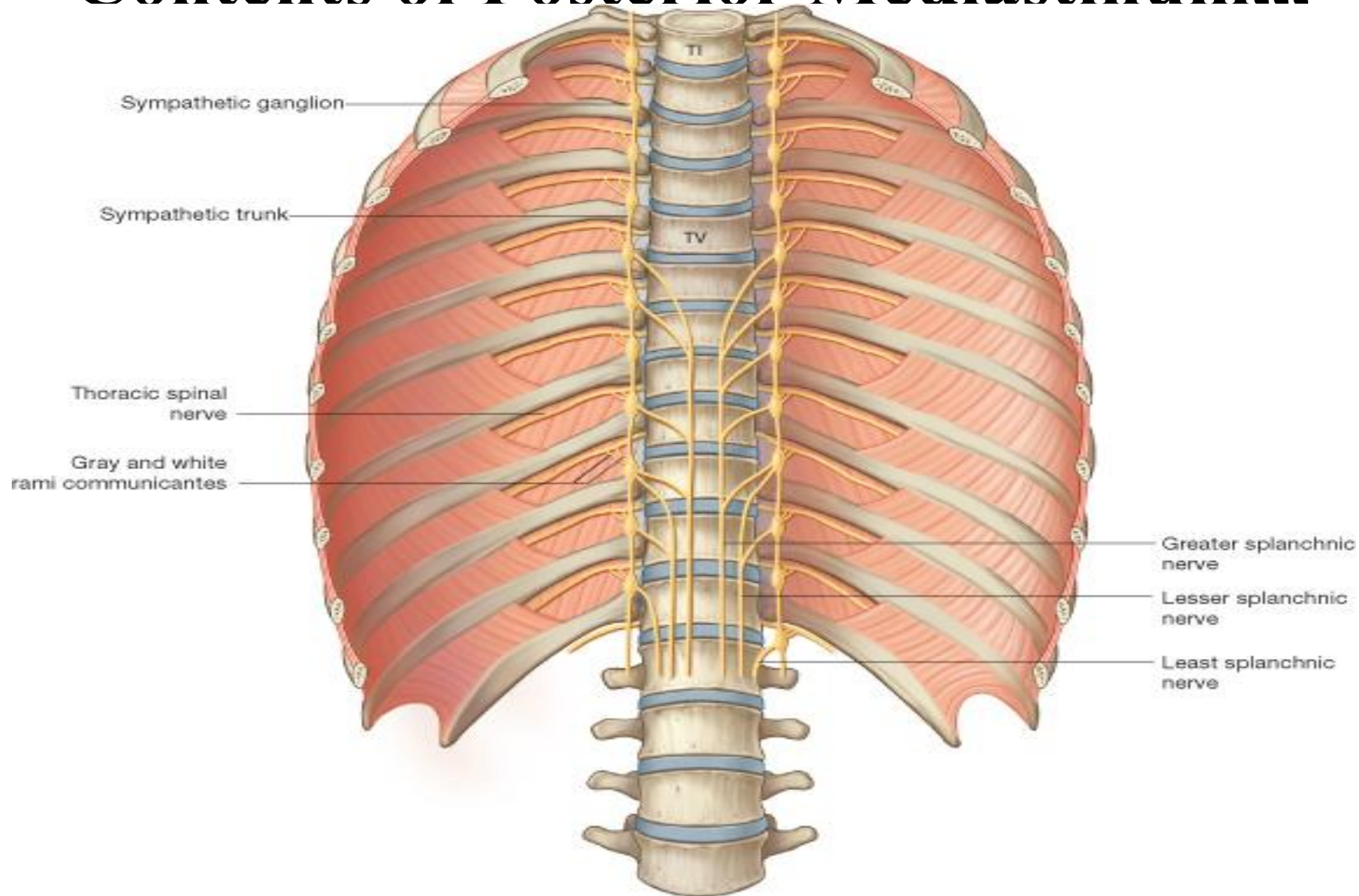
❖ Nerves of the Posterior Mediastinum

- The sympathetic trunks and their associated ganglia form a major portion of the autonomic nervous system
- **The thoracic sympathetic trunks** are in continuity with the cervical and lumbar sympathetic trunks
- The thoracic trunks lie against the heads of the ribs in the superior part of the thorax, the costovertebral joints in the midthoracic level, and the sides of the vertebral bodies in the inferior part of the thorax

Contents of Posterior Mediastinum,,,

- The lower thoracic splanchnic nerves also known as *greater, lesser, and least splanchnic nerves* are part of the abdominopelvic splanchnic nerves because they supply viscera inferior to the diaphragm
- They supply sympathetic innervation for most of the abdominal viscera

Contents of Posterior Mediastinum...





**Thank you
for your attention**